

# Sid Mathur's Thematic ETF Screener – Daily Report

2026-05-12

## Contents

|                           |    |
|---------------------------|----|
| Action Center             | 2  |
| Full Sample PC1           | 4  |
| Position Sheet            | 6  |
| Cluster Overview          | 9  |
| Net Signal Strength (NSS) | 10 |
| Cluster Details           | 12 |
| Long ETF Positions (full) | 66 |
| Diversifier Candidates    | 78 |
| Run Parameters            | 79 |

**Action Center****Run date:** 2026-05-12   **Universe:** 668 ETFs / 26 clusters**Full PC1:** LONG since 2026-04-13 (D+20, 4.0%)   **Clusters long:** 17/26   **ETFs long:** 502/668 (75%)**Today's events****BUY (21)**

| level   | id        | name                             | category               | cluster_signal | spread_buy | prob_buy | nss    |
|---------|-----------|----------------------------------|------------------------|----------------|------------|----------|--------|
| etf     | SIVR      | abrdn Physical Silver Shares ... | Silver                 | FLAT           | 2.057      | 0.95     | 0.796  |
| etf     | SLV       | iShares Silver Trust             | Silver                 | FLAT           | 1.939      | 0.95     | 0.794  |
| etf     | KWEB      | KraneShares CSI China Interne... | China Equity           | LONG           | 0.209      | 0.94     | 0.656  |
| etf     | LTPZ      | Pimco 15 Year U.S. TIPS Index... | Inflation-Protected... | LONG           | 0.047      | 0.87     | 0.292  |
| etf     | PPLT      | abrdn Physical Platinum Share... | Platinum               | FLAT           | 0.746      | 0.66     | 0.155  |
| etf     | SILJ      | Amplify Junior Silver Miners ... | Materials              | FLAT           | 0.041      | 0.66     | 0.141  |
| etf     | BAB       | Invesco Taxable Municipal Bon... | Taxable Municipal B... | LONG           | 0.004      | 0.74     | 0.098  |
| etf     | SIL       | Global X Silver Miners ETF       | Materials              | FLAT           | 0.735      | 0.68     | 0.096  |
| etf     | SUB       | iShares Short-Term National M... | Municipal Bond         | FLAT           | 0.019      | 0.90     | 0.003  |
| etf     | VMBS      | Vanguard Mortgage-Backed Secu... | Mortgage-Backed Fix... | LONG           | 0.055      | 0.78     | -0.083 |
| etf     | BOND      | PIMCO Active Bond Exchange-Tr... | Active Fixed Income    | LONG           | 0.090      | 0.81     | -0.105 |
| etf     | MBB       | iShares MBS ETF                  | Mortgage-Backed Fix... | LONG           | 0.078      | 0.70     | -0.169 |
| cluster | cluster_5 | Mixed: BND, AGG, SCHZ            | Mixed                  |                | 0.035      | 0.70     | -0.177 |
| etf     | GVI       | iShares Intermediate Governme... | Fixed Income           | LONG           | 0.034      | 0.70     | -0.224 |
| etf     | AGG       | iShares Core U.S. Aggregate B... | Aggregate Fixed Inc... | LONG           | 0.028      | 0.65     | -0.232 |
| etf     | BND       | Vanguard Total Bond Market ETF   | Aggregate Fixed Inc... | LONG           | 0.017      | 0.64     | -0.256 |
| etf     | SPAB      | State Street SPDR Portfolio A... | Aggregate Fixed Inc... | LONG           | 0.003      | 0.59     | -0.299 |
| etf     | SCHZ      | Schwab US Aggregate Bond ETF     | Aggregate Fixed Inc... | LONG           | 0.002      | 0.58     | -0.343 |
| etf     | NANR      | State Street SPDR S&P North A... | Materials              | FLAT           | 0.328      | 0.35     | -0.351 |
| etf     | SPMB      | State Street SPDR Portfolio M... | Mortgage-Backed Fix... | LONG           | 0.005      | 0.53     | -0.358 |
| etf     | AGZ       | iShares Agency Bond ETF          | Agency Fixed Income    | LONG           | 0.013      | 0.45     | -0.526 |

**SELL (4)**

| level | id   | name                             | category               | cluster_signal | spread_sell | prob_sell | nss    |
|-------|------|----------------------------------|------------------------|----------------|-------------|-----------|--------|
| etf   | LVHD | Franklin U.S. Low Volatility ... | U.S. Low Volatility... | LONG           | -0.260      | 0.99      | -0.952 |
| etf   | FDL  | First Trust Morningstar ETF      | U.S. Dividend Equity   | LONG           | -0.277      | 0.91      | -0.883 |
| etf   | PKW  | Invesco BuyBack Achievers ETF    | U.S. Buyback Equity    | LONG           | -0.691      | 0.97      | 0.033  |
| etf   | IPAY | Amplify Digital Payments ETF     | Financials             | LONG           | -0.544      | 0.82      | 0.165  |

**Imminent (cluster level,  $p > 50\%$ )****Imminent BUY (top 5)**

| id         | name                             | category                 | spread_buy | prob_buy | nss   |
|------------|----------------------------------|--------------------------|------------|----------|-------|
| cluster_23 | Cash / Ultra-Short Fixed Inco... | Cash / Ultra-Short Fi... | 0.393      | 1        | 1.000 |
| cluster_22 | Mixed: SRLN, BKLN, FTSL          | Mixed                    | 1.305      | 1        | 1.000 |
| cluster_3  | Technology (34/63)               | Technology               | 17.497     | 1        | 1.000 |
| cluster_8  | Emerging Markets Equity (19/2... | Emerging Markets Equi... | 11.512     | 1        | 1.000 |
| cluster_4  | Mixed: IWB, ITOT, SCHB           | Mixed                    | 8.861      | 1        | 0.987 |

**Imminent SELL (top 5)**

| id         | name                    | category         | spread_sell | prob_sell | nss    |
|------------|-------------------------|------------------|-------------|-----------|--------|
| cluster_26 | Currency (1/2)          | Currency         | -1.324      | 1.00      | -1.000 |
| cluster_17 | Consumer Staples (6/11) | Consumer Staples | -0.034      | 0.74      | -0.232 |
| cluster_15 | Mixed: DBP, GLTR, GDXJ  | Mixed            | 0.401       | 0.61      | -0.193 |
| cluster_11 | Healthcare (16/17)      | Healthcare       | -0.362      | 0.87      | 0.021  |
| cluster_18 | Mixed: ILF, EWZ, ECH    | Mixed            | -0.544      | 0.91      | 0.072  |

## Full Sample PC1



Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

**Current state:** LONG since 2026-04-13 (20 days, 4.0%)

5dma: 140.590 20dma: 138.541 50dma: 134.180

Spread vs 50dma: +6.410 Spread vs 20dma: +2.050

P(buy crossover within 5 days): 100.0% P(sell): 4.3%

## Top 10 PC1 weights

|     | ticker | weight | name                                       | category              |
|-----|--------|--------|--------------------------------------------|-----------------------|
| 102 | ACWI   | 0.0021 | iShares MSCI ACWI ETF                      | Global Equity         |
| 113 | AOA    | 0.0021 | iShares Core 80/20 Aggressive Allocatio... | Asset Allocation      |
| 133 | SPGM   | 0.0021 | State Street SPDR Portfolio MSCI Global... | Global Equity         |
| 137 | VT     | 0.0021 | Vanguard Total World Stock Index ETF       | Global Equity         |
| 46  | SPY    | 0.0020 | State Street SPDR S&P 500 ETF Trust        | U.S. Large Cap Equity |
| 116 | AOR    | 0.0020 | iShares Core 60/40 Balanced Allocation ... | Asset Allocation      |

|     |      |        |                                            |                          |
|-----|------|--------|--------------------------------------------|--------------------------|
| 120 | URTH | 0.0020 | iShares MSCI World ETF                     | Global Equity            |
| 122 | CRBN | 0.0020 | iShares Low Carbon Optimized MSCI ACWI ... | Global Equity            |
| 180 | RSP  | 0.0020 | Invesco S&P 500 Equal Weight ETF           | U.S. Equal Weight Equity |
| 264 | IMCG | 0.0020 | iShares Morningstar Mid-Cap Growth ETF     | U.S. Mid Cap Growth      |

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## Position Sheet

Replicate each currently-long cluster via its 3-5 member basket. OLS weights sum to ~1 per cluster for unit-variance replication; scale each cluster's basket to your target cluster allocation. sticky=yes means members are unchanged from the prior run (only weights drift); sticky=no means a fresh greedy selection.

### Active positions (long clusters)

| ticker | cluster    | theme                          | cluster_nss | days_long | cum_return | basket_weight | basket_cor | sticky | etf_signal |
|--------|------------|--------------------------------|-------------|-----------|------------|---------------|------------|--------|------------|
| IEMG   | cluster_8  | Emerging Markets Equity (19... | 1.000       | 21        | 9.9%       | 0.776         | 0.99       | no     | LONG       |
| QQQ    | cluster_3  | Technology (34/63)             | 1.000       | 20        | 15.3%      | 0.398         | 0.98       | no     | LONG       |
| VGT    | cluster_3  | Technology (34/63)             | 1.000       | 20        | 15.3%      | 0.396         | 0.98       | no     | LONG       |
| SRLN   | cluster_22 | Mixed: SRLN, BKLN, FTSL        | 1.000       | 24        | 1.2%       | 0.311         | 0.89       | no     | LONG       |
| FLOT   | cluster_22 | Mixed: SRLN, BKLN, FTSL        | 1.000       | 24        | 1.2%       | 0.292         | 0.89       | no     | FLAT       |
| BKLN   | cluster_22 | Mixed: SRLN, BKLN, FTSL        | 1.000       | 24        | 1.2%       | 0.209         | 0.89       | no     | LONG       |
| VUG    | cluster_3  | Technology (34/63)             | 1.000       | 20        | 15.3%      | 0.198         | 0.98       | no     | LONG       |
| VWO    | cluster_8  | Emerging Markets Equity (19... | 1.000       | 21        | 9.9%       | 0.178         | 0.99       | no     | LONG       |
| FLRN   | cluster_22 | Mixed: SRLN, BKLN, FTSL        | 1.000       | 24        | 1.2%       | 0.127         | 0.89       | no     | FLAT       |
| FLTR   | cluster_22 | Mixed: SRLN, BKLN, FTSL        | 1.000       | 24        | 1.2%       | 0.034         | 0.89       | no     | FLAT       |
| EEM    | cluster_8  | Emerging Markets Equity (19... | 1.000       | 21        | 9.9%       | -0.012        | 0.99       | no     | LONG       |
| VOO    | cluster_4  | Mixed: IWB, ITOT, SCHB         | 0.987       | 20        | 5.9%       | 0.804         | 0.99       | no     | LONG       |
| IVV    | cluster_4  | Mixed: IWB, ITOT, SCHB         | 0.987       | 20        | 5.9%       | 0.309         | 0.99       | no     | LONG       |
| SPY    | cluster_4  | Mixed: IWB, ITOT, SCHB         | 0.987       | 20        | 5.9%       | -0.154        | 0.99       | no     | LONG       |
| VTV    | cluster_6  | Mixed: IWD, SCHV, VONV         | 0.919       | 20        | 2.6%       | 0.546         | 1.00       | no     | LONG       |
| VO     | cluster_6  | Mixed: IWD, SCHV, VONV         | 0.919       | 20        | 2.6%       | 0.278         | 1.00       | no     | LONG       |
| VIG    | cluster_6  | Mixed: IWD, SCHV, VONV         | 0.919       | 20        | 2.6%       | 0.158         | 1.00       | no     | LONG       |
| DBC    | cluster_21 | Broad Commodities (6/9)        | 0.918       | 84        | 47.8%      | 0.662         | 0.98       | no     | LONG       |
| PDBC   | cluster_21 | Broad Commodities (6/9)        | 0.918       | 84        | 47.8%      | 0.535         | 0.98       | no     | LONG       |
| FTGC   | cluster_21 | Broad Commodities (6/9)        | 0.918       | 84        | 47.8%      | -0.029        | 0.98       | no     | FLAT       |
| IGOV   | cluster_24 | Currency (4/6)                 | 0.873       | 7         | 0.3%       | 0.315         | 0.98       | no     | LONG       |
| FXF    | cluster_24 | Currency (4/6)                 | 0.873       | 7         | 0.3%       | 0.306         | 0.98       | no     | LONG       |
| BWX    | cluster_24 | Currency (4/6)                 | 0.873       | 7         | 0.3%       | 0.269         | 0.98       | no     | LONG       |
| MCHI   | cluster_13 | China Equity (13/15)           | 0.858       | 18        | 2.8%       | 1.376         | 0.98       | no     | LONG       |
| KWEB   | cluster_13 | China Equity (13/15)           | 0.858       | 18        | 2.8%       | -0.038        | 0.98       | no     | FLAT       |
| FXI    | cluster_13 | China Equity (13/15)           | 0.858       | 18        | 2.8%       | -0.395        | 0.98       | no     | LONG       |
| VB     | cluster_2  | Mixed: VB, SCHM, IJH           | 0.849       | 20        | 3.7%       | 0.582         | 0.99       | no     | LONG       |
| IJH    | cluster_2  | Mixed: VB, SCHM, IJH           | 0.849       | 20        | 3.7%       | 0.331         | 0.99       | no     | LONG       |
| IWM    | cluster_2  | Mixed: VB, SCHM, IJH           | 0.849       | 20        | 3.7%       | 0.067         | 0.99       | no     | LONG       |
| SCHH   | cluster_14 | Real Estate (14/14)            | 0.816       | 21        | 4.2%       | 0.537         | 1.00       | no     | LONG       |
| VNQ    | cluster_14 | Real Estate (14/14)            | 0.816       | 21        | 4.2%       | 0.515         | 1.00       | no     | LONG       |
| XLRE   | cluster_14 | Real Estate (14/14)            | 0.816       | 21        | 4.2%       | -0.075        | 1.00       | no     | LONG       |
| SCHP   | cluster_19 | Inflation-Protected Fixed I... | 0.787       | 22        | 0.8%       | 0.748         | 0.99       | no     | LONG       |
| DVY    | cluster_10 | U.S. Dividend Equity (11/20)   | 0.787       | 14        | 0.6%       | 0.615         | 0.98       | no     | LONG       |

|      |            |                                |        |    |       |        |      |    |      |
|------|------------|--------------------------------|--------|----|-------|--------|------|----|------|
| HDV  | cluster_10 | U.S. Dividend Equity (11/20)   | 0.787  | 14 | 0.6%  | 0.236  | 0.98 | no | LONG |
| STIP | cluster_19 | Inflation-Protected Fixed I... | 0.787  | 22 | 0.8%  | 0.168  | 0.99 | no | FLAT |
| SCHD | cluster_10 | U.S. Dividend Equity (11/20)   | 0.787  | 14 | 0.6%  | 0.131  | 0.98 | no | LONG |
| VTIP | cluster_19 | Inflation-Protected Fixed I... | 0.787  | 22 | 0.8%  | 0.127  | 0.99 | no | FLAT |
| HYG  | cluster_9  | High Yield Bond (10) + Pref... | 0.489  | 20 | 0.7%  | 0.476  | 0.99 | no | LONG |
| PFF  | cluster_9  | High Yield Bond (10) + Pref... | 0.489  | 20 | 0.7%  | 0.260  | 0.99 | no | LONG |
| EMB  | cluster_9  | High Yield Bond (10) + Pref... | 0.489  | 20 | 0.7%  | 0.157  | 0.99 | no | FLAT |
| VBR  | cluster_7  | Mixed: VBR, MDYV, IVOV         | 0.380  | 21 | 2.9%  | 0.430  | 1.00 | no | LONG |
| IJR  | cluster_7  | Mixed: VBR, MDYV, IVOV         | 0.380  | 21 | 2.9%  | 0.363  | 1.00 | no | LONG |
| XLF  | cluster_7  | Mixed: VBR, MDYV, IVOV         | 0.380  | 21 | 2.9%  | 0.219  | 1.00 | no | LONG |
| VHT  | cluster_11 | Healthcare (16/17)             | 0.021  | 3  | -2.0% | 1.989  | 0.99 | no | FLAT |
| XBI  | cluster_11 | Healthcare (16/17)             | 0.021  | 3  | -2.0% | 0.128  | 0.99 | no | FLAT |
| XLV  | cluster_11 | Healthcare (16/17)             | 0.021  | 3  | -2.0% | -1.104 | 0.99 | no | FLAT |
| BND  | cluster_5  | Mixed: BND, AGG, SCHZ          | -0.177 | 0  | 0.0%  | 0.629  | 1.00 | no | LONG |
| AGG  | cluster_5  | Mixed: BND, AGG, SCHZ          | -0.177 | 0  | 0.0%  | 0.354  | 1.00 | no | LONG |
| BNDX | cluster_5  | Mixed: BND, AGG, SCHZ          | -0.177 | 0  | 0.0%  | 0.033  | 1.00 | no | LONG |
| VPU  | cluster_17 | Consumer Staples (6/11)        | -0.232 | 6  | -2.3% | 0.703  | 0.99 | no | FLAT |
| XLP  | cluster_17 | Consumer Staples (6/11)        | -0.232 | 6  | -2.3% | 0.584  | 0.99 | no | LONG |
| XLU  | cluster_17 | Consumer Staples (6/11)        | -0.232 | 6  | -2.3% | -0.324 | 0.99 | no | FLAT |

### Watchlist (imminent-buy clusters, p > 80%, not yet long)

| ticker | cluster    | theme                          | prob_buy | cluster_nss | basket_weight | basket_cor | sticky | etf_signal |
|--------|------------|--------------------------------|----------|-------------|---------------|------------|--------|------------|
| GUNR   | cluster_12 | Materials (13/17)              | 1.00     | 0.550       | 0.681         | 0.97       | no     | FLAT       |
| EWJ    | cluster_20 | Japan Equity (7/9)             | 1.00     | 0.947       | 0.481         | 0.97       | no     | LONG       |
| VTEB   | cluster_16 | Municipal Bond (13/13)         | 1.00     | 0.244       | 0.468         | 0.97       | no     | FLAT       |
| IEFA   | cluster_1  | Mixed: VEA, SPDW, SCHF         | 1.00     | 0.622       | 0.421         | 0.99       | no     | FLAT       |
| ICSH   | cluster_23 | Cash / Ultra-Short Fixed In... | 1.00     | 1.000       | 0.418         | 0.84       | no     | FLAT       |
| MUB    | cluster_16 | Municipal Bond (13/13)         | 1.00     | 0.244       | 0.380         | 0.97       | no     | LONG       |
| SHV    | cluster_23 | Cash / Ultra-Short Fixed In... | 1.00     | 1.000       | 0.377         | 0.84       | no     | FLAT       |
| BBJP   | cluster_20 | Japan Equity (7/9)             | 1.00     | 0.947       | 0.365         | 0.97       | no     | LONG       |
| VEA    | cluster_1  | Mixed: VEA, SPDW, SCHF         | 1.00     | 0.622       | 0.339         | 0.99       | no     | LONG       |
| SUB    | cluster_16 | Municipal Bond (13/13)         | 1.00     | 0.244       | 0.332         | 0.97       | no     | LONG       |
| MINT   | cluster_23 | Cash / Ultra-Short Fixed In... | 1.00     | 1.000       | 0.176         | 0.84       | no     | LONG       |
| VXUS   | cluster_1  | Mixed: VEA, SPDW, SCHF         | 1.00     | 0.622       | 0.172         | 0.99       | no     | LONG       |
| COPX   | cluster_12 | Materials (13/17)              | 1.00     | 0.550       | 0.161         | 0.97       | no     | LONG       |
| GBIL   | cluster_23 | Cash / Ultra-Short Fixed In... | 1.00     | 1.000       | 0.118         | 0.84       | no     | LONG       |
| VPL    | cluster_20 | Japan Equity (7/9)             | 1.00     | 0.947       | 0.072         | 0.97       | no     | LONG       |
| URA    | cluster_12 | Materials (13/17)              | 1.00     | 0.550       | 0.069         | 0.97       | no     | LONG       |
| BIL    | cluster_23 | Cash / Ultra-Short Fixed In... | 1.00     | 1.000       | -0.040        | 0.84       | no     | LONG       |
| ILF    | cluster_18 | Mixed: ILF, EWZ, ECH           | 0.98     | 0.072       | 0.382         | 0.93       | no     | FLAT       |
| EPI    | cluster_25 | India Equity (3/4)             | 0.98     | 0.084       | 0.256         | 1.00       | no     | LONG       |

|      |            |                      |      |       |       |      |    |      |
|------|------------|----------------------|------|-------|-------|------|----|------|
| INDA | cluster_25 | India Equity (3/4)   | 0.98 | 0.084 | 0.250 | 1.00 | no | FLAT |
| PIN  | cluster_25 | India Equity (3/4)   | 0.98 | 0.084 | 0.249 | 1.00 | no | FLAT |
| INDY | cluster_25 | India Equity (3/4)   | 0.98 | 0.084 | 0.246 | 1.00 | no | FLAT |
| ECH  | cluster_18 | Mixed: ILF, EWZ, ECH | 0.98 | 0.072 | 0.158 | 0.93 | no | LONG |
| DBA  | cluster_18 | Mixed: ILF, EWZ, ECH | 0.98 | 0.072 | 0.056 | 0.93 | no | FLAT |
| EWZ  | cluster_18 | Mixed: ILF, EWZ, ECH | 0.98 | 0.072 | 0.047 | 0.93 | no | FLAT |

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## Cluster Overview

change shows the signal-state shift vs the prior run. dNSS is today's NSS minus the prior run's.

|    | id         | name                           | category             | basket                         | n  | signal | change  | days | cum_return | nss    | dNSS   |
|----|------------|--------------------------------|----------------------|--------------------------------|----|--------|---------|------|------------|--------|--------|
| 3  | cluster_3  | Technology (34/63)             | Technology           | QQQ+VUG+VGT (cor=0.98)         | 63 | LONG   | shifted | 20   | 15.3%      | 1.000  | +0.585 |
| 8  | cluster_8  | Emerging Markets Equity (19... | Emerging Markets ... | IEMG+VWO+EEM (cor=0.99)        | 25 | LONG   | shifted | 21   | 9.9%       | 1.000  | +0.595 |
| 22 | cluster_22 | Mixed: SRLN, BKLN, FTSL        | Mixed                | FLOT+BKLN+SRLN+FLRN+FLTR (c... | 9  | LONG   | shifted | 24   | 1.2%       | 1.000  | +0.186 |
| 23 | cluster_23 | Cash / Ultra-Short Fixed In... | Cash / Ultra-Shor... | BIL+SHV+MINT+GBIL+ICSH (cor... | 9  | FLAT   | shifted |      |            | 1.000  | +0.000 |
| 4  | cluster_4  | Mixed: IWB, ITOT, SCHB         | Mixed                | VOO+IVV+SPY (cor=0.99)         | 61 | LONG   | shifted | 20   | 5.9%       | 0.987  | +1.183 |
| 20 | cluster_20 | Japan Equity (7/9)             | Japan Equity         | EWJ+BBJP+VPL (cor=0.97)        | 9  | FLAT   | shifted |      |            | 0.947  | -0.014 |
| 6  | cluster_6  | Mixed: IWD, SCHV, VONV         | Mixed                | VTV+VIG+VO (cor=1.00)          | 54 | LONG   | shifted | 20   | 2.6%       | 0.919  | -0.070 |
| 21 | cluster_21 | Broad Commodities (6/9)        | Broad Commodities    | PDBC+FTGC+DBC (cor=0.98)       | 9  | LONG   | shifted | 84   | 47.8%      | 0.918  | +0.102 |
| 24 | cluster_24 | Currency (4/6)                 | Currency             | BWX+IGOV+FXF (cor=0.98)        | 6  | LONG   | shifted | 7    | 0.3%       | 0.873  | -0.127 |
| 13 | cluster_13 | China Equity (13/15)           | China Equity         | KWEB+MCHI+FXI (cor=0.98)       | 15 | LONG   | shifted | 18   | 2.8%       | 0.858  | +1.200 |
| 2  | cluster_2  | Mixed: VB, SCHM, IJH           | Mixed                | IJH+IWM+VB (cor=0.99)          | 66 | LONG   | shifted | 20   | 3.7%       | 0.849  | +0.110 |
| 14 | cluster_14 | Real Estate (14/14)            | Real Estate          | VNQ+SCHH+XLRE (cor=1.00)       | 14 | LONG   | shifted | 21   | 4.2%       | 0.816  | +0.852 |
| 10 | cluster_10 | U.S. Dividend Equity (11/20)   | U.S. Dividend Equ... | SCHD+DVY+HDV (cor=0.98)        | 20 | LONG   | shifted | 14   | 0.6%       | 0.787  | +0.644 |
| 19 | cluster_19 | Inflation-Protected Fixed I... | Inflation-Protect... | VTIP+SCHP+STIP (cor=0.99)      | 10 | LONG   | shifted | 22   | 0.8%       | 0.787  | +0.249 |
| 1  | cluster_1  | Mixed: VEA, SPDW, SCHF         | Mixed                | VEA+IEFA+VXUS (cor=0.99)       | 82 | FLAT   | shifted |      |            | 0.622  | -0.378 |
| 12 | cluster_12 | Materials (13/17)              | Materials            | URA+GUNR+COPX (cor=0.97)       | 17 | FLAT   | shifted |      |            | 0.550  | +0.517 |
| 9  | cluster_9  | High Yield Bond (10) + Pref... | High Yield Bond      | HYG+EMB+PFF (cor=0.99)         | 22 | LONG   | shifted | 20   | 0.7%       | 0.489  | +1.459 |
| 7  | cluster_7  | Mixed: VBR, MDYV, IVOV         | Mixed                | IJR+XLF+VBR (cor=1.00)         | 48 | LONG   | shifted | 21   | 2.9%       | 0.380  | -0.619 |
| 16 | cluster_16 | Municipal Bond (13/13)         | Municipal Bond       | MUB+VTEB+SUB (cor=0.97)        | 13 | FLAT   | shifted |      |            | 0.244  | -0.051 |
| 25 | cluster_25 | India Equity (3/4)             | India Equity         | INDA+INDY+EPI+PIN (cor=1.00)   | 4  | FLAT   | shifted |      |            | 0.084  | -0.916 |
| 18 | cluster_18 | Mixed: ILF, EWZ, ECH           | Mixed                | EWZ+ILF+DBA+ECH (cor=0.93)     | 11 | FLAT   | shifted |      |            | 0.072  | -0.138 |
| 11 | cluster_11 | Healthcare (16/17)             | Healthcare           | XLV+VHT+XBI (cor=0.99)         | 17 | LONG   | shifted | 3    | -2.0%      | 0.021  | -0.188 |
| 5  | cluster_5  | Mixed: BND, AGG, SCHZ          | Mixed                | BND+AGG+BNDX (cor=1.00)        | 58 | LONG   | shifted | 0    | 0.0%       | -0.177 | -1.089 |
| 15 | cluster_15 | Mixed: DBP, GLTR, GDXJ         | Mixed                | GLD+IAU+SLV (cor=0.95)         | 13 | FLAT   | shifted |      |            | -0.193 | -0.964 |
| 17 | cluster_17 | Consumer Staples (6/11)        | Consumer Staples     | XLU+XLP+VPU (cor=0.99)         | 11 | LONG   | shifted | 6    | -2.3%      | -0.232 | -1.025 |
| 26 | cluster_26 | Currency (1/2)                 | Currency             | SQQQ+UUP (cor=1.00)            | 2  | FLAT   | shifted |      |            | -1.000 | -1.791 |

## Net Signal Strength (NSS)

NSS = P(buy crossover within 5 days) - P(sell crossover within 5 days). Range [-1, 1]; positive = bullish lean, negative = bearish lean.

ETF rows only (Cluster Overview already covers clusters), cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive 126d trailing return — i.e. would not be blocked by the strategy's entry filter.

### Top 20 most bullish (NSS desc, abs-mom approved, cluster-deduped)

| id   | name                               | category               | cluster    | cluster_name               | cluster_signal | prob_buy | prob_sell | nss   |
|------|------------------------------------|------------------------|------------|----------------------------|----------------|----------|-----------|-------|
| NXTG | First Trust Indxx NextG ETF        | Technology             | cluster_3  | Technology (34/63)         | LONG           | 1        | 0.00      | 1.000 |
| VLUE | iShares MSCI USA Value Factor E... | U.S. Value Factor E... | cluster_6  | Mixed: IWD, SCHV, VONV     | LONG           | 1        | 0.00      | 1.000 |
| IBDR | iShares iBonds Dec 2026 Term Co... | Target Maturity Bond   | cluster_5  | Mixed: BND, AGG, SCHZ      | LONG           | 1        | 0.00      | 1.000 |
| MINT | PIMCO Enhanced Short Maturity A... | Cash / Ultra-Short ... | cluster_23 | Cash / Ultra-Short Fixe... | FLAT           | 1        | 0.00      | 1.000 |
| FLTR | VanEck IG Floating Rate ETF        | Floating Rate Fixed... | cluster_22 | Mixed: SRLN, BKLN, FTSL    | LONG           | 1        | 0.00      | 1.000 |
| IHF  | iShares U.S. Health Care Provid... | Healthcare             | cluster_11 | Healthcare (16/17)         | LONG           | 1        | 0.00      | 1.000 |
| EWT  | iShares MSCI Taiwan ETF            | Taiwan Equity          | cluster_8  | Emerging Markets Equity... | LONG           | 1        | 0.00      | 1.000 |
| EWM  | iShares MSCI Malaysia Index Fund   | Malaysia Equity        | cluster_18 | Mixed: ILF, EWZ, ECH       | FLAT           | 1        | 0.00      | 1.000 |
| FV   | First Trust Dorsey Wright Focus... | Momentum Equity        | cluster_2  | Mixed: VB, SCHM, IJH       | LONG           | 1        | 0.00      | 1.000 |
| EIS  | iShares MSCI Israel ETF            | Israel Equity          | cluster_4  | Mixed: IWB, ITOT, SCHB     | LONG           | 1        | 0.00      | 1.000 |
| CQQQ | Invesco China Technology ETF       | China Equity           | cluster_13 | China Equity (13/15)       | LONG           | 1        | 0.00      | 1.000 |
| VPL  | Vanguard FTSE Pacific ETF          | Asia-Pacific Equity    | cluster_20 | Japan Equity (7/9)         | FLAT           | 1        | 0.00      | 1.000 |
| SLX  | VanEck Steel ETF                   | Materials              | cluster_12 | Materials (13/17)          | FLAT           | 1        | 0.00      | 1.000 |
| FXV  | Invesco CurrencyShares Japanese... | Currency               | cluster_24 | Currency (4/6)             | LONG           | 1        | 0.00      | 1.000 |
| FTGC | First Trust Global Tactical Com... | Broad Commodities      | cluster_21 | Broad Commodities (6/9)    | LONG           | 1        | 0.00      | 1.000 |
| XLP  | State Street Consumer Staples S... | Consumer Staples       | cluster_17 | Consumer Staples (6/11)    | LONG           | 1        | 0.00      | 0.999 |
| NURE | Nuveen Short-Term REIT ETF         | Real Estate            | cluster_14 | Real Estate (14/14)        | LONG           | 1        | 0.00      | 0.999 |
| SCHD | Schwab US Dividend Equity ETF      | U.S. Dividend Equity   | cluster_10 | U.S. Dividend Equity (1... | LONG           | 1        | 0.00      | 0.999 |
| PFXF | VanEck Preferred Securities ex ... | Preferred Stock        | cluster_9  | High Yield Bond (10) + ... | LONG           | 1        | 0.00      | 0.998 |
| ENZL | iShares MSCI New Zealand ETF       | New Zealand Equity     | cluster_1  | Mixed: VEA, SPDW, SCHF     | FLAT           | 1        | 0.01      | 0.993 |

### Top 20 most bearish (NSS asc, cluster-deduped)

| id   | name                               | category               | cluster    | cluster_name            | cluster_signal | prob_buy | prob_sell | nss    |
|------|------------------------------------|------------------------|------------|-------------------------|----------------|----------|-----------|--------|
| IHI  | iShares U.S. Medical Devices ETF   | Healthcare             | cluster_11 | Healthcare (16/17)      | LONG           | 0.00     | 1.00      | -1.000 |
| SQQQ | ProShares UltraPro Short QQQ       | Leveraged / Inverse    | cluster_26 | Currency (1/2)          | FLAT           | 0.00     | 1.00      | -1.000 |
| SPLV | Invesco S&P 500 Low Volatility ... | U.S. Low Volatility... | cluster_6  | Mixed: IWD, SCHV, VONV  | LONG           | 0.00     | 1.00      | -1.000 |
| EIDO | iShares MSCI Indonesia ETF         | Indonesia Equity       | cluster_18 | Mixed: ILF, EWZ, ECH    | FLAT           | 0.00     | 1.00      | -0.997 |
| IDU  | iShares U.S. Utilities ETF         | Utilities              | cluster_17 | Consumer Staples (6/11) | LONG           | 0.00     | 0.99      | -0.992 |

|      |                                    |                        |            |                            |      |      |      |        |
|------|------------------------------------|------------------------|------------|----------------------------|------|------|------|--------|
| LVHD | Franklin U.S. Low Volatility Hi... | U.S. Low Volatility... | cluster_10 | U.S. Dividend Equity (1... | LONG | 0.04 | 0.99 | -0.952 |
| PPA  | Invesco Aerospace & Defense ETF    | Industrials            | cluster_2  | Mixed: VB, SCHM, IJH       | LONG | 0.00 | 0.93 | -0.926 |
| IEF  | iShares 7-10 Year Treasury Bond... | Fixed Income           | cluster_5  | Mixed: BND, AGG, SCHZ      | LONG | 0.04 | 0.96 | -0.926 |
| MOO  | VanEck Agribusiness ETF            | Consumer Staples       | cluster_12 | Materials (13/17)          | FLAT | 0.05 | 0.86 | -0.816 |
| GOAU | US Global GO Gold and Precious ... | Materials              | cluster_15 | Mixed: DBP, GLTR, GDXJ     | FLAT | 0.10 | 0.88 | -0.787 |
| EFAV | iShares MSCI EAFE Min Vol Facto... | International Equity   | cluster_1  | Mixed: VEA, SPDW, SCHF     | FLAT | 0.22 | 0.94 | -0.724 |
| SHM  | State Street SPDR Nuveen ICE Sh... | Municipal Bond         | cluster_16 | Municipal Bond (13/13)     | FLAT | 0.26 | 0.96 | -0.697 |
| ESPO | VanEck Video Gaming and eSports... | Consumer Discretion... | cluster_4  | Mixed: IWB, ITOT, SCHB     | LONG | 0.40 | 0.97 | -0.564 |
| KIE  | State Street SPDR S&P Insurance... | Financials             | cluster_7  | Mixed: VBR, MDYV, IVOV     | LONG | 0.56 | 1.00 | -0.436 |
| CHIQ | Global X MSCI China Consumer Di... | China Equity           | cluster_13 | China Equity (13/15)       | LONG | 0.57 | 0.90 | -0.333 |
| INDY | iShares India 50 ETF               | India Equity           | cluster_25 | India Equity (3/4)         | FLAT | 0.67 | 0.98 | -0.306 |
| IBUY | Amplify Online Retail ETF          | Consumer Discretion... | cluster_3  | Technology (34/63)         | LONG | 0.87 | 0.98 | -0.110 |
| ANGL | VanEck Fallen Angel High Yield ... | High Yield Bond        | cluster_9  | High Yield Bond (10) + ... | LONG | 1.00 | 0.95 | 0.053  |
| LTPZ | Pimco 15 Year U.S. TIPS Index E... | Inflation-Protected... | cluster_19 | Inflation-Protected Fix... | LONG | 0.87 | 0.58 | 0.292  |
| FXE  | Invesco CurrencyShares Euro Cur... | Currency               | cluster_24 | Currency (4/6)             | LONG | 1.00 | 0.65 | 0.346  |

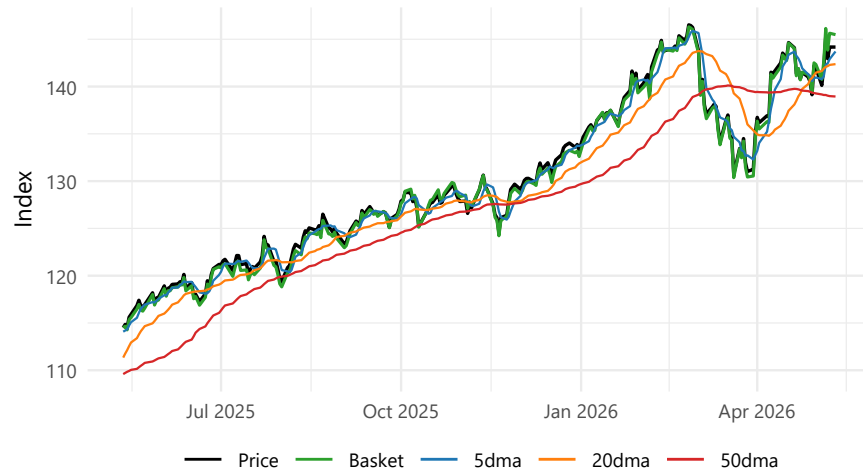
## Cluster Details

### Cluster 1 – Mixed: VEA, SPDW, SCHF

**Members:** 82   **Var explained:** 0.839   **Signal:** FLAT

#### Signal logic:

- Latest buy\_cross: 2026-04-10 – filter **PASS** (126d at date = +44.8%)
- Latest sell\_cross: 2026-05-04
- Current 126d return: +27.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-04 more recent than last effective buy 2026-04-10)



#### Category breakdown

| category                      | n_members | abs_weight |
|-------------------------------|-----------|------------|
| International Equity          | 21        | 0.2686     |
| Developed ex-US Equity        | 9         | 0.1179     |
| International Dividend Equity | 6         | 0.0748     |
| Global ex-US Equity           | 5         | 0.0656     |
| Europe Equity                 | 4         | 0.0518     |
| Alternatives                  | 4         | 0.0464     |
| Eurozone Equity               | 3         | 0.0372     |
| Asia-Pacific Equity           | 2         | 0.0242     |

|                               |   |        |
|-------------------------------|---|--------|
| Real Estate                   | 2 | 0.0225 |
| Global Equity                 | 2 | 0.0222 |
| Emerging Markets Fixed Income | 2 | 0.0193 |
| Currency                      | 2 | 0.0181 |
| Commodity                     | 1 | 0.0132 |
| Unknown                       | 1 | 0.0130 |
| Global ex-US Small Cap Equity | 1 | 0.0129 |
| Germany Equity                | 1 | 0.0124 |
| France Equity                 | 1 | 0.0124 |
| United Kingdom Equity         | 1 | 0.0123 |
| Italy Equity                  | 1 | 0.0122 |
| Europe Financials             | 1 | 0.0120 |
| Sweden Equity                 | 1 | 0.0120 |
| Netherlands Equity            | 1 | 0.0120 |
| Australia Equity              | 1 | 0.0119 |
| Spain Equity                  | 1 | 0.0116 |
| Canada Equity                 | 1 | 0.0114 |
| Switzerland Equity            | 1 | 0.0114 |
| Belgium Equity                | 1 | 0.0113 |
| Austria Equity                | 1 | 0.0112 |
| Poland Equity                 | 1 | 0.0104 |
| South Africa Equity           | 1 | 0.0099 |
| New Zealand Equity            | 1 | 0.0091 |
| Mexico Equity                 | 1 | 0.0089 |

**Replication basket** – VEA+IEFA+VXUS (cor=0.99) basket lead at lag 1: -0.002

| ticker | weight_ols | name                                   | category               | etf_signal |
|--------|------------|----------------------------------------|------------------------|------------|
| VEA    | 0.339      | Vanguard FTSE Developed Markets ETF    | Developed ex-US Equity | LONG       |
| IEFA   | 0.421      | iShares Core MSCI EAFE ETF             | Developed ex-US Equity | FLAT       |
| VXUS   | 0.172      | Vanguard Total International Stock ETF | Global ex-US Equity    | LONG       |

## Members

|    | ticker | weight | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 2  | EFA    | 0.0133 | iShares MSCI EAFE ETF                      | Developed ex-US Equity      | FLAT       |
| 3  | VEA    | 0.0133 | Vanguard FTSE Developed Markets ETF        | Developed ex-US Equity      | LONG       |
| 63 | IEFA   | 0.0133 | iShares Core MSCI EAFE ETF                 | Developed ex-US Equity      | FLAT       |
| 64 | SCHF   | 0.0133 | Schwab International Equity ETF            | Developed ex-US Equity      | LONG       |
| 65 | SPDW   | 0.0133 | State Street SPDR Portfolio Developed W... | Developed ex-US Equity      | LONG       |
| 21 | DWM    | 0.0132 | WisdomTree International Equity Fund       | International Dividend E... | FLAT       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 34 | TLTD | 0.0132 | FlexShares Morningstar Developed Market... | International Equity        | FLAT |
| 49 | GSIE | 0.0132 | Goldman Sachs ActiveBeta International ... | Commodity                   | FLAT |
| 52 | ESGD | 0.0132 | iShares ESG Aware MSCI EAFE ETF            | International Equity        | FLAT |
| 28 | PXF  | 0.0131 | Invesco RAFI Developed Markets ex-U.S. ... | International Equity        | LONG |
| 38 | ACWX | 0.0131 | iShares MSCI ACWI ex U.S. ETF              | Global ex-US Equity         | LONG |
| 39 | CWI  | 0.0131 | State Street SPDR MSCI ACWI ex-US ETF      | Global ex-US Equity         | LONG |
| 41 | INTF | 0.0131 | iShares International Equity Factor ETF    | International Equity        | FLAT |
| 42 | IQLT | 0.0131 | iShares MSCI Intl Quality Factor ETF       | International Equity        | FLAT |
| 44 | QEFA | 0.0131 | State Street SPDR MSCI EAFE StrategicFa... | International Equity        | FLAT |
| 58 | IEUR | 0.0131 | iShares Core MSCI Europe ETF               | Europe Equity               | FLAT |
| 59 | IXUS | 0.0131 | iShares Core MSCI Total International S... | Global ex-US Equity         | LONG |
| 60 | VEU  | 0.0131 | Vanguard FTSE All World Ex US ETF          | Global ex-US Equity         | LONG |
| 61 | VGK  | 0.0131 | Vanguard FTSEEuropean ETF                  | Europe Equity               | FLAT |
| 62 | VXUS | 0.0131 | Vanguard Total International Stock ETF     | Global ex-US Equity         | LONG |
| 6  | IEV  | 0.0130 | iShares Europe ETF                         | Europe Equity               | FLAT |
| 32 | EFV  | 0.0130 | iShares MSCI EAFE Value ETF                | Developed ex-US Equity      | FLAT |
| 35 | JPIN | 0.0130 | JPMorgan Diversified Return Internation... | International Equity        | FLAT |
| 46 | RODM | 0.0130 | Hartford Multifactor Developed Markets ... | International Equity        | FLAT |
| 47 | VYMI | 0.0130 | Vanguard International High Dividend Yi... | International Dividend E... | LONG |
| 48 | FNDF | 0.0130 | Schwab Fundamental International Equity... | International Equity        | LONG |
| 56 | SCHC | 0.0130 | Schwab International Small-Cap Equity E... | Developed ex-US Equity      | LONG |
| 75 | FEU  | 0.0130 |                                            |                             | FLAT |
| 24 | IQDF | 0.0129 | FlexShares International Quality Divide... | International Equity        | LONG |
| 31 | SCZ  | 0.0129 | iShares MSCI EAFE Small-Cap ETF            | Developed ex-US Equity      | LONG |
| 33 | FNDC | 0.0129 | Schwab Fundamental International Small ... | International Equity        | LONG |
| 57 | VSS  | 0.0129 | Vanguard FTSE All-Wld ex-US SmCp Idx ETF   | Global ex-US Small Cap E... | LONG |
| 8  | DLS  | 0.0128 | WisdomTree International SmallCap Fund     | International Equity        | LONG |
| 19 | EZU  | 0.0128 | iShares MSCI Eurozone ETF                  | Eurozone Equity             | FLAT |
| 43 | IVLU | 0.0128 | iShares MSCI Intl Value Factor ETF         | International Equity        | FLAT |
| 72 | HFXI | 0.0128 | NYLI FTSE International Equity Currency... | International Equity        | LONG |
| 29 | EFG  | 0.0127 | iShares MSCI EAFE Growth ETF               | Developed ex-US Equity      | FLAT |
| 45 | FEZ  | 0.0127 | State Street SPDR EURO STOXX 50 ETF        | Eurozone Equity             | FLAT |
| 54 | VIGI | 0.0127 | Vanguard International Dividend Appreci... | International Dividend E... | LONG |
| 4  | FEP  | 0.0126 | First Trust Europe AlphaDEX Fund           | Europe Equity               | LONG |
| 30 | GWX  | 0.0126 | State Street SPDR S&P International Sma... | International Equity        | LONG |
| 40 | IMTM | 0.0126 | iShares MSCI Intl Momentum Factor ETF      | International Equity        | LONG |
| 1  | EWG  | 0.0124 | iShares MSCI Germany Index Fund            | Germany Equity              | LONG |
| 5  | VIDI | 0.0124 | Vident International Equity Strategy ETF   | International Equity        | LONG |
| 16 | EWQ  | 0.0124 | iShares MSCI France Index Fund             | France Equity               | FLAT |
| 20 | IDV  | 0.0124 | iShares International Select Dividend E... | International Dividend E... | LONG |
| 53 | HDEF | 0.0124 | Xtrackers MSCI EAFE High Dividend Yield... | International Equity        | LONG |
| 17 | EWU  | 0.0123 | iShares MSCI United Kingdom ETF            | United Kingdom Equity       | LONG |
| 50 | IDLV | 0.0123 | Invesco S&P International Developed Low... | International Equity        | LONG |
| 79 | IPKW | 0.0123 | Invesco International BuyBack Achievers... | International Equity        | LONG |
| 82 | EWI  | 0.0122 | iShares MSCI Italy ETF                     | Italy Equity                | LONG |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 22 | EPP  | 0.0121 | iShares MSCI Pacific Ex-Japan Index Fund   | Asia-Pacific Equity         | FLAT |
| 51 | EFAV | 0.0121 | iShares MSCI EAFE Min Vol Factor ETF       | International Equity        | FLAT |
| 23 | EUFN | 0.0120 | iShares MSCI Europe Financials ETF         | Europe Financials           | FLAT |
| 71 | EWD  | 0.0120 | iShares MSCI Sweden ETF                    | Sweden Equity               | LONG |
| 77 | EWN  | 0.0120 | iShares MSCI Netherlands Index Fund        | Netherlands Equity          | LONG |
| 81 | BBAX | 0.0120 | JPMorgan BetaBuilders Developed Asia Pa... | Asia-Pacific Equity         | LONG |
| 13 | EWA  | 0.0119 | iShares MSCI Australia Index Fund          | Australia Equity            | FLAT |
| 27 | DWX  | 0.0119 | State Street SPDR S&P International Div... | International Dividend E... | LONG |
| 12 | HEZU | 0.0118 | iShares Currency Hedged MSCI Eurozone E... | Eurozone Equity             | FLAT |
| 36 | DBEF | 0.0118 | Xtrackers MSCI EAFE Hedged Equity ETF      | Alternatives                | FLAT |
| 37 | HEFA | 0.0118 | iShares Currency Hedged MSCI EAFE ETF      | Alternatives                | FLAT |
| 11 | PID  | 0.0117 | Invesco International Dividend Achiever... | International Dividend E... | LONG |
| 15 | EWP  | 0.0116 | iShares MSCI Spain ETF                     | Spain Equity                | FLAT |
| 9  | HEDJ | 0.0114 | WisdomTree Europe Hedged Equity Fund       | Alternatives                | FLAT |
| 10 | IHDG | 0.0114 | WisdomTree International Hedged Quality... | Alternatives                | LONG |
| 14 | EWL  | 0.0114 | iShares MSCI Switzerland ETF               | Switzerland Equity          | LONG |
| 18 | EWG  | 0.0114 | iShares MSCI Canada Index Fund             | Canada Equity               | LONG |
| 25 | NFRA | 0.0114 | FlexShares STOXX Global Broad Infrastru... | Global Equity               | LONG |
| 55 | VNQI | 0.0114 | Vanguard Global ex-U.S. Real Estate ETF    | Real Estate                 | LONG |
| 80 | EWK  | 0.0113 | iShares MSCI Belgium ETF                   | Belgium Equity              | FLAT |
| 76 | EWO  | 0.0112 | iShares MSCI Austria ETF                   | Austria Equity              | LONG |
| 7  | RWX  | 0.0111 | State Street SPDR Dow Jones Internation... | Real Estate                 | LONG |
| 26 | IGF  | 0.0108 | iShares Global Infrastructure ETF          | Global Equity               | LONG |
| 70 | EPOL | 0.0104 | iShares MSCI Poland ETF                    | Poland Equity               | FLAT |
| 74 | EZA  | 0.0099 | iShares MSCI South Africa Index Fund       | South Africa Equity         | LONG |
| 69 | FXA  | 0.0097 | Invesco CurrencyShares Australian Dolla... | Currency                    | LONG |
| 66 | EBND | 0.0096 | State Street SPDR Bloomberg Emerging Ma... | Emerging Markets Fixed I... | LONG |
| 67 | EMLC | 0.0096 | VanEck J. P. Morgan EM Local Currency B... | Emerging Markets Fixed I... | LONG |
| 73 | ENZL | 0.0091 | iShares MSCI New Zealand ETF               | New Zealand Equity          | LONG |
| 68 | EWG  | 0.0089 | iShares MSCI Mexico ETF                    | Mexico Equity               | LONG |
| 78 | FXC  | 0.0084 | Invesco CurrencyShares Canadian Dollar ... | Currency                    | LONG |

**Cluster 2 – Mixed: VB, SCHM, IJH**

**Members:** 66   **Var explained:** 0.848   **Signal:** LONG since 2026-04-13 (20 days, 3.7%)

**Signal logic:**

- Latest buy\_cross: 2026-04-13 – filter **PASS** (126d at date = +31.7%)
- Latest sell\_cross: 2026-03-06
- Current 126d return: +28.4% – filter **ALLOWS** new entries today

- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



### Category breakdown

| category                          | n_members | abs_weight |
|-----------------------------------|-----------|------------|
| Industrials                       | 15        | 0.2179     |
| U.S. Small Cap Growth             | 6         | 0.0954     |
| U.S. Mid Cap Growth               | 6         | 0.0949     |
| Consumer Discretionary            | 6         | 0.0845     |
| U.S. Mid Cap Equity               | 5         | 0.0813     |
| U.S. Small Cap Equity             | 5         | 0.0803     |
| Utilities                         | 3         | 0.0428     |
| U.S. Momentum Equity              | 3         | 0.0428     |
| Growth Equity                     | 2         | 0.0307     |
| U.S. Small Cap Momentum Equity    | 2         | 0.0304     |
| Momentum Equity                   | 2         | 0.0296     |
| Unknown                           | 1         | 0.0163     |
| U.S. Small Cap ESG Equity         | 1         | 0.0162     |
| U.S. Small Cap Multifactor Equity | 1         | 0.0161     |
| U.S. Extended Market Equity       | 1         | 0.0161     |
| U.S. Mid Cap Quality Equity       | 1         | 0.0158     |
| U.S. Mid Cap Momentum Equity      | 1         | 0.0154     |
| U.S. Small Cap Industrials        | 1         | 0.0151     |
| Technology                        | 1         | 0.0150     |
| U.S. Micro Cap Equity             | 1         | 0.0150     |



|                                       |   |        |
|---------------------------------------|---|--------|
| IPO Equity                            | 1 | 0.0143 |
| U.S. Small Cap Consumer Discretionary | 1 | 0.0142 |

**Replication basket** – IJH+IWM+VB (cor=0.99) basket lead at lag 1: -0.006

| ticker | weight_ols | name                            | category              | etf_signal |
|--------|------------|---------------------------------|-----------------------|------------|
| IJH    | 0.331      | iShares Core S&P Mid-Cap ETF    | U.S. Mid Cap Equity   | LONG       |
| IWM    | 0.067      | iShares Russell 2000 Index Fund | U.S. Small Cap Equity | LONG       |
| VB     | 0.582      | Vanguard Small-Cap ETF          | U.S. Small Cap Equity | LONG       |

## Members

|    | ticker | weight | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 23 | SCHM   | 0.0163 | Schwab U.S. Mid Cap ETF                    | U.S. Mid Cap Equity         | LONG       |
| 29 | MDY    | 0.0163 | State Street SPDR S&P MIDCAP 400 ETF Tr... | U.S. Mid Cap Equity         | LONG       |
| 37 | IVOO   | 0.0163 | Vanguard S&P Mid-Cap 400 ETF               |                             | LONG       |
| 42 | IJH    | 0.0163 | iShares Core S&P Mid-Cap ETF               | U.S. Mid Cap Equity         | LONG       |
| 43 | SPMD   | 0.0163 | State Street SPDR Portfolio S&P 400 Mid... | U.S. Mid Cap Equity         | LONG       |
| 44 | VB     | 0.0163 | Vanguard Small-Cap ETF                     | U.S. Small Cap Equity       | LONG       |
| 20 | JHMM   | 0.0162 | John Hancock Multifactor Mid Cap ETF       | U.S. Mid Cap Equity         | LONG       |
| 26 | NUSC   | 0.0162 | Nuveen ESG Small-Cap ETF                   | U.S. Small Cap ESG Equity   | LONG       |
| 31 | IJK    | 0.0162 | iShares S&P Mid-Cap 400 Growth ETF         | U.S. Mid Cap Growth         | LONG       |
| 33 | IVOG   | 0.0162 | Vanguard S&P Mid-Cap 400 Growth ETF        | U.S. Mid Cap Growth         | LONG       |
| 34 | MDYG   | 0.0162 | State Street SPDR S&P 400 Mid Cap Growt... | U.S. Mid Cap Growth         | LONG       |
| 45 | SCHA   | 0.0162 | Schwab U.S. Small-Cap ETF                  | U.S. Small Cap Equity       | LONG       |
| 27 | SMLF   | 0.0161 | iShares U.S. Small-Cap Equity Factor ETF   | U.S. Small Cap Multifact... | LONG       |
| 41 | VXF    | 0.0161 | Vanguard Extended Market ETF               | U.S. Extended Market Equ... | LONG       |
| 4  | IWM    | 0.0160 | iShares Russell 2000 Index Fund            | U.S. Small Cap Equity       | LONG       |
| 30 | IJT    | 0.0160 | iShares S&P SmallCap 600 Growth ETF        | U.S. Small Cap Growth       | LONG       |
| 32 | VIOG   | 0.0160 | Vanguard S&P Small-Cap 600 Growth ETF      | U.S. Small Cap Growth       | LONG       |
| 35 | SLYG   | 0.0160 | State Street SPDR S&P 600 Small Cap Gro... | U.S. Small Cap Growth       | LONG       |
| 38 | VTWO   | 0.0160 | Vanguard Russell 2000 ETF                  | U.S. Small Cap Equity       | LONG       |
| 40 | ISCG   | 0.0159 | iShares Morningstar Small-Cap Growth ETF   | Growth Equity               | LONG       |
| 15 | FXR    | 0.0158 | First Trust Industrials AlphaDEX           | Industrials                 | LONG       |
| 28 | IWO    | 0.0158 | iShares Russell 2000 Growth Fund           | U.S. Small Cap Growth       | LONG       |
| 36 | VTWG   | 0.0158 | Vanguard Russell 2000 Growth ETF           | U.S. Small Cap Growth       | LONG       |
| 39 | VBK    | 0.0158 | Vanguard Small-Cap Growth ETF              | U.S. Small Cap Growth       | LONG       |
| 54 | PSC    | 0.0158 | Principal U.S. Small-Cap ETF               | U.S. Small Cap Equity       | LONG       |
| 61 | XMHQ   | 0.0158 | Invesco S&P MidCap Quality ETF             | U.S. Mid Cap Quality Equ... | LONG       |
| 14 | IMCG   | 0.0157 | iShares Morningstar Mid-Cap Growth ETF     | U.S. Mid Cap Growth         | LONG       |
| 21 | VIS    | 0.0157 | Vanguard Industrials ETF                   | Industrials                 | LONG       |

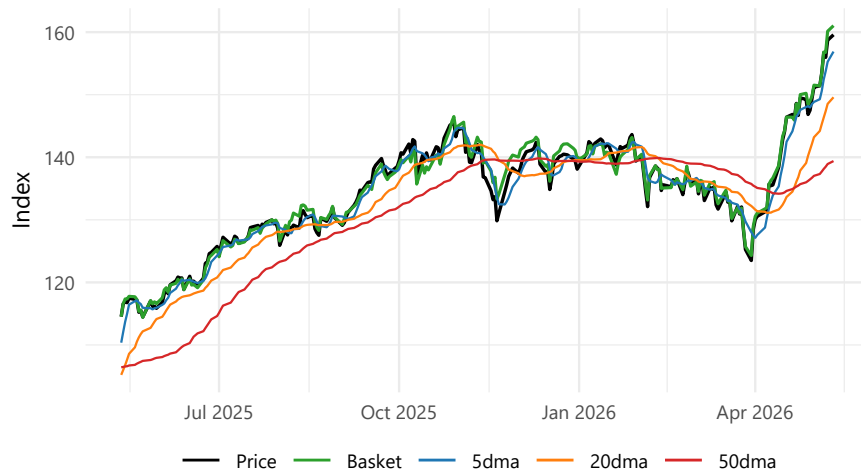
|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 19 | IYJ  | 0.0156 | iShares U.S. Industrials ETF               | Industrials                 | LONG |
| 22 | FIDU | 0.0156 | Fidelity MSCI Industrials Index ETF        | Industrials                 | LONG |
| 60 | PAVE | 0.0155 | Global X U.S. Infrastructure Developmen... | Industrials                 | LONG |
| 10 | XMMO | 0.0154 | Invesco S&P MidCap Momentum ETF            | U.S. Mid Cap Momentum Eq... | LONG |
| 11 | IWP  | 0.0153 | iShares Russell Midcap Growth ETF          | U.S. Mid Cap Growth         | LONG |
| 13 | VOT  | 0.0153 | Vanguard Mid-Cap Growth ETF                | U.S. Mid Cap Growth         | LONG |
| 46 | FXD  | 0.0153 | First Trust Cons. Discret. AlphaDEX        | Consumer Discretionary      | LONG |
| 62 | VFMO | 0.0153 | Vanguard U.S. Momentum Factor ETF          | U.S. Momentum Equity        | LONG |
| 66 | XSMO | 0.0153 | Invesco S&P SmallCap Momentum ETF          | U.S. Small Cap Momentum ... | LONG |
| 1  | XLI  | 0.0152 | State Street Industrial Select Sector S... | Industrials                 | LONG |
| 59 | PSCI | 0.0151 | Invesco S&P SmallCap Industrials ETF       | U.S. Small Cap Industria... | LONG |
| 6  | FTC  | 0.0150 | First Trust Large Cap Growth AlphaDEX F... | Technology                  | LONG |
| 24 | DWAS | 0.0150 | Invesco Dorsey Wright SmallCap Momentum... | U.S. Small Cap Momentum ... | LONG |
| 25 | IWC  | 0.0150 | iShares Microcap ETF                       | U.S. Micro Cap Equity       | LONG |
| 53 | RSPD | 0.0150 | Invesco S&P 500 Equal Weight Consumer D... | Consumer Discretionary      | FLAT |
| 7  | PDP  | 0.0149 | Invesco Dorsey Wright Momentum ETF         | Momentum Equity             | LONG |
| 9  | RPG  | 0.0148 | Invesco S&P 500 Pure Growth ETF            | Growth Equity               | LONG |
| 18 | FIW  | 0.0148 | First Trust Water ETF                      | Utilities                   | LONG |
| 48 | FTXR | 0.0148 | First Trust Nasdaq Transportation ETF      | Industrials                 | LONG |
| 5  | FV   | 0.0147 | First Trust Dorsey Wright Focus 5 ETF      | Momentum Equity             | LONG |
| 63 | PKB  | 0.0147 | Invesco Building & Construction ETF        | Industrials                 | LONG |
| 64 | PRN  | 0.0146 | Invesco Dorsey Wright Industrials Momen... | Industrials                 | LONG |
| 16 | PHO  | 0.0145 | Invesco Water Resources ETF                | Utilities                   | FLAT |
| 8  | FPX  | 0.0143 | First Trust US Equity Opportunities ETF    | IPO Equity                  | LONG |
| 57 | XTN  | 0.0143 | State Street SPDR S&P Transportation ETF   | Industrials                 | FLAT |
| 47 | PEJ  | 0.0142 | Invesco Leisure and Entertainment ETF      | Consumer Discretionary      | FLAT |
| 65 | PSCD | 0.0142 | Invesco S&P SmallCap Consumer Discretio... | U.S. Small Cap Consumer ... | FLAT |
| 52 | IYT  | 0.0141 | iShares U.S. Transportation ETF            | Industrials                 | LONG |
| 2  | XRT  | 0.0139 | State Street SPDR S&P Retail ETF           | Consumer Discretionary      | FLAT |
| 58 | QMOM | 0.0138 | Alpha Architect U.S. Quantitative Momen... | U.S. Momentum Equity        | LONG |
| 12 | MTUM | 0.0137 | iShares MSCI USA Momentum Factor ETF       | U.S. Momentum Equity        | LONG |
| 56 | XHB  | 0.0137 | State Street SPDR S&P Homebuilders ETF     | Consumer Discretionary      | LONG |
| 17 | CGW  | 0.0135 | Invesco S&P Global Water Index ETF         | Utilities                   | FLAT |
| 55 | XAR  | 0.0135 | State Street SPDR S&P Aerospace & Defen... | Industrials                 | FLAT |
| 49 | PPA  | 0.0133 | Invesco Aerospace & Defense ETF            | Industrials                 | FLAT |
| 3  | JETS | 0.0127 | U.S. Global Jets ETF                       | Industrials                 | LONG |
| 50 | ITA  | 0.0125 | iShares U.S. Aerospace & Defense ETF       | Industrials                 | FLAT |
| 51 | ITB  | 0.0124 | iShares U.S. Home Construction ETF         | Consumer Discretionary      | FLAT |

## Cluster 3 – Technology (34/63)

Members: 63 Var explained: 0.840 Signal: LONG since 2026-04-13 (20 days, 15.3%)

**Signal logic:**

- Latest buy\_cross: 2026-04-13 – filter **PASS** (126d at date = +34.2%)
- Latest sell\_cross: 2026-03-06
- Current 126d return: +44.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)

**Category breakdown**

| category                  | n_members | abs_weight |
|---------------------------|-----------|------------|
| Technology                | 34        | 0.5348     |
| U.S. Large Cap Growth     | 11        | 0.1854     |
| U.S. Equity               | 2         | 0.0333     |
| Growth Equity             | 2         | 0.0332     |
| Unknown                   | 2         | 0.0315     |
| Convertible Bond          | 2         | 0.0306     |
| Financials                | 2         | 0.0298     |
| U.S. Mega Cap Growth      | 1         | 0.0167     |
| U.S. Equal Weight Equity  | 1         | 0.0167     |
| U.S. Small Cap Technology | 1         | 0.0153     |
| IPO Equity                | 1         | 0.0152     |
| Options Income            | 1         | 0.0152     |
| Trend-Following Equity    | 1         | 0.0148     |
| Consumer Discretionary    | 1         | 0.0147     |

U.S. Trend-Following Equity 1 0.0128

**Replication basket** – QQQ+VUG+VGT (cor=0.98) basket lead at lag 1: -0.010

| ticker | weight_ols | name                          | category              | etf_signal |
|--------|------------|-------------------------------|-----------------------|------------|
| QQQ    | 0.398      | Invesco QQQ Trust, Series 1   | U.S. Large Cap Growth | LONG       |
| VUG    | 0.198      | Vanguard Growth ETF           | U.S. Large Cap Growth | LONG       |
| VGT    | 0.396      | Vanguard Information Tech ETF | Technology            | LONG       |

## Members

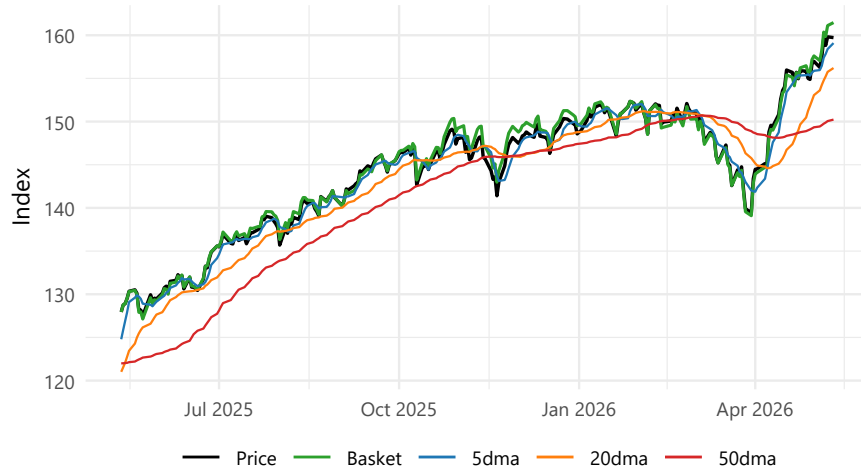
|    | ticker | weight | name                                       | category                 | etf_signal |
|----|--------|--------|--------------------------------------------|--------------------------|------------|
| 14 | QQQ    | 0.0170 | Invesco QQQ Trust, Series 1                | U.S. Large Cap Growth    | LONG       |
| 35 | IGM    | 0.0170 | iShares Expanded Tech Sector ETF           | Technology               | LONG       |
| 40 | ONEQ   | 0.0170 | Fidelity Nasdaq Composite Index ETF        | U.S. Equity              | LONG       |
| 50 | ILCG   | 0.0170 | iShares Morningstar Large-Cap Growth ETF   | U.S. Large Cap Growth    | LONG       |
| 29 | XNTK   | 0.0169 | State Street SPDR NYSE Technology ETF      | Technology               | LONG       |
| 39 | NULG   | 0.0169 | Nuveen ESG Large-Cap Growth ETF            | U.S. Large Cap Growth    | LONG       |
| 43 | IWF    | 0.0169 | iShares Russell 1000 Growth Fund           | U.S. Large Cap Growth    | LONG       |
| 45 | VGT    | 0.0169 | Vanguard Information Tech ETF              | Technology               | LONG       |
| 47 | FTEC   | 0.0169 | Fidelity MSCI Information Technology In... | Technology               | LONG       |
| 48 | VONG   | 0.0169 | Vanguard Russell 1000 Growth ETF           | U.S. Large Cap Growth    | LONG       |
| 52 | SCHG   | 0.0169 | Schwab U.S. Large-Cap Growth ETF           | U.S. Large Cap Growth    | LONG       |
| 54 | VUG    | 0.0169 | Vanguard Growth ETF                        | U.S. Large Cap Growth    | LONG       |
| 8  | AIQ    | 0.0168 | Global X Artificial Intelligence & Tech... | Technology               | LONG       |
| 18 | QTEC   | 0.0168 | First Trust NASDAQ-100-Technology Secto... | Technology               | LONG       |
| 24 | FXL    | 0.0168 | First Trust Technology AlphaDEX            | Technology               | LONG       |
| 37 | IYW    | 0.0168 | iShares U.S. Technology ETF                | Technology               | LONG       |
| 44 | IVW    | 0.0168 | iShares S&P 500 Growth ETF                 | U.S. Large Cap Growth    | LONG       |
| 51 | IUSG   | 0.0168 | iShares Core S&P U.S. Growth ETF           | U.S. Large Cap Growth    | LONG       |
| 17 | QQEW   | 0.0167 | First Trust Nasdaq-100 Select Equal Wei... | U.S. Equal Weight Equity | LONG       |
| 36 | IXN    | 0.0167 | iShares Global Tech ETF                    | Technology               | LONG       |
| 42 | IWY    | 0.0167 | iShares Russell Top 200 Growth ETF         | Growth Equity            | LONG       |
| 46 | VOOG   | 0.0167 | Vanguard S&P 500 Growth ETF                | U.S. Large Cap Growth    | LONG       |
| 49 | MGK    | 0.0167 | Vanguard Mega Cap Growth ETF               | U.S. Mega Cap Growth     | LONG       |
| 53 | SPYG   | 0.0167 | State Street SPDR Portfolio S&P 500 Gro... | U.S. Large Cap Growth    | LONG       |
| 1  | XLK    | 0.0166 | State Street Technology Select Sector S... | Technology               | LONG       |
| 21 | XT     | 0.0165 | iShares Future Exponential Technologies... | Technology               | LONG       |
| 34 | PWB    | 0.0165 | Invesco Large Cap Growth ETF               | Growth Equity            | LONG       |
| 55 | SPHB   | 0.0162 | Invesco S&P 500 High Beta ETF              | U.S. Equity              | LONG       |
| 20 | TDIV   | 0.0160 | First Trust NASDAQ Technology Dividend ... | Technology               | LONG       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 28 | FDN  | 0.0160 | First Trust DJ Internet Index Fund         | Technology                  | LONG |
| 60 | FIVG | 0.0160 |                                            |                             | FLAT |
| 6  | BOTZ | 0.0159 | Global X Robotics & Artificial Intellig... | Technology                  | LONG |
| 7  | ROBO | 0.0159 | ROBO Global Robotics and Automation Ind... | Technology                  | LONG |
| 63 | QTUM | 0.0159 | Defiance Quantum ETF                       | Technology                  | LONG |
| 27 | PNQI | 0.0158 | Invesco Nasdaq Internet ETF                | Technology                  | LONG |
| 5  | SKYY | 0.0157 | First Trust Cloud Computing ETF            | Technology                  | LONG |
| 2  | SOXX | 0.0156 | iShares PHLX SOX Semiconductor Sector I... | Technology                  | LONG |
| 3  | SMH  | 0.0156 | VanEck Semiconductor ETF                   | Technology                  | LONG |
| 4  | IGV  | 0.0155 | iShares Expanded Tech-Software Sector E... | Technology                  | LONG |
| 22 | XSD  | 0.0155 | State Street SPDR S&P Semiconductor ETF    | Technology                  | LONG |
| 38 | CWB  | 0.0155 | State Street SPDR Bloomberg Convertible... | Convertible Bond            | LONG |
| 61 | PBS  | 0.0155 |                                            |                             | LONG |
| 62 | PTF  | 0.0155 | Invesco Dorsey Wright Technology Moment... | Technology                  | LONG |
| 13 | ARKQ | 0.0154 | ARK Autonomous Technology & Robotics ETF   | Technology                  | LONG |
| 15 | NXTG | 0.0154 | First Trust Indxx NextG ETF                | Technology                  | LONG |
| 19 | PSI  | 0.0153 | Invesco Semiconductors ETF                 | Technology                  | LONG |
| 57 | PSCT | 0.0153 | Invesco S&P SmallCap Information Techno... | U.S. Small Cap Technology   | LONG |
| 26 | IPO  | 0.0152 | Renaissance IPO ETF                        | IPO Equity                  | LONG |
| 31 | FINX | 0.0152 | Global X FinTech ETF                       | Financials                  | LONG |
| 33 | QYLD | 0.0152 | Global X NASDAQ 100 Covered Call ETF       | Options Income              | LONG |
| 59 | FTXL | 0.0152 | First Trust Nasdaq Semiconductor ETF       | Technology                  | LONG |
| 12 | ARKW | 0.0151 | ARK Next Generation Internet ETF           | Technology                  | LONG |
| 41 | ICVT | 0.0151 | iShares Convertible Bond ETF               | Convertible Bond            | LONG |
| 11 | ARKK | 0.0148 | ARK Innovation ETF                         | Technology                  | LONG |
| 25 | CIBR | 0.0148 | First Trust NASDAQ Cybersecurity ETF       | Technology                  | LONG |
| 32 | PTNQ | 0.0148 | Pacer Trendpilot 100 ETF                   | Trend-Following Equity      | LONG |
| 23 | IBUY | 0.0147 | Amplify Online Retail ETF                  | Consumer Discretionary      | FLAT |
| 10 | HACK | 0.0146 | Amplify Cybersecurity ETF                  | Technology                  | LONG |
| 30 | IPAY | 0.0146 | Amplify Digital Payments ETF               | Financials                  | FLAT |
| 16 | PRNT | 0.0144 | 3D Printing (The) ETF                      | Technology                  | LONG |
| 9  | BLOK | 0.0137 | Amplify Blockchain Technology ETF          | Technology                  | LONG |
| 56 | PTLC | 0.0128 | Pacer Trendpilot US Large Cap ETF          | U.S. Trend-Following Equ... | LONG |
| 58 | BLCN | 0.0123 | Siren NexGen Economy ETF                   | Technology                  | LONG |

**Cluster 4 – Mixed: IWB, ITOT, SCHB****Members:** 61 **Var explained:** 0.892 **Signal:** LONG since 2026-04-13 (20 days, 5.9%)**Signal logic:**

- Latest buy\_cross: 2026-04-13 – filter **PASS** (126d at date = +26.5%)

- Latest sell\_cross: 2026-03-06
- Current 126d return: +28.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



### Category breakdown

| category                     | n_members | abs_weight |
|------------------------------|-----------|------------|
| U.S. Large Cap Equity        | 9         | 0.1558     |
| U.S. Total Market Equity     | 7         | 0.1214     |
| Global Equity                | 7         | 0.1191     |
| Consumer Discretionary       | 6         | 0.0914     |
| Asset Allocation             | 4         | 0.0631     |
| Communication Services       | 4         | 0.0592     |
| U.S. Mega Cap Equity         | 3         | 0.0510     |
| U.S. Quality / Factor Equity | 3         | 0.0505     |
| U.S. Multifactor Equity      | 3         | 0.0504     |
| U.S. ESG Equity              | 2         | 0.0345     |
| U.S. Equity                  | 2         | 0.0345     |
| Unknown                      | 2         | 0.0312     |
| Commodity                    | 1         | 0.0173     |
| Large Cap Value Equities     | 1         | 0.0172     |
| Large Cap Blend Equities     | 1         | 0.0171     |
| Dividend Equity              | 1         | 0.0168     |
| U.S. Dividend Equity         | 1         | 0.0165     |
| Alternatives                 | 1         | 0.0149     |

|                          |   |        |
|--------------------------|---|--------|
| International IPO Equity | 1 | 0.0133 |
| Israel Equity            | 1 | 0.0126 |
| Singapore Equity         | 1 | 0.0121 |

**Replication basket** – VOO+IVV+SPY (cor=0.99) basket lead at lag 1: -0.005

| ticker | weight_ols | name                                | category              | etf_signal |
|--------|------------|-------------------------------------|-----------------------|------------|
| VOO    | 0.804      | Vanguard S&P 500 ETF                | U.S. Large Cap Equity | LONG       |
| IVV    | 0.309      | iShares Core S&P 500 ETF            | U.S. Large Cap Equity | LONG       |
| SPY    | -0.154     | State Street SPDR S&P 500 ETF Trust | U.S. Large Cap Equity | LONG       |

## Members

|    | ticker | weight | name                                       | category                 | etf_signal |
|----|--------|--------|--------------------------------------------|--------------------------|------------|
| 37 | IWB    | 0.0174 | iShares Russell 1000 ETF                   | U.S. Large Cap Equity    | LONG       |
| 49 | ITOT   | 0.0174 | iShares Core S&P Total U.S. Stock Marke... | U.S. Total Market Equity | LONG       |
| 51 | SCHB   | 0.0174 | Schwab U.S. Broad Market ETF               | U.S. Total Market Equity | LONG       |
| 4  | SPY    | 0.0173 | State Street SPDR S&P 500 ETF Trust        | U.S. Large Cap Equity    | LONG       |
| 12 | URTH   | 0.0173 | iShares MSCI World ETF                     | Global Equity            | LONG       |
| 30 | IWV    | 0.0173 | iShares Russell 3000 Fund                  | U.S. Total Market Equity | LONG       |
| 31 | IYY    | 0.0173 | iShares Dow Jones U.S. ETF                 | U.S. Total Market Equity | LONG       |
| 34 | SPYX   | 0.0173 | State Street SPDR S&P 500 Fossil Fuel R... | U.S. Equity              | LONG       |
| 36 | ESGU   | 0.0173 | iShares ESG Aware MSCI USA ETF             | U.S. ESG Equity          | LONG       |
| 42 | VTHR   | 0.0173 | Vanguard Russell 3000 ETF                  | U.S. Total Market Equity | LONG       |
| 43 | GSLC   | 0.0173 | Goldman Sachs ActiveBeta U.S. Large Cap... | Commodity                | LONG       |
| 45 | VONE   | 0.0173 | Vanguard Russell 1000 ETF                  | U.S. Large Cap Equity    | LONG       |
| 47 | VV     | 0.0173 | Vanguard Large-Cap ETF                     | U.S. Large Cap Equity    | LONG       |
| 48 | ILCB   | 0.0173 | iShares Morningstar Large-Cap ETF          | U.S. Large Cap Equity    | LONG       |
| 50 | IVV    | 0.0173 | iShares Core S&P 500 ETF                   | U.S. Large Cap Equity    | LONG       |
| 52 | SCHX   | 0.0173 | Schwab U.S. Large-Cap ETF                  | U.S. Large Cap Equity    | LONG       |
| 53 | SPLG   | 0.0173 |                                            |                          | LONG       |
| 54 | SPTM   | 0.0173 | State Street SPDR Portfolio S&P 1500 Co... | U.S. Total Market Equity | LONG       |
| 55 | VOO    | 0.0173 | Vanguard S&P 500 ETF                       | U.S. Large Cap Equity    | LONG       |
| 56 | VTI    | 0.0173 | Vanguard Total Stock Market ETF            | U.S. Total Market Equity | LONG       |
| 26 | JHML   | 0.0172 | John Hancock Multifactor Large Cap ETF     | U.S. Large Cap Equity    | LONG       |
| 29 | SUSA   | 0.0172 | iShares ESG Optimized MSCI USA ETF         | U.S. ESG Equity          | LONG       |
| 38 | IWL    | 0.0172 | iShares Russell Top 200 ETF                | U.S. Equity              | LONG       |
| 44 | EPS    | 0.0172 | WisdomTree U.S. LargeCap Fund              | Large Cap Value Equities | LONG       |
| 46 | MGC    | 0.0172 | Vanguard Mega Cap ETF                      | U.S. Mega Cap Equity     | LONG       |
| 7  | ACWI   | 0.0171 | iShares MSCI ACWI ETF                      | Global Equity            | LONG       |
| 13 | CRBN   | 0.0171 | iShares Low Carbon Optimized MSCI ACWI ... | Global Equity            | LONG       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 15 | VT   | 0.0171 | Vanguard Total World Stock Index ETF       | Global Equity               | LONG |
| 22 | TILT | 0.0171 | FlexShares Mornigstar US Market Factors... | U.S. Multifactor Equity     | LONG |
| 28 | DSI  | 0.0171 | iShares ESG MSCI KLD 400 ETF               | Large Cap Blend Equities    | LONG |
| 32 | LRGF | 0.0171 | iShares U.S. Equity Factor ETF             | U.S. Multifactor Equity     | LONG |
| 8  | AOA  | 0.0170 | iShares Core 80/20 Aggressive Allocatio... | Asset Allocation            | LONG |
| 14 | SPGM | 0.0170 | State Street SPDR Portfolio MSCI Global... | Global Equity               | LONG |
| 33 | OEF  | 0.0170 | iShares S&P 100 Fund                       | U.S. Mega Cap Equity        | LONG |
| 39 | QUAL | 0.0170 | iShares MSCI USA Quality Factor ETF        | U.S. Quality / Factor Eq... | LONG |
| 24 | QDF  | 0.0168 | FlexShares Quality Dividend Index Fund     | Dividend Equity             | LONG |
| 40 | QUS  | 0.0168 | State Street SPDR MSCI USA StrategicFac... | U.S. Quality / Factor Eq... | LONG |
| 23 | IOO  | 0.0167 | iShares Global 100 ETF                     | Global Equity               | LONG |
| 25 | CATH | 0.0167 | Global X S&P 500 Catholic Values ETF       | Global Equity               | LONG |
| 35 | XLG  | 0.0167 | Invesco S&P 500 Top 50 ETF                 | U.S. Mega Cap Equity        | LONG |
| 41 | SPHQ | 0.0166 | Invesco S&P 500 Quality ETF                | U.S. Quality / Factor Eq... | LONG |
| 11 | AOR  | 0.0165 | iShares Core 60/40 Balanced Allocation ... | Asset Allocation            | LONG |
| 27 | DGRW | 0.0165 | WisdomTree U.S. Quality Dividend Growth... | U.S. Dividend Equity        | LONG |
| 17 | IYC  | 0.0162 | iShares U.S. Consumer Discretionary ETF    | Consumer Discretionary      | LONG |
| 60 | OMFL | 0.0161 | Invesco Russell 1000 Dynamic Multifacto... | U.S. Multifactor Equity     | LONG |
| 21 | FDIS | 0.0159 | Fidelity MSCI Consumer Discretionary In... | Consumer Discretionary      | LONG |
| 18 | VCR  | 0.0158 | Vanguard Consumer Discretion ETF           | Consumer Discretionary      | LONG |
| 1  | XLY  | 0.0156 | State Street Consumer Discretionary Sel... | Consumer Discretionary      | LONG |
| 10 | AOM  | 0.0155 | iShares Core 40/60 Moderate Allocation ... | Asset Allocation            | LONG |
| 19 | VOX  | 0.0153 | Vanguard Communication Services ETF        | Communication Services      | LONG |
| 20 | FCOM | 0.0153 | Fidelity MSCI Communication Services In... | Communication Services      | LONG |
| 2  | XLC  | 0.0149 | State Street Communication Services Sel... | Communication Services      | LONG |
| 5  | QAI  | 0.0149 | NYLI Hedge Multi-Strategy Tracker ETF      | Alternatives                | LONG |
| 59 | RTH  | 0.0147 | VanEck Retail ETF                          | Consumer Discretionary      | LONG |
| 9  | AOK  | 0.0142 | iShares Core 30/70 Conservative Allocat... | Asset Allocation            | LONG |
| 61 | IGN  | 0.0139 |                                            |                             | FLAT |
| 57 | IYZ  | 0.0137 | iShares U.S. Telecommunications ETF        | Communication Services      | FLAT |
| 16 | FPXI | 0.0133 | First Trust International Equity Opport... | International IPO Equity    | LONG |
| 3  | ESPO | 0.0132 | VanEck Video Gaming and eSports ETF        | Consumer Discretionary      | FLAT |
| 58 | EIS  | 0.0126 | iShares MSCI Israel ETF                    | Israel Equity               | LONG |
| 6  | EWS  | 0.0121 | iShares MSCI Singapore ETF                 | Singapore Equity            | LONG |

**Cluster 5 – Mixed: BND, AGG, SCHZ****Members:** 58   **Var explained:** 0.832   **Signal:** LONG since 2026-05-11 (0 days, 0.0%)**Signal logic:**

- Latest buy\_cross: 2026-05-11 – filter **PASS** (126d at date = +5.1%)



- Latest sell\_cross: 2026-04-30
- Current 126d return: +5.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-11 > last sell at 2026-04-30)



### Category breakdown

| category                        | n_members | abs_weight |
|---------------------------------|-----------|------------|
| Fixed Income                    | 19        | 0.3282     |
| Investment Grade Corporate Bond | 13        | 0.2301     |
| Aggregate Fixed Income          | 6         | 0.1120     |
| Mortgage-Backed Fixed Income    | 6         | 0.0982     |
| Active Fixed Income             | 5         | 0.0890     |
| Cash / Ultra-Short Fixed Income | 3         | 0.0470     |
| International Fixed Income      | 2         | 0.0313     |
| Target Maturity Bond            | 2         | 0.0309     |
| Agency Fixed Income             | 1         | 0.0166     |
| Taxable Municipal Bond          | 1         | 0.0165     |

**Replication basket** – BND+AGG+BNDX (cor=1.00) basket lead at lag 1: 0.010

| ticker | weight_ols | name                           | category               | etf_signal |
|--------|------------|--------------------------------|------------------------|------------|
| BND    | 0.629      | Vanguard Total Bond Market ETF | Aggregate Fixed Income | LONG       |

|      |       |                                       |                          |      |
|------|-------|---------------------------------------|--------------------------|------|
| AGG  | 0.354 | iShares Core U.S. Aggregate Bond ETF  | Aggregate Fixed Income   | LONG |
| BNDX | 0.033 | Vanguard Total International Bond ETF | International Fixed I... | LONG |

## Members

|    | ticker | weight | name                                        | category                    | etf_signal |
|----|--------|--------|---------------------------------------------|-----------------------------|------------|
| 21 | BIV    | 0.0188 | Vanguard Intermediate-Term Bond ETF         | Fixed Income                | FLAT       |
| 26 | AGG    | 0.0188 | iShares Core U.S. Aggregate Bond ETF        | Aggregate Fixed Income      | LONG       |
| 28 | BND    | 0.0188 | Vanguard Total Bond Market ETF              | Aggregate Fixed Income      | LONG       |
| 30 | SCHZ   | 0.0188 | Schwab US Aggregate Bond ETF                | Aggregate Fixed Income      | LONG       |
| 20 | IUSB   | 0.0187 | iShares Core Universal USD Bond ETF         | Aggregate Fixed Income      | LONG       |
| 31 | SPAB   | 0.0187 | State Street SPDR Portfolio Aggregate B...  | Aggregate Fixed Income      | LONG       |
| 2  | IEF    | 0.0185 | iShares 7-10 Year Treasury Bond ETF         | Fixed Income                | FLAT       |
| 17 | IGIB   | 0.0184 | iShares 5-10 Year Investment Grade Corp...  | Investment Grade Corpora... | LONG       |
| 24 | VCIT   | 0.0184 | Vanguard Intermediate-Term Corporate Bo...  | Investment Grade Corpora... | LONG       |
| 7  | FBND   | 0.0183 | Fidelity Total Bond ETF                     | Active Fixed Income         | LONG       |
| 43 | SPTI   | 0.0183 | State Street SPDR Portfolio Intermediat...  | Fixed Income                | FLAT       |
| 47 | VGIT   | 0.0183 | Vanguard Intermediate-Term Treasury ETF     | Fixed Income                | FLAT       |
| 13 | AGGY   | 0.0182 | WisdomTree Yield Enhanced U.S. Aggregat...  | Aggregate Fixed Income      | LONG       |
| 15 | SPIB   | 0.0182 | State Street SPDR Portfolio Intermediat...  | Investment Grade Corpora... | LONG       |
| 46 | SCHR   | 0.0182 | Schwab Intermediate-Term U.S. Treasury ...  | Fixed Income                | FLAT       |
| 6  | GTO    | 0.0181 | Invesco Total Return Bond ETF               | Active Fixed Income         | LONG       |
| 11 | GVI    | 0.0181 | iShares Intermediate Government/Credit ...  | Fixed Income                | LONG       |
| 40 | IEI    | 0.0181 | iShares 3-7 Year Treasury Bond ETF          | Fixed Income                | FLAT       |
| 12 | QLTA   | 0.0180 | iShares Aaa A Rated Corporate Bond ETF      | Investment Grade Corpora... | LONG       |
| 27 | USIG   | 0.0180 | iShares Broad USD Investment Grade Corp...  | Investment Grade Corpora... | LONG       |
| 50 | MBB    | 0.0180 | iShares MBS ETF                             | Mortgage-Backed Fixed In... | LONG       |
| 51 | VMBS   | 0.0180 | Vanguard Mortgage-Backed Securities ETF     | Mortgage-Backed Fixed In... | LONG       |
| 3  | LQD    | 0.0179 | iShares iBoxx \$ Investment Grade Corpor... | Investment Grade Corpora... | LONG       |
| 5  | BOND   | 0.0179 | PIMCO Active Bond Exchange-Traded Fund ...  | Active Fixed Income         | LONG       |
| 52 | SPMB   | 0.0179 | State Street SPDR Portfolio Mortgage Ba...  | Mortgage-Backed Fixed In... | LONG       |
| 19 | ISTB   | 0.0178 | iShares Core 1-5 Year USD Bond ETF          | Fixed Income                | FLAT       |
| 8  | FLCO   | 0.0177 | Franklin Investment Grade Corporate ETF     | Active Fixed Income         | LONG       |
| 10 | CORP   | 0.0177 | PIMCO Investment Grade Corporate Bond I...  | Investment Grade Corpora... | LONG       |
| 22 | BLV    | 0.0176 | Vanguard Long-Term Bond ETF                 | Fixed Income                | FLAT       |
| 41 | TLH    | 0.0176 | iShares 10-20 Year Treasury Bond ETF        | Fixed Income                | FLAT       |
| 45 | GOVT   | 0.0176 | iShares U.S. Treasury Bond ETF              | Fixed Income                | FLAT       |
| 35 | IGSB   | 0.0175 | iShares 1-5 Year Investment Grade Corpo...  | Investment Grade Corpora... | FLAT       |
| 37 | VCSH   | 0.0175 | Vanguard Short-Term Corporate Bond ETF      | Investment Grade Corpora... | FLAT       |
| 23 | BSV    | 0.0174 | Vanguard Short-Term Bond ETF                | Fixed Income                | LONG       |
| 49 | GNMA   | 0.0174 | iShares GNMA Bond ETF                       | Mortgage-Backed Fixed In... | LONG       |
| 18 | IGLB   | 0.0172 | iShares 10 Year Investment Grade Corpor...  | Investment Grade Corpora... | LONG       |
| 44 | SPTL   | 0.0172 | State Street SPDR Portfolio Long Term T...  | Fixed Income                | FLAT       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 16 | SPLB | 0.0171 | State Street SPDR Portfolio Long Term C... | Investment Grade Corpora... | LONG |
| 25 | VCLT | 0.0171 | Vanguard Long-Term Corporate Bond ETF      | Investment Grade Corpora... | LONG |
| 36 | SLQD | 0.0171 | iShares 0-5 Year Investment Grade Corpo... | Investment Grade Corpora... | FLAT |
| 48 | VGLT | 0.0171 | Vanguard Long-Term Treasury ETF            | Fixed Income                | FLAT |
| 29 | TOTL | 0.0170 | State Street DoubleLine Total Return Ta... | Active Fixed Income         | FLAT |
| 1  | TLT  | 0.0169 | iShares 20+ Year Treasury Bond ETF         | Fixed Income                | FLAT |
| 39 | AGZ  | 0.0166 | iShares Agency Bond ETF                    | Agency Fixed Income         | LONG |
| 9  | BAB  | 0.0165 | Invesco Taxable Municipal Bond ETF         | Taxable Municipal Bond      | LONG |
| 42 | EDV  | 0.0163 | Vanguard Extended Duration Treasury ETF    | Fixed Income                | FLAT |
| 34 | SPSB | 0.0162 | State Street SPDR Portfolio Short Term ... | Fixed Income                | LONG |
| 53 | SHY  | 0.0162 | iShares 1-3 Year Treasury Bond ETF         | Fixed Income                | FLAT |
| 56 | VGSH | 0.0159 | Vanguard Short-Term Treasury ETF           | Cash / Ultra-Short Fixed... | FLAT |
| 57 | BNDX | 0.0158 | Vanguard Total International Bond ETF      | International Fixed Inco... | LONG |
| 33 | BSCQ | 0.0157 | Invesco BulletShares 2026 Corporate Bon... | Target Maturity Bond        | FLAT |
| 55 | SCHO | 0.0156 | Schwab Short-Term U.S. Treasury ETF        | Cash / Ultra-Short Fixed... | FLAT |
| 58 | IAGG | 0.0156 | iShares International Aggregate Bond Fu... | International Fixed Inco... | LONG |
| 54 | SPTS | 0.0155 | State Street SPDR Portfolio Short Term ... | Cash / Ultra-Short Fixed... | FLAT |
| 14 | IBDR | 0.0152 | iShares iBonds Dec 2026 Term Corporate ... | Target Maturity Bond        | FLAT |
| 38 | CMBS | 0.0136 | iShares CMBS Bond ETF                      | Mortgage-Backed Fixed In... | LONG |
| 4  | LMBS | 0.0133 | First Trust Low Duration Opportunities ... | Mortgage-Backed Fixed In... | FLAT |
| 32 | LDUR | 0.0120 | PIMCO Enhanced Low Duration Active Exch... | Fixed Income                | LONG |

### Cluster 6 – Mixed: IWD, SCHV, VONV

**Members:** 54   **Var explained:** 0.922   **Signal:** LONG since 2026-04-13 (20 days, 2.6%)

#### Signal logic:

- Latest buy\_cross: 2026-04-13 – filter **PASS** (126d at date = +23.0%)
- Latest sell\_cross: 2026-03-06
- Current 126d return: +21.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



### Category breakdown

| category                     | n_members | abs_weight |
|------------------------------|-----------|------------|
| U.S. Dividend Equity         | 11        | 0.2040     |
| U.S. Large Cap Value         | 10        | 0.1903     |
| U.S. Multifactor Equity      | 3         | 0.0559     |
| U.S. Mid Cap Equity          | 3         | 0.0556     |
| U.S. Value Factor Equity     | 3         | 0.0553     |
| U.S. Low Volatility Equity   | 3         | 0.0514     |
| U.S. Fundamental Equity      | 2         | 0.0380     |
| U.S. Equal Weight Equity     | 2         | 0.0377     |
| U.S. Mid Cap Value           | 2         | 0.0376     |
| U.S. Equity                  | 2         | 0.0375     |
| Value Equity                 | 2         | 0.0369     |
| U.S. Revenue Weighted Equity | 1         | 0.0188     |
| Large Cap Blend Equities     | 1         | 0.0188     |
| U.S. Mega Cap Value          | 1         | 0.0188     |
| U.S. Size Factor Equity      | 1         | 0.0187     |
| U.S. Large Cap Equity        | 1         | 0.0184     |
| U.S. Buyback Equity          | 1         | 0.0183     |
| Dividend Equity              | 1         | 0.0180     |
| U.S. Quality / Factor Equity | 1         | 0.0179     |
| U.S. Moat Equity             | 1         | 0.0177     |
| U.S. ESG Equity              | 1         | 0.0176     |

Global Equity 1 0.0168

**Replication basket** – VTV+VIG+VO (cor=1.00) basket lead at lag 1: 0.012

| ticker | weight_ols | name                          | category             | etf_signal |
|--------|------------|-------------------------------|----------------------|------------|
| VTV    | 0.546      | Vanguard Value ETF            | U.S. Large Cap Value | LONG       |
| VIG    | 0.158      | Vanguard Div Appreciation ETF | U.S. Dividend Equity | LONG       |
| VO     | 0.278      | Vanguard Mid-Cap ETF          | U.S. Mid Cap Equity  | LONG       |

## Members

|    | ticker | weight | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 22 | IWD    | 0.0192 | iShares Russell 1000 Value ETF             | U.S. Large Cap Value        | LONG       |
| 2  | RSP    | 0.0191 | Invesco S&P 500 Equal Weight ETF           | U.S. Equal Weight Equity    | LONG       |
| 9  | PRF    | 0.0191 | Invesco RAFI US 1000 ETF                   | U.S. Fundamental Equity     | LONG       |
| 29 | VONV   | 0.0191 | Vanguard Russell 1000 Value ETF            | U.S. Large Cap Value        | LONG       |
| 32 | ILCV   | 0.0191 | iShares Morningstar Large-Cap Value ETF    | U.S. Large Cap Value        | LONG       |
| 35 | SCHV   | 0.0191 | Schwab U.S. Large-Cap Value ETF            | U.S. Large Cap Value        | LONG       |
| 12 | CFA    | 0.0190 | VictoryShares US 500 Volatility Wtd ETF    | U.S. Multifactor Equity     | LONG       |
| 18 | FNDX   | 0.0190 | Schwab Fundamental U.S. Large Company E... | U.S. Fundamental Equity     | LONG       |
| 24 | IVE    | 0.0190 | iShares S&P 500 Value ETF                  | U.S. Large Cap Value        | LONG       |
| 25 | JPUS   | 0.0190 | JPMorgan Diversified Return U.S. Equity... | U.S. Multifactor Equity     | LONG       |
| 34 | IUSV   | 0.0190 | iShares Core S&P U.S. Value ETF            | U.S. Large Cap Value        | LONG       |
| 38 | VTV    | 0.0190 | Vanguard Value ETF                         | U.S. Large Cap Value        | LONG       |
| 45 | DTD    | 0.0190 | WisdomTree U.S. Total Dividend Fund        | U.S. Dividend Equity        | LONG       |
| 50 | DGRO   | 0.0190 | iShares Core Dividend Growth ETF           | U.S. Dividend Equity        | LONG       |
| 15 | NULV   | 0.0189 | Nuveen ESG Large-Cap Value ETF             | U.S. Large Cap Value        | LONG       |
| 20 | IWX    | 0.0189 | iShares Russell Top 200 Value ETF          | Value Equity                | LONG       |
| 28 | VOOV   | 0.0189 | Vanguard S&P 500 Value ETF                 | U.S. Large Cap Value        | LONG       |
| 36 | SPYV   | 0.0189 | State Street SPDR Portfolio S&P 500 Val... | U.S. Large Cap Value        | LONG       |
| 44 | DLN    | 0.0189 | WisdomTree U.S. LargeCap Dividend Fund     | U.S. Dividend Equity        | LONG       |
| 5  | FEX    | 0.0188 | First Trust Large Cap Core AlphaDEX Fund   | Large Cap Blend Equities    | LONG       |
| 10 | RWL    | 0.0188 | Invesco S&P 500 Revenue ETF                | U.S. Revenue Weighted Eq... | LONG       |
| 30 | MGV    | 0.0188 | Vanguard Mega Cap Value ETF                | U.S. Mega Cap Value         | LONG       |
| 31 | VYM    | 0.0188 | Vanguard High Dividend Yield ETF           | U.S. Dividend Equity        | LONG       |
| 39 | IWS    | 0.0188 | iShares Russell Mid-Cap Value ETF          | U.S. Mid Cap Value          | LONG       |
| 40 | VOE    | 0.0188 | Vanguard Mid-Cap Value ETF                 | U.S. Mid Cap Value          | LONG       |
| 42 | PFM    | 0.0188 | Invesco Dividend Achievers ETF             | U.S. Dividend Equity        | LONG       |
| 11 | AIVL   | 0.0187 | WisdomTree U.S. AI Enhanced Value Fund     | U.S. Value Factor Equity    | LONG       |
| 21 | ONEV   | 0.0187 | State Street SPDR Russell 1000 Low Vola... | U.S. Equity                 | LONG       |
| 26 | SIZE   | 0.0187 | iShares MSCI USA Size Factor ETF           | U.S. Size Factor Equity     | LONG       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 51 | VIG  | 0.0187 | Vanguard Div Appreciation ETF              | U.S. Dividend Equity        | LONG |
| 53 | JPME | 0.0187 | JPMorgan Diversified Return U.S. Mid Ca... | U.S. Equity                 | LONG |
| 19 | EQAL | 0.0186 | Invesco Russell 1000 Equal Weight ETF      | U.S. Equal Weight Equity    | LONG |
| 33 | IMCB | 0.0186 | iShares Morningstar Mid-Cap ETF            | U.S. Mid Cap Equity         | LONG |
| 54 | JVAL | 0.0186 | JPMorgan U.S. Value Factor ETF             | U.S. Value Factor Equity    | LONG |
| 17 | FDVV | 0.0185 | Fidelity High Dividend ETF                 | U.S. Dividend Equity        | LONG |
| 23 | IWR  | 0.0185 | iShares Russell Mid-Cap ETF                | U.S. Mid Cap Equity         | LONG |
| 37 | VO   | 0.0185 | Vanguard Mid-Cap ETF                       | U.S. Mid Cap Equity         | LONG |
| 47 | DIA  | 0.0184 | State Street SPDR Dow Jones Industrial ... | U.S. Large Cap Equity       | LONG |
| 4  | PKW  | 0.0183 | Invesco BuyBack Achievers ETF              | U.S. Buyback Equity         | FLAT |
| 14 | NOBL | 0.0182 | ProShares S&P 500 Dividend Aristocrats ... | U.S. Dividend Equity        | FLAT |
| 6  | PWV  | 0.0181 | Invesco Large Cap Value ETF                | Value Equity                | LONG |
| 7  | RDVY | 0.0181 | First Trust Rising Dividend Achievers E... | U.S. Dividend Equity        | LONG |
| 3  | FVD  | 0.0180 | First Trust VL Dividend                    | U.S. Dividend Equity        | FLAT |
| 16 | SDY  | 0.0180 | State Street SPDR S&P Dividend ETF         | U.S. Dividend Equity        | LONG |
| 27 | VLUE | 0.0180 | iShares MSCI USA Value Factor ETF          | U.S. Value Factor Equity    | LONG |
| 43 | OUSA | 0.0180 | ALPS O'Shares U.S. Quality Dividend ETF    | Dividend Equity             | LONG |
| 41 | FTCS | 0.0179 | First Trust Capital Strength ETF           | U.S. Quality / Factor Eq... | FLAT |
| 49 | LGLV | 0.0179 | State Street SPDR US Large Cap Low Vola... | U.S. Low Volatility Equi... | FLAT |
| 13 | CFO  | 0.0178 | VictoryShares US 500 Enhanced Volatilit... | U.S. Multifactor Equity     | LONG |
| 8  | MOAT | 0.0177 | VanEck Morningstar Wide Moat ETF           | U.S. Moat Equity            | LONG |
| 48 | USMV | 0.0176 | iShares MSCI USA Min Vol Factor ETF        | U.S. Low Volatility Equi... | LONG |
| 52 | ETHO | 0.0176 | Amplify Etho Climate Leadership U.S. ETF   | U.S. ESG Equity             | LONG |
| 1  | ACWV | 0.0168 | iShares MSCI Global Min Vol Factor ETF     | Global Equity               | LONG |
| 46 | SPLV | 0.0159 | Invesco S&P 500 Low Volatility ETF         | U.S. Low Volatility Equi... | FLAT |

### Cluster 7 – Mixed: VBR, MDYV, IVOV

**Members:** 48   **Var explained:** 0.880   **Signal:** LONG since 2026-04-10 (21 days, 2.9%)

#### Signal logic:

- Latest buy\_cross: 2026-04-10 – filter **PASS** (126d at date = +35.3%)
- Latest sell\_cross: 2026-02-27
- Current 126d return: +22.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-02-27)



### Category breakdown

| category                              | n_members | abs_weight |
|---------------------------------------|-----------|------------|
| Financials                            | 13        | 0.2576     |
| U.S. Small Cap Value                  | 6         | 0.1304     |
| U.S. Small Cap Equity                 | 5         | 0.1082     |
| U.S. Mid Cap Value                    | 3         | 0.0657     |
| Dividend Equity                       | 2         | 0.0424     |
| Leveraged / Inverse                   | 2         | 0.0422     |
| U.S. Small Cap Low Volatility         | 2         | 0.0407     |
| U.S. Small Cap Fundamental Equity     | 1         | 0.0218     |
| Mid Cap Value Equities                | 1         | 0.0218     |
| U.S. Mid Cap Dividend Equity          | 1         | 0.0217     |
| Small Cap Value Equities              | 1         | 0.0216     |
| U.S. Small Cap Dividend Equity        | 1         | 0.0216     |
| U.S. Mid/Small Cap Equity             | 1         | 0.0216     |
| U.S. Small/Mid Cap Fundamental Equity | 1         | 0.0215     |
| Value Equity                          | 1         | 0.0215     |
| U.S. Mid Cap Equity                   | 1         | 0.0214     |
| U.S. Small/Mid Cap Equity             | 1         | 0.0213     |
| U.S. Equity                           | 1         | 0.0210     |
| U.S. Dividend Equity                  | 1         | 0.0206     |
| U.S. Mid Cap Low Volatility           | 1         | 0.0198     |
| Global Equity                         | 1         | 0.0181     |

Real Estate 1 0.0173

**Replication basket** – IJR+XLF+VBR (cor=1.00) basket lead at lag 1: -0.003

| ticker | weight_ols | name                                     | category              | etf_signal |
|--------|------------|------------------------------------------|-----------------------|------------|
| IJR    | 0.363      | iShares Core S&P Small-Cap ETF           | U.S. Small Cap Equity | LONG       |
| XLF    | 0.219      | State Street Financial Select Sector ... | Financials            | LONG       |
| VBR    | 0.430      | Vanguard Small-Cap Value ETF             | U.S. Small Cap Value  | LONG       |

## Members

|    | ticker | weight | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 37 | VBR    | 0.0220 | Vanguard Small-Cap Value ETF               | U.S. Small Cap Value        | LONG       |
| 28 | IJJ    | 0.0219 | iShares S&P Mid-Cap 400 Value ETF          | U.S. Mid Cap Value          | LONG       |
| 30 | IVOV   | 0.0219 | Vanguard S&P Mid-Cap 400 Value ETF         | U.S. Mid Cap Value          | LONG       |
| 31 | MDYV   | 0.0219 | State Street SPDR S&P 400 Mid Cap Value... | U.S. Mid Cap Value          | LONG       |
| 21 | EZM    | 0.0218 | WisdomTree U.S. MidCap Fund                | Mid Cap Value Equities      | LONG       |
| 23 | FNDA   | 0.0218 | Schwab Fundamental U.S. Small Company E... | U.S. Small Cap Fundament... | LONG       |
| 8  | FYX    | 0.0217 | First Trust Small Cap Core AlphaDEX Fund   | U.S. Small Cap Equity       | LONG       |
| 10 | VI00   | 0.0217 | Vanguard S&P Small-Cap 600 ETF             | U.S. Small Cap Equity       | LONG       |
| 11 | IJR    | 0.0217 | iShares Core S&P Small-Cap ETF             | U.S. Small Cap Equity       | LONG       |
| 12 | SPSM   | 0.0217 | State Street SPDR Portfolio S&P 600 Sma... | U.S. Small Cap Equity       | LONG       |
| 19 | DON    | 0.0217 | WisdomTree U.S. MidCap Dividend Fund       | U.S. Mid Cap Dividend Eq... | LONG       |
| 26 | IWN    | 0.0217 | iShares Russell 2000 Value ETF             | U.S. Small Cap Value        | LONG       |
| 29 | IJS    | 0.0217 | iShares S&P SmallCap 600 Value ETF         | U.S. Small Cap Value        | LONG       |
| 32 | SLYV   | 0.0217 | State Street SPDR S&P 600 Small Cap Val... | U.S. Small Cap Value        | LONG       |
| 34 | VTWV   | 0.0217 | Vanguard Russell 2000 Value ETF            | U.S. Small Cap Value        | LONG       |
| 47 | SDVY   | 0.0217 | First Trust SMID Cap Rising Dividend Ac... | Dividend Equity             | LONG       |
| 7  | FNX    | 0.0216 | First Trust Mid Cap Core AlphaDEX Fund     | U.S. Mid/Small Cap Equity   | LONG       |
| 18 | DES    | 0.0216 | WisdomTree U.S. SmallCap Dividend Fund     | U.S. Small Cap Dividend ... | LONG       |
| 20 | EES    | 0.0216 | WisdomTree U.S. SmallCap Fund              | Small Cap Value Equities    | LONG       |
| 33 | VIOV   | 0.0216 | Vanguard S&P Small-Cap 600 Value ETF       | U.S. Small Cap Value        | LONG       |
| 9  | PRFZ   | 0.0215 | Invesco RAFI US 1500 Small-Mid ETF         | U.S. Small/Mid Cap Funda... | LONG       |
| 46 | FYT    | 0.0215 | First Trust Small Cap Value AlphaDEX Fu... | Value Equity                | LONG       |
| 41 | RWK    | 0.0214 | Invesco S&P MidCap 400 Revenue ETF         | U.S. Mid Cap Equity         | LONG       |
| 38 | RWJ    | 0.0213 | Invesco S&P SmallCap 600 Revenue ETF       | U.S. Small Cap Equity       | LONG       |
| 48 | ISMD   | 0.0213 | Inspire Small/Mid Cap ETF                  | U.S. Small/Mid Cap Equity   | LONG       |
| 39 | RFV    | 0.0212 | Invesco S&P MidCap 400 Pure Value ETF      | Leveraged / Inverse         | LONG       |
| 42 | OMFS   | 0.0210 | Invesco Russell 2000 Dynamic Multifact...  | U.S. Equity                 | LONG       |
| 13 | FXO    | 0.0209 | First Trust Financials AlphaDEX            | Financials                  | FLAT       |
| 43 | RZV    | 0.0209 | Invesco S&P SmallCap 600 Pure Value ETF    | Leveraged / Inverse         | LONG       |



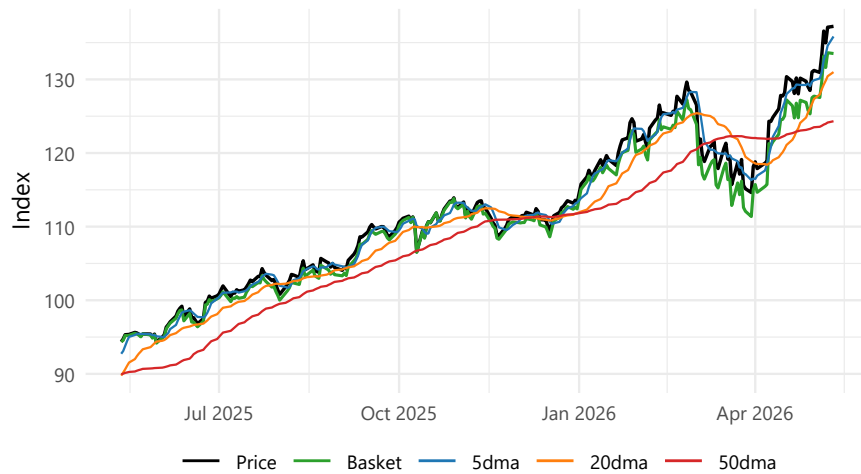
|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 6  | REGL | 0.0207 | ProShares S&P MidCap 400 Dividend Arist... | Dividend Equity             | LONG |
| 17 | SMDV | 0.0206 | ProShares Russell 2000 Dividend Growers... | U.S. Dividend Equity        | LONG |
| 25 | XSLV | 0.0204 | Invesco S&P SmallCap Low Volatility ETF    | U.S. Small Cap Low Volat... | LONG |
| 35 | VFH  | 0.0204 | Vanguard Financials ETF                    | Financials                  | FLAT |
| 36 | FNCL | 0.0204 | Fidelity MSCI Financials Index ETF         | Financials                  | LONG |
| 27 | SMMV | 0.0203 | iShares MSCI USA Small-Cap Min Vol Fact... | U.S. Small Cap Low Volat... | LONG |
| 3  | KBE  | 0.0202 | State Street SPDR S&P Bank ETF             | Financials                  | LONG |
| 15 | IYF  | 0.0202 | iShares U.S. Financial ETF                 | Financials                  | FLAT |
| 45 | KCE  | 0.0202 | State Street SPDR S&P Capital Markets E... | Financials                  | LONG |
| 16 | IYG  | 0.0201 | iShares U.S. Financial Services ETF        | Financials                  | FLAT |
| 24 | XMLV | 0.0198 | Invesco S&P MidCap Low Volatility ETF      | U.S. Mid Cap Low Volatil... | FLAT |
| 1  | XLF  | 0.0197 | State Street Financial Select Sector SP... | Financials                  | LONG |
| 22 | KBWB | 0.0197 | Invesco KBW Bank ETF                       | Financials                  | FLAT |
| 2  | KRE  | 0.0196 | State Street SPDR S&P Regional Banking ... | Financials                  | LONG |
| 14 | IAT  | 0.0194 | iShares U.S. Regional Banks ETF            | Financials                  | LONG |
| 44 | IAI  | 0.0189 | iShares U.S. Broker-Dealers & Securitie... | Financials                  | LONG |
| 40 | PSP  | 0.0181 | Invesco Global Listed Private Equity ETF   | Global Equity               | LONG |
| 4  | KIE  | 0.0177 | State Street SPDR S&P Insurance ETF        | Financials                  | FLAT |
| 5  | REM  | 0.0173 | iShares Mortgage Real Estate ETF           | Real Estate                 | LONG |

### Cluster 8 – Emerging Markets Equity (19/25)

**Members:** 25   **Var explained:** 0.875   **Signal:** LONG since 2026-04-10 (21 days, 9.9%)

#### Signal logic:

- Latest buy\_cross: 2026-04-10 – filter **PASS** (126d at date = +57.0%)
- Latest sell\_cross: 2026-03-05
- Current 126d return: +49.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-05)



### Category breakdown

| category                | n_members | abs_weight |
|-------------------------|-----------|------------|
| Emerging Markets Equity | 19        | 0.7724     |
| Asia Equity             | 2         | 0.0812     |
| Asia ex-Japan Equity    | 1         | 0.0418     |
| Taiwan Equity           | 1         | 0.0357     |
| Global Dividend Equity  | 1         | 0.0348     |
| South Korea Equity      | 1         | 0.0341     |

**Replication basket** – IEMG+VWO+EEM (cor=0.99) basket lead at lag 1: -0.021

| ticker | weight_ols | name                                   | category                 | etf_signal |
|--------|------------|----------------------------------------|--------------------------|------------|
| IEMG   | 0.776      | iShares Core MSCI Emerging Markets ETF | Emerging Markets Equi... | LONG       |
| VWO    | 0.178      | Vanguard FTSE Emerging Markets ETF     | Emerging Markets Equi... | LONG       |
| EEM    | -0.012     | iShares MSCI Emerging Index Fund       | Emerging Markets Equi... | LONG       |

### Members

|    | ticker | weight | name                                   | category                | etf_signal |
|----|--------|--------|----------------------------------------|-------------------------|------------|
| 21 | IEMG   | 0.0425 | iShares Core MSCI Emerging Markets ETF | Emerging Markets Equity | LONG       |

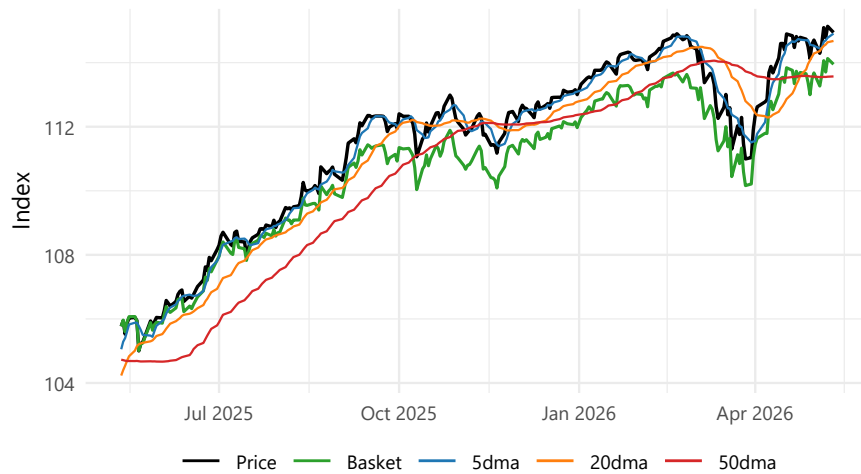
|    |      |        |                                            |                         |      |
|----|------|--------|--------------------------------------------|-------------------------|------|
| 2  | EEM  | 0.0424 | iShares MSCI Emerging Index Fund           | Emerging Markets Equity | LONG |
| 16 | GEM  | 0.0423 | Goldman Sachs ActiveBeta Emerging Marke... | Emerging Markets Equity | LONG |
| 23 | SPEM | 0.0423 | State Street SPDR Portfolio Emerging Ma... | Emerging Markets Equity | LONG |
| 3  | VWO  | 0.0422 | Vanguard FTSE Emerging Markets ETF         | Emerging Markets Equity | LONG |
| 20 | ESGE | 0.0422 | iShares ESG Aware MSCI EM ETF              | Emerging Markets Equity | LONG |
| 22 | SCHE | 0.0421 | Schwab Emerging Markets Equity ETF         | Emerging Markets Equity | LONG |
| 15 | EMGF | 0.0419 | iShares Emerging Markets Equity Factor ... | Emerging Markets Equity | LONG |
| 18 | XSOE | 0.0419 | WisdomTree Emerging Markets Ex-State Ow... | Emerging Markets Equity | LONG |
| 5  | AAXJ | 0.0418 | iShares MSCI All Country Asia ex Japan ... | Asia ex-Japan Equity    | LONG |
| 11 | EEMA | 0.0413 | iShares MSCI Emerging Markets Asia ETF     | Emerging Markets Equity | LONG |
| 14 | GMF  | 0.0411 | State Street SPDR S&P Emerging Asia Pac... | Asia Equity             | LONG |
| 12 | PXH  | 0.0410 | Invesco RAFI Emerging Markets ETF          | Emerging Markets Equity | LONG |
| 17 | FNDE | 0.0410 | Schwab Fundamental Emerging Markets Equ... | Emerging Markets Equity | LONG |
| 19 | EEMV | 0.0404 | iShares MSCI Emerging Markets Min Vol F... | Emerging Markets Equity | LONG |
| 8  | DGS  | 0.0402 | WisdomTree Emerging Market SmallCap Fund   | Emerging Markets Equity | LONG |
| 10 | AIA  | 0.0401 | iShares Asia 50 ETF                        | Asia Equity             | LONG |
| 7  | DEM  | 0.0396 | WisdomTree Emerging Markets High Divide... | Emerging Markets Equity | LONG |
| 6  | EWX  | 0.0390 | State Street SPDR S&P Emerging Markets ... | Emerging Markets Equity | LONG |
| 4  | FEM  | 0.0385 | First Trust Emerging Markets AlphaDEX F... | Emerging Markets Equity | LONG |
| 13 | DVYE | 0.0368 | iShares Emerging Markets Dividend Index... | Emerging Markets Equity | FLAT |
| 9  | EWT  | 0.0357 | iShares MSCI Taiwan ETF                    | Taiwan Equity           | LONG |
| 24 | SDIV | 0.0348 | Global X SuperDividend ETF                 | Global Dividend Equity  | LONG |
| 25 | FEMS | 0.0347 | First Trust Emerging Markets Small Cap ... | Emerging Markets Equity | FLAT |
| 1  | EWY  | 0.0341 | iShares MSCI South Korea ETF               | South Korea Equity      | LONG |

### Cluster 9 – High Yield Bond (10) + Preferred Stock (7)

**Members:** 22   **Var explained:** 0.772   **Signal:** LONG since 2026-04-13 (20 days, 0.7%)

#### Signal logic:

- Latest buy\_cross: 2026-04-13 – filter **PASS** (126d at date = +10.1%)
- Latest sell\_cross: 2026-03-05
- Current 126d return: +9.6% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-05)



### Category breakdown

| category                      | n_members | abs_weight |
|-------------------------------|-----------|------------|
| High Yield Bond               | 10        | 0.4918     |
| Preferred Stock               | 7         | 0.2893     |
| Emerging Markets Fixed Income | 4         | 0.1695     |
| Unknown                       | 1         | 0.0493     |

**Replication basket** – HYG+EMB+PFF (cor=0.99) basket lead at lag 1: 0.032

| ticker | weight_ols | name                                      | category                 | etf_signal |
|--------|------------|-------------------------------------------|--------------------------|------------|
| HYG    | 0.476      | iShares iBoxx \$ High Yield Corporate ... | High Yield Bond          | LONG       |
| EMB    | 0.157      | iShares J.P. Morgan USD Emerging Mark...  | Emerging Markets Fixe... | FLAT       |
| PFF    | 0.260      | iShares Preferred and Income Securit...   | Preferred Stock          | LONG       |

### Members

|    | ticker | weight | name                                        | category        | etf_signal |
|----|--------|--------|---------------------------------------------|-----------------|------------|
| 14 | JNK    | 0.0502 | State Street SPDR Bloomberg High Yield ...  | High Yield Bond | LONG       |
| 1  | HYG    | 0.0501 | iShares iBoxx \$ High Yield Corporate Bo... | High Yield Bond | LONG       |
| 20 | HYLB   | 0.0500 | Xtrackers USD High Yield Corporate Bond...  | High Yield Bond | LONG       |

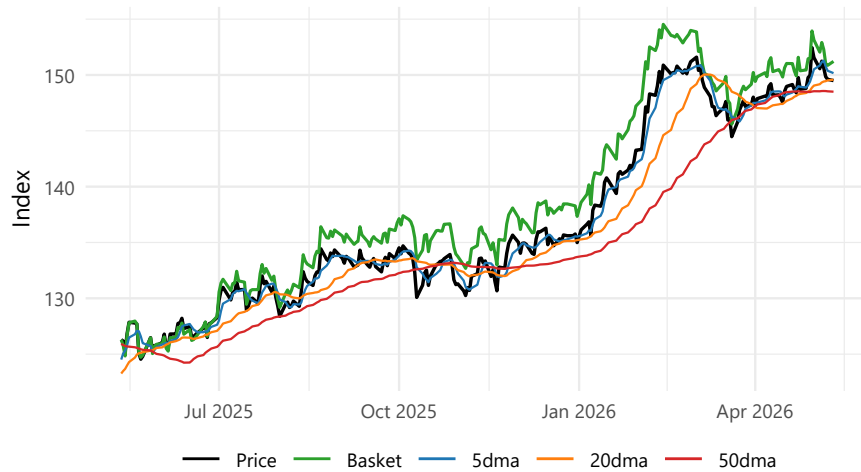
|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 19 | BBHY | 0.0499 | JPMorgan BetaBuilders USD High Yield Co... | High Yield Bond             | LONG |
| 17 | SHYG | 0.0497 | iShares 0-5 Year High Yield Corporate B... | High Yield Bond             | LONG |
| 15 | SJNK | 0.0496 | State Street SPDR Bloomberg Short Term ... | High Yield Bond             | LONG |
| 8  | PHB  | 0.0493 |                                            |                             | LONG |
| 16 | ANGL | 0.0492 | VanEck Fallen Angel High Yield Bond ETF    | High Yield Bond             | LONG |
| 18 | FALN | 0.0492 | iShares Fallen Angels USD Bond ETF         | High Yield Bond             | LONG |
| 6  | HYS  | 0.0488 | PIMCO 0-5 Year High Yield Corporat Bond... | High Yield Bond             | LONG |
| 10 | PFF  | 0.0453 | iShares Preferred and Income Securities... | Preferred Stock             | LONG |
| 3  | HYLS | 0.0450 | First Trust Tactical High Yield ETF        | High Yield Bond             | LONG |
| 2  | EMB  | 0.0444 | iShares J.P. Morgan USD Emerging Market... | Emerging Markets Fixed I... | FLAT |
| 22 | VWOB | 0.0443 | Vanguard Emerging Markets Government Bo... | Emerging Markets Fixed I... | FLAT |
| 12 | PFXF | 0.0438 | VanEck Preferred Securities ex Financia... | Preferred Stock             | LONG |
| 21 | PCY  | 0.0435 | Invesco Emerging Markets Sovereign Debt... | Emerging Markets Fixed I... | FLAT |
| 7  | PGX  | 0.0421 | Invesco Preferred ETF                      | Preferred Stock             | LONG |
| 11 | PSK  | 0.0416 | State Street SPDR ICE Preferred Securit... | Preferred Stock             | LONG |
| 4  | FPE  | 0.0410 | First Trust Preferred Securities and In... | Preferred Stock             | LONG |
| 5  | PGF  | 0.0407 | Invesco Financial Preferred ETF            | Preferred Stock             | FLAT |
| 13 | HYEM | 0.0373 | VanEck Emerging Markets High Yield Bond... | Emerging Markets Fixed I... | LONG |
| 9  | VRP  | 0.0348 | Invesco Variable Rate Preferred ETF        | Preferred Stock             | LONG |

## Cluster 10 – U.S. Dividend Equity (11/20)

**Members:** 20   **Var explained:** 0.832   **Signal:** LONG since 2026-04-21 (14 days, 0.6%)

### Signal logic:

- Latest buy\_cross: 2026-04-21 – filter **PASS** (126d at date = +21.9%)
- Latest sell\_cross: 2026-03-09
- Current 126d return: +20.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-21 > last sell at 2026-03-09)



### Category breakdown

| category                    | n_members | abs_weight |
|-----------------------------|-----------|------------|
| U.S. Dividend Equity        | 11        | 0.5763     |
| MLP / Energy Infrastructure | 4         | 0.1667     |
| Value Equity                | 2         | 0.1037     |
| Dividend Equity             | 1         | 0.0524     |
| U.S. Defensive Equity       | 1         | 0.0515     |
| U.S. Low Volatility Equity  | 1         | 0.0495     |

**Replication basket** – SCHD+DVY+HDV (cor=0.98) basket lead at lag 1: 0.027

| ticker | weight_ols | name                           | category             | etf_signal |
|--------|------------|--------------------------------|----------------------|------------|
| SCHD   | 0.131      | Schwab US Dividend Equity ETF  | U.S. Dividend Equity | LONG       |
| DVY    | 0.615      | iShares Select Dividend ETF    | U.S. Dividend Equity | LONG       |
| HDV    | 0.236      | iShares Core High Dividend ETF | U.S. Dividend Equity | LONG       |

### Members

| ticker | weight | name                        | category             | etf_signal |
|--------|--------|-----------------------------|----------------------|------------|
| 15 DVY | 0.0538 | iShares Select Dividend ETF | U.S. Dividend Equity | LONG       |

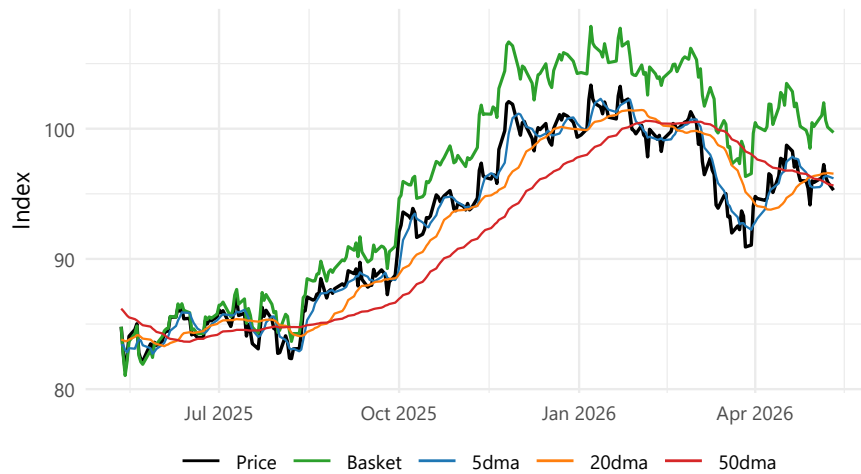
|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 20 | SPYD | 0.0535 | State Street SPDR Portfolio S&P 500 Hig... | U.S. Dividend Equity        | LONG |
| 14 | SDOG | 0.0534 | ALPS Sector Dividend Dogs ETF              | U.S. Dividend Equity        | LONG |
| 6  | DHS  | 0.0533 | WisdomTree U.S. High Dividend Fund         | U.S. Dividend Equity        | LONG |
| 19 | ONEY | 0.0528 | State Street SPDR Russell 1000 Yield Fo... | U.S. Dividend Equity        | LONG |
| 11 | FTA  | 0.0526 | First Trust Large Cap Value AlphaDEX Fu... | Value Equity                | FLAT |
| 5  | FDL  | 0.0524 | First Trust Morningstar ETF                | U.S. Dividend Equity        | FLAT |
| 18 | SPHD | 0.0524 | Invesco S&P 500 High Dividend Low Volat... | Dividend Equity             | LONG |
| 10 | SCHD | 0.0521 | Schwab US Dividend Equity ETF              | U.S. Dividend Equity        | LONG |
| 13 | DIV  | 0.0517 | Global X Super Dividend ETF                | U.S. Dividend Equity        | LONG |
| 16 | RDIV | 0.0516 | Invesco S&P Ultra Dividend Revenue ETF     | U.S. Dividend Equity        | LONG |
| 7  | CDC  | 0.0515 | VictoryShares US EQ Income Enhanced Vol... | U.S. Defensive Equity       | LONG |
| 17 | RPV  | 0.0511 | Invesco S&P 500 Pure Value ETF             | Value Equity                | LONG |
| 12 | PEY  | 0.0510 | Invesco High Yield Equity Dividend Achi... | U.S. Dividend Equity        | LONG |
| 9  | HDV  | 0.0508 | iShares Core High Dividend ETF             | U.S. Dividend Equity        | LONG |
| 8  | LVHD | 0.0495 | Franklin U.S. Low Volatility High Divid... | U.S. Low Volatility Equi... | FLAT |
| 1  | EMLP | 0.0472 | First Trust North American Energy Infra... | MLP / Energy Infrastruct... | LONG |
| 4  | MLPX | 0.0419 | Global X MLP & Energy Infrastructure ETF   | MLP / Energy Infrastruct... | LONG |
| 2  | AMLP | 0.0393 | Alerian MLP ETF                            | MLP / Energy Infrastruct... | LONG |
| 3  | MLPA | 0.0383 | Global X MLP ETF                           | MLP / Energy Infrastruct... | LONG |

## Cluster 11 – Healthcare (16/17)

**Members:** 17   **Var explained:** 0.764   **Signal:** LONG since 2026-05-06 (3 days, -2.0%)

### Signal logic:

- Latest buy\_cross: 2026-05-06 – filter **PASS** (126d at date = +13.9%)
- Latest sell\_cross: 2026-04-29
- Current 126d return: +14.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-06 > last sell at 2026-04-29)



### Category breakdown

| category                  | n_members | abs_weight |
|---------------------------|-----------|------------|
| Healthcare                | 16        | 0.9419     |
| U.S. Small Cap Healthcare | 1         | 0.0581     |

**Replication basket** – XLV+VHT+XBI (cor=0.99) basket lead at lag 1: -0.007

| ticker | weight_ols | name                                     | category   | etf_signal |
|--------|------------|------------------------------------------|------------|------------|
| XLV    | -1.104     | State Street Health Care Select Secto... | Healthcare | FLAT       |
| VHT    | 1.989      | Vanguard Health Care ETF                 | Healthcare | FLAT       |
| XBI    | 0.128      | State Street SPDR S&P Biotech ETF        | Healthcare | FLAT       |

### Members

|    | ticker | weight | name                                       | category   | etf_signal |
|----|--------|--------|--------------------------------------------|------------|------------|
| 15 | VHT    | 0.0653 | Vanguard Health Care ETF                   | Healthcare | FLAT       |
| 16 | FHLC   | 0.0652 | Fidelity MSCI Health Care Index ETF        | Healthcare | FLAT       |
| 9  | FXH    | 0.0645 | First Trust Health Care AlphaDEX           | Healthcare | LONG       |
| 14 | RSPH   | 0.0639 | Invesco S&P 500 Equal Weight Health Car... | Healthcare | FLAT       |
| 13 | IYH    | 0.0637 | iShares U.S. Healthcare ETF                | Healthcare | FLAT       |



|    |      |        |                                            |                           |      |
|----|------|--------|--------------------------------------------|---------------------------|------|
| 10 | IXJ  | 0.0622 | iShares Global Healthcare ETF              | Healthcare                | FLAT |
| 1  | XLV  | 0.0620 | State Street Health Care Select Sector ... | Healthcare                | FLAT |
| 3  | IBB  | 0.0605 | iShares Biotechnology ETF                  | Healthcare                | LONG |
| 6  | FBT  | 0.0595 | First Trust Amex Biotech Index Fund        | Healthcare                | FLAT |
| 8  | PSCH | 0.0581 | Invesco S&P SmallCap Health Care ETF       | U.S. Small Cap Healthcare | LONG |
| 7  | XHE  | 0.0574 | State Street SPDR S&P Health Care Equip... | Healthcare                | FLAT |
| 12 | IHI  | 0.0573 | iShares U.S. Medical Devices ETF           | Healthcare                | FLAT |
| 5  | PTH  | 0.0545 | Invesco Dorsey Wright Healthcare Moment... | Healthcare                | FLAT |
| 17 | PPH  | 0.0542 | VanEck Pharmaceutical ETF                  | Healthcare                | FLAT |
| 4  | XBI  | 0.0535 | State Street SPDR S&P Biotech ETF          | Healthcare                | FLAT |
| 2  | ARKG | 0.0511 | ARK Genomic Revolution ETF                 | Healthcare                | LONG |
| 11 | IHF  | 0.0470 | iShares U.S. Health Care Providers ETF     | Healthcare                | LONG |

## Cluster 12 – Materials (13/17)

**Members:** 17   **Var explained:** 0.756   **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-04-08 – filter **PASS** (126d at date = +66.8%)
- Latest sell\_cross: 2026-05-04
- Current 126d return: +54.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-04 more recent than last effective buy 2026-04-08)



### Category breakdown

| category          | n_members | abs_weight |
|-------------------|-----------|------------|
| Materials         | 13        | 0.7907     |
| Consumer Staples  | 1         | 0.0589     |
| Energy            | 1         | 0.0571     |
| Norway Equity     | 1         | 0.0519     |
| Industrial Metals | 1         | 0.0414     |

**Replication basket** – URA+GUNR+COPX (cor=0.97) basket lead at lag 1: 0.037

| ticker | weight_ols | name                                     | category  | etf_signal |
|--------|------------|------------------------------------------|-----------|------------|
| URA    | 0.069      | Global X Uranium ETF                     | Materials | LONG       |
| GUNR   | 0.681      | FlexShares Global Upstream Natural Re... | Materials | FLAT       |
| COPX   | 0.161      | Global X Copper Miners ETF               | Materials | LONG       |

### Members

|    | ticker | weight | name                                       | category  | etf_signal |
|----|--------|--------|--------------------------------------------|-----------|------------|
| 11 | GNR    | 0.0653 | State Street SPDR S&P Global Natural Re... | Materials | FLAT       |
| 6  | MXI    | 0.0648 | iShares Global Materials ETF               | Materials | FLAT       |

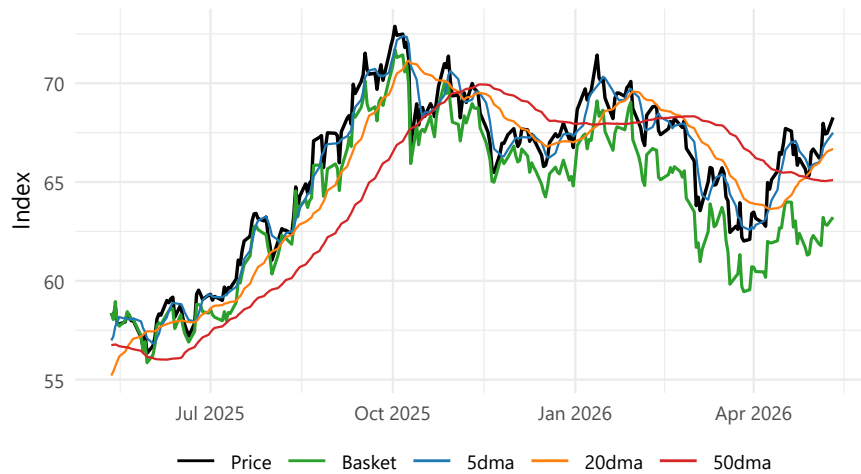
|    |      |        |                                            |                   |      |
|----|------|--------|--------------------------------------------|-------------------|------|
| 10 | GUNR | 0.0643 | FlexShares Global Upstream Natural Reso... | Materials         | FLAT |
| 7  | IYM  | 0.0641 | iShares U.S. Basic Materials ETF           | Materials         | FLAT |
| 12 | PICK | 0.0627 | iShares MSCI Global Select Metals & Min... | Materials         | FLAT |
| 9  | VAW  | 0.0622 | Vanguard Materials ETF                     | Materials         | FLAT |
| 17 | NANR | 0.0614 | State Street SPDR S&P North American Na... | Materials         | LONG |
| 1  | XLB  | 0.0612 | State Street Materials Select Sector SP... | Materials         | FLAT |
| 8  | RSPM | 0.0611 | Invesco S&P 500 Equal Weight Materials ... | Materials         | FLAT |
| 13 | SLX  | 0.0606 | VanEck Steel ETF                           | Materials         | LONG |
| 4  | XME  | 0.0605 | State Street SPDR S&P Metals & Mining E... | Materials         | LONG |
| 5  | MOO  | 0.0589 | VanEck Agribusiness ETF                    | Consumer Staples  | LONG |
| 3  | COPX | 0.0587 | Global X Copper Miners ETF                 | Materials         | LONG |
| 16 | IGE  | 0.0571 | iShares North American Natural Resource... | Energy            | LONG |
| 14 | NORW | 0.0519 | Global X MSCI Norway ETF                   | Norway Equity     | LONG |
| 2  | URA  | 0.0438 | Global X Uranium ETF                       | Materials         | LONG |
| 15 | DBB  | 0.0414 | Invesco DB Base Metals Fund                | Industrial Metals | LONG |

### Cluster 13 – China Equity (13/15)

**Members:** 15   **Var explained:** 0.804   **Signal:** LONG since 2026-04-15 (18 days, 2.8%)

#### Signal logic:

- Latest buy\_cross: 2026-04-15 – filter **PASS** (126d at date = +24.3%)
- Latest sell\_cross: 2026-02-27
- Current 126d return: +20.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-15 > last sell at 2026-02-27)



### Category breakdown

| category                | n_members | abs_weight |
|-------------------------|-----------|------------|
| China Equity            | 13        | 0.8766     |
| Emerging Markets Equity | 1         | 0.0642     |
| Hong Kong Equity        | 1         | 0.0591     |

**Replication basket** – KWEB+MCHI+FXI (cor=0.98) basket lead at lag 1: -0.040

| ticker | weight_ols | name                               | category     | etf_signal |
|--------|------------|------------------------------------|--------------|------------|
| KWEB   | -0.038     | KraneShares CSI China Internet ETF | China Equity | FLAT       |
| MCHI   | 1.376      | iShares MSCI China ETF             | China Equity | LONG       |
| FXI    | -0.395     | iShares China Large-Cap ETF        | China Equity | LONG       |

### Members

|    | ticker | weight | name                                       | category     | etf_signal |
|----|--------|--------|--------------------------------------------|--------------|------------|
| 11 | CXSE   | 0.0736 | WisdomTree China ex-State-Owned Enterpr... | China Equity | LONG       |
| 9  | GXC    | 0.0735 | State Street SPDR S&P China ETF            | China Equity | LONG       |
| 1  | MCHI   | 0.0729 | iShares MSCI China ETF                     | China Equity | LONG       |
| 5  | CQQQ   | 0.0712 | Invesco China Technology ETF               | China Equity | LONG       |

|    |      |        |                                            |                         |      |
|----|------|--------|--------------------------------------------|-------------------------|------|
| 2  | FXI  | 0.0711 | iShares China Large-Cap ETF                | China Equity            | LONG |
| 7  | CHIQ | 0.0706 | Global X MSCI China Consumer Discretion... | China Equity            | FLAT |
| 4  | KWEB | 0.0679 | KraneShares CSI China Internet ETF         | China Equity            | FLAT |
| 6  | ASHR | 0.0669 | Xtrackers Harvest CSI 300 China A-Share... | China Equity            | LONG |
| 8  | CNYA | 0.0661 | iShares MSCI China A ETF                   | China Equity            | LONG |
| 13 | KGRN | 0.0651 | KraneShares MSCI China Clean Technology... | China Equity            | LONG |
| 3  | EMQQ | 0.0642 | EMQQ The Emerging Markets Internet ETF     | Emerging Markets Equity | FLAT |
| 14 | CNXT | 0.0595 | VanEck ChiNext Innovators ETF              | China Equity            | LONG |
| 10 | KBA  | 0.0594 | KraneShares Boser MSCI China A 50 Conn...  | China Equity            | LONG |
| 12 | EWI  | 0.0591 | iShares MSCI Hong Kong Index Fund          | Hong Kong Equity        | LONG |
| 15 | KURE | 0.0587 | KraneShares MSCI All China Health Care ... | China Equity            | FLAT |

### Cluster 14 – Real Estate (14/14)

**Members:** 14   **Var explained:** 0.962   **Signal:** LONG since 2026-04-10 (21 days, 4.2%)

#### Signal logic:

- Latest buy\_cross: 2026-04-10 – filter **PASS** (126d at date = +21.2%)
- Latest sell\_cross: 2026-03-10
- Current 126d return: +13.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-10)



**Category breakdown**

| category    | n_members | abs_weight |
|-------------|-----------|------------|
| Real Estate | 14        | 1          |

**Replication basket** – VNQ+SCHH+XLRE (cor=1.00) basket lead at lag 1: -0.001

| ticker | weight_ols | name                                     | category    | etf_signal |
|--------|------------|------------------------------------------|-------------|------------|
| VNQ    | 0.515      | Vanguard Real Estate ETF                 | Real Estate | LONG       |
| SCHH   | 0.537      | Schwab U.S. REIT ETF                     | Real Estate | LONG       |
| XLRE   | -0.075     | State Street Real Estate Select Secto... | Real Estate | LONG       |

**Members**

|    | ticker | weight | name                                       | category    | etf_signal |
|----|--------|--------|--------------------------------------------|-------------|------------|
| 8  | VNQ    | 0.0725 | Vanguard Real Estate ETF                   | Real Estate | LONG       |
| 9  | FREL   | 0.0725 | Fidelity MSCI Real Estate Index ETF        | Real Estate | LONG       |
| 11 | SCHH   | 0.0725 | Schwab U.S. REIT ETF                       | Real Estate | LONG       |
| 2  | IYR    | 0.0724 | iShares U.S. Real Estate ETF               | Real Estate | LONG       |
| 6  | RWR    | 0.0723 | State Street SPDR Dow Jones REIT ETF       | Real Estate | LONG       |
| 10 | USRT   | 0.0723 | iShares Core U.S. REIT ETF                 | Real Estate | LONG       |
| 13 | PSR    | 0.0721 | Invesco Active U.S. Real Estate Fund       | Real Estate | LONG       |
| 1  | XLRE   | 0.0719 | State Street Real Estate Select Sector ... | Real Estate | LONG       |
| 5  | ICF    | 0.0719 | iShares Select U.S. REIT ETF               | Real Estate | LONG       |
| 7  | REET   | 0.0715 | iShares Global REIT ETF                    | Real Estate | LONG       |
| 3  | RWO    | 0.0711 | State Street SPDR Dow Jones Global Real... | Real Estate | LONG       |
| 4  | REZ    | 0.0698 | iShares Residential and Multisector Rea... | Real Estate | LONG       |
| 14 | NURE   | 0.0689 | Nuveen Short-Term REIT ETF                 | Real Estate | LONG       |
| 12 | INDS   | 0.0683 | Pacer Industrial Real Estate ETF           | Real Estate | LONG       |

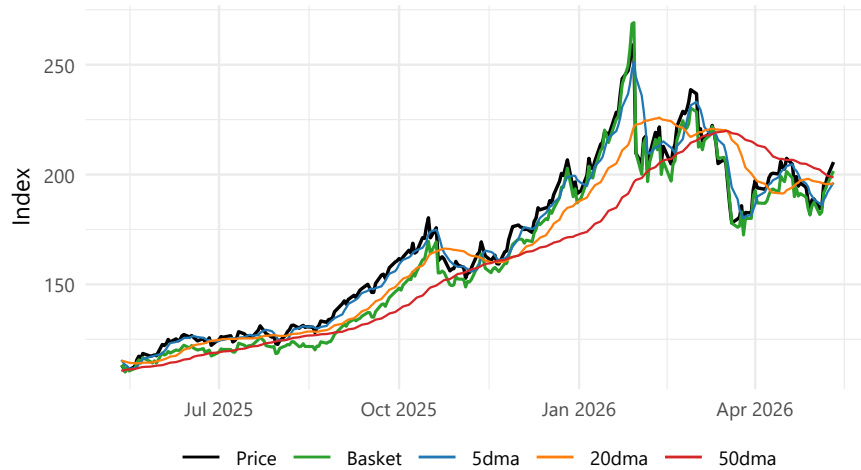
**Cluster 15 – Mixed: DBP, GLTR, GDXJ**

**Members:** 13 **Var explained:** 0.831 **Signal:** FLAT

**Signal logic:**

- Latest buy\_cross: 2025-11-11 – filter **PASS** (126d at date = +60.6%)

- Latest sell\_cross: 2026-04-28
- Current 126d return: +79.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-04-28 more recent than last effective buy 2025-11-11)



### Category breakdown

| category        | n_members | abs_weight |
|-----------------|-----------|------------|
| Materials       | 5         | 0.3889     |
| Gold            | 3         | 0.2341     |
| Precious Metals | 2         | 0.1624     |
| Silver          | 2         | 0.1525     |
| Platinum        | 1         | 0.0622     |

**Replication basket** – GLD+IAU+SLV (cor=0.95)    basket lead at lag 1: -0.017

| ticker | weight_ols | name                      | category | etf_signal |
|--------|------------|---------------------------|----------|------------|
| GLD    | 0.115      | SPDR Gold Shares          | Gold     | FLAT       |
| IAU    | 0.623      | iShares Gold Trust Shares | Gold     | FLAT       |
| SLV    | 0.418      | iShares Silver Trust      | Silver   | LONG       |

### Members

|    | ticker | weight | name                                       | category        | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------|------------|
| 12 | DBP    | 0.0814 | Invesco DB Precious Metals Fund            | Precious Metals | FLAT       |
| 7  | GLTR   | 0.0810 | abrdn Physical Precious Metals Basket S... | Precious Metals | FLAT       |
| 2  | GDXJ   | 0.0786 | VanEck Junior Gold Miners ETF              | Materials       | FLAT       |
| 1  | GDX    | 0.0785 | VanEck Gold Miners ETF                     | Materials       | FLAT       |
| 3  | GLD    | 0.0781 | SPDR Gold Shares                           | Gold            | FLAT       |
| 10 | IAU    | 0.0781 | iShares Gold Trust Shares                  | Gold            | FLAT       |
| 11 | SGOL   | 0.0779 | abrdn Physical Gold Shares ETF             | Gold            | FLAT       |
| 13 | GOAU   | 0.0777 | US Global GO Gold and Precious Metal Mi... | Materials       | FLAT       |
| 6  | SIL    | 0.0776 | Global X Silver Miners ETF                 | Materials       | LONG       |
| 5  | SILJ   | 0.0765 | Amplify Junior Silver Miners ETF           | Materials       | LONG       |
| 4  | SLV    | 0.0763 | iShares Silver Trust                       | Silver          | LONG       |
| 9  | SIVR   | 0.0761 | abrdn Physical Silver Shares ETF           | Silver          | LONG       |
| 8  | PPLT   | 0.0622 | abrdn Physical Platinum Shares ETF         | Platinum        | LONG       |

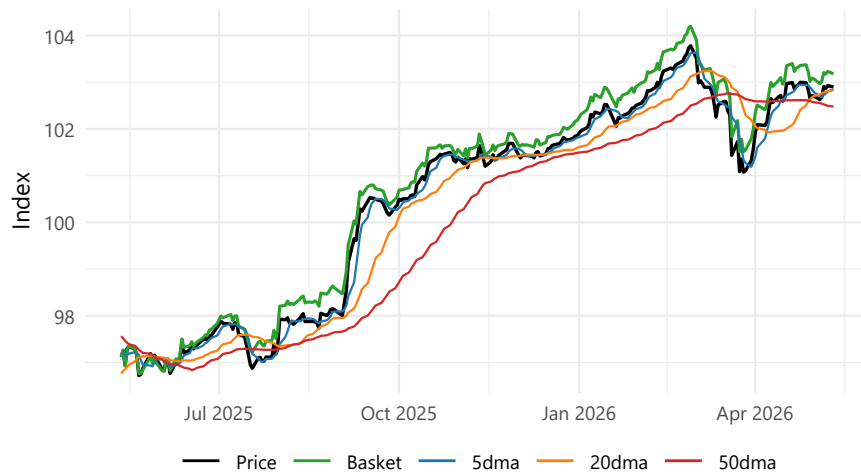
### Cluster 16 – Municipal Bond (13/13)

**Members:** 13   **Var explained:** 0.819   **Signal:** FLAT

#### Signal logic:

- Latest buy\_cross: 2026-04-14 – filter **PASS** (126d at date = +6.9%)
- Latest sell\_cross: 2026-05-05
- Current 126d return: +5.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-05 more recent than last effective buy 2026-04-14)





### Category breakdown

| category       | n_members | abs_weight |
|----------------|-----------|------------|
| Municipal Bond | 13        | 1          |

**Replication basket** – MUB+VTEB+SUB (cor=0.97) basket lead at lag 1: 0.189

| ticker | weight_ols | name                                     | category       | etf_signal |
|--------|------------|------------------------------------------|----------------|------------|
| MUB    | 0.380      | iShares National Muni Bond ETF           | Municipal Bond | LONG       |
| VTEB   | 0.468      | Vanguard Tax-Exempt Bond ETF             | Municipal Bond | FLAT       |
| SUB    | 0.332      | iShares Short-Term National Muni Bond... | Municipal Bond | LONG       |

### Members

|    | ticker | weight | name                                       | category       | etf_signal |
|----|--------|--------|--------------------------------------------|----------------|------------|
| 13 | VTEB   | 0.0817 | Vanguard Tax-Exempt Bond ETF               | Municipal Bond | FLAT       |
| 11 | MUB    | 0.0813 | iShares National Muni Bond ETF             | Municipal Bond | LONG       |
| 6  | CMF    | 0.0801 | iShares California Muni Bond ETF           | Municipal Bond | LONG       |
| 9  | TFI    | 0.0801 | State Street SPDR Nuveen ICE Municipal ... | Municipal Bond | LONG       |
| 8  | ITM    | 0.0800 | VanEck Intermediate Muni ETF               | Municipal Bond | LONG       |
| 1  | FMB    | 0.0796 | First Trust Managed Municipal ETF          | Municipal Bond | LONG       |

|    |      |        |                                            |                |      |
|----|------|--------|--------------------------------------------|----------------|------|
| 7  | NYF  | 0.0793 | iShares New York Muni Bond ETF             | Municipal Bond | LONG |
| 5  | PZA  | 0.0780 | Invesco National AMT-Free Municipal Bon... | Municipal Bond | LONG |
| 4  | MUNI | 0.0769 | PIMCO Intermediate Municipal Bond Activ... | Municipal Bond | FLAT |
| 2  | HYD  | 0.0748 | VanEck High Yield Muni ETF                 | Municipal Bond | LONG |
| 3  | HYMB | 0.0720 | State Street SPDR Nuveen ICE High Yield... | Municipal Bond | LONG |
| 12 | SUB  | 0.0694 | iShares Short-Term National Muni Bond E... | Municipal Bond | LONG |
| 10 | SHM  | 0.0668 | State Street SPDR Nuveen ICE Short Term... | Municipal Bond | FLAT |

### Cluster 17 – Consumer Staples (6/11)

**Members:** 11   **Var explained:** 0.748   **Signal:** LONG since 2026-05-01 (6 days, -2.3%)

#### Signal logic:

- Latest buy\_cross: 2026-05-01 – filter **PASS** (126d at date = +9.8%)
- Latest sell\_cross: 2026-03-09
- Current 126d return: +7.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-01 > last sell at 2026-03-09)



#### Category breakdown

| category                        | n_members | abs_weight |
|---------------------------------|-----------|------------|
| Consumer Staples                | 6         | 0.5726     |
| Utilities                       | 4         | 0.3528     |
| U.S. Small Cap Consumer Staples | 1         | 0.0746     |

**Replication basket** – XLU+XLP+VPU (cor=0.99) basket lead at lag 1: -0.011

| ticker | weight_ols | name                                     | category         | etf_signal |
|--------|------------|------------------------------------------|------------------|------------|
| XLU    | -0.324     | State Street Utilities Select Sector ... | Utilities        | FLAT       |
| XLP    | 0.584      | State Street Consumer Staples Select ... | Consumer Staples | LONG       |
| VPU    | 0.703      | Vanguard Utilities ETF                   | Utilities        | FLAT       |

## Members

|    | ticker | weight | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 5  | VDC    | 0.0983 | Vanguard Consumer Staples ETF              | Consumer Staples            | LONG       |
| 6  | FSTA   | 0.0983 | Fidelity MSCI Consumer Staples Index ETF   | Consumer Staples            | LONG       |
| 1  | XLP    | 0.0973 | State Street Consumer Staples Select Se... | Consumer Staples            | LONG       |
| 4  | IYK    | 0.0947 | iShares U.S. Consumer Staples ETF          | Consumer Staples            | LONG       |
| 3  | KXI    | 0.0944 | iShares Global Consumer Staples ETF        | Consumer Staples            | LONG       |
| 11 | PBJ    | 0.0897 | Invesco Food & Beverage ETF                | Consumer Staples            | FLAT       |
| 7  | IDU    | 0.0892 | iShares U.S. Utilities ETF                 | Utilities                   | FLAT       |
| 8  | VPU    | 0.0880 | Vanguard Utilities ETF                     | Utilities                   | FLAT       |
| 9  | FUTY   | 0.0880 | Fidelity MSCI Utilities Index ETF          | Utilities                   | FLAT       |
| 2  | XLU    | 0.0875 | State Street Utilities Select Sector SP... | Utilities                   | FLAT       |
| 10 | PSCC   | 0.0746 | Invesco S&P SmallCap Consumer Staples E... | U.S. Small Cap Consumer ... | FLAT       |

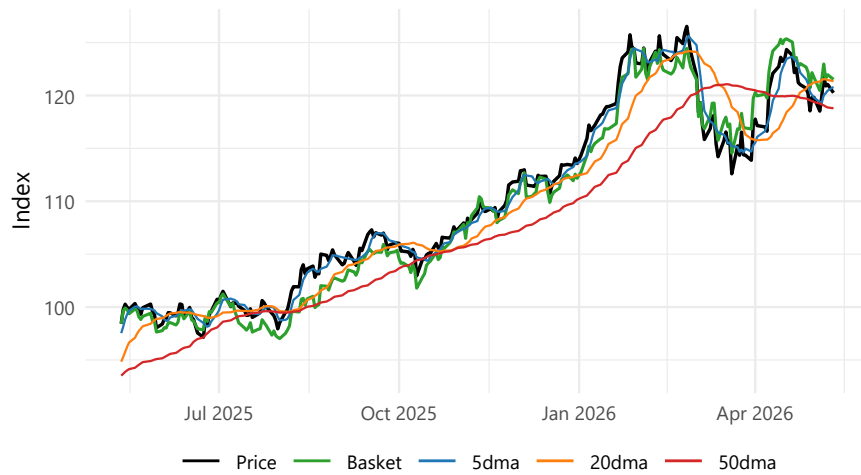
## Cluster 18 – Mixed: ILF, EWZ, ECH

**Members:** 11 **Var explained:** 0.325 **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-04-13 – filter **PASS** (126d at date = +36.9%)
- Latest sell\_cross: 2026-04-29
- Current 126d return: +23.2% – filter **ALLOWS** new entries today

- Decision: **FLAT** (last sell 2026-04-29 more recent than last effective buy 2026-04-13)



### Category breakdown

| category                 | n_members | abs_weight |
|--------------------------|-----------|------------|
| Latin America Equity     | 1         | 0.1565     |
| Brazil Equity            | 1         | 0.1432     |
| Chile Equity             | 1         | 0.1273     |
| Malaysia Equity          | 1         | 0.1242     |
| Indonesia Equity         | 1         | 0.1137     |
| Philippines Equity       | 1         | 0.1051     |
| Vietnam Equity           | 1         | 0.0792     |
| Agricultural Commodities | 1         | 0.0598     |
| Turkey Equity            | 1         | 0.0575     |
| Unknown                  | 1         | 0.0189     |
| Natural Gas              | 1         | 0.0147     |

**Replication basket** – EWZ+ILF+DBA+ECH (cor=0.93) basket lead at lag 1: 0.035

| ticker | weight_ols | name                         | category                 | etf_signal |
|--------|------------|------------------------------|--------------------------|------------|
| EWZ    | 0.047      | iShares MSCI Brazil ETF      | Brazil Equity            | FLAT       |
| ILF    | 0.382      | iShares Latin America 40 ETF | Latin America Equity     | FLAT       |
| DBA    | 0.056      | Invesco DB Agriculture Fund  | Agricultural Commodit... | FLAT       |

|     |       |                        |              |      |
|-----|-------|------------------------|--------------|------|
| ECH | 0.158 | iShares MSCI Chile ETF | Chile Equity | LONG |
|-----|-------|------------------------|--------------|------|

**Members**

|    | ticker | weight | name                              | category                 | etf_signal |
|----|--------|--------|-----------------------------------|--------------------------|------------|
| 4  | ILF    | 0.1565 | iShares Latin America 40 ETF      | Latin America Equity     | FLAT       |
| 1  | EWZ    | 0.1432 | iShares MSCI Brazil ETF           | Brazil Equity            | FLAT       |
| 3  | ECH    | 0.1273 | iShares MSCI Chile ETF            | Chile Equity             | LONG       |
| 8  | EWM    | 0.1242 | iShares MSCI Malaysia Index Fund  | Malaysia Equity          | LONG       |
| 6  | EIDO   | 0.1137 | iShares MSCI Indonesia ETF        | Indonesia Equity         | FLAT       |
| 7  | EPHE   | 0.1051 | iShares MSCI Philippines ETF      | Philippines Equity       | FLAT       |
| 11 | VNM    | 0.0792 | VanEck Vietnam ETF                | Vietnam Equity           | LONG       |
| 2  | DBA    | 0.0598 | Invesco DB Agriculture Fund       | Agricultural Commodities | FLAT       |
| 10 | TUR    | 0.0575 | iShares MSCI Turkey ETF           | Turkey Equity            | LONG       |
| 9  | RSX    | 0.0189 |                                   |                          | LONG       |
| 5  | UNG    | 0.0147 | United States Natural Gas Fund LP | Natural Gas              | FLAT       |

**Cluster 19 – Inflation-Protected Fixed Income (10/10)**

**Members:** 10   **Var explained:** 0.908   **Signal:** LONG since 2026-04-09 (22 days, 0.8%)

**Signal logic:**

- Latest buy\_cross: 2026-04-09 – filter **PASS** (126d at date = +4.1%)
- Latest sell\_cross: 2026-03-20
- Current 126d return: +4.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-09 > last sell at 2026-03-20)



### Category breakdown

| category                         | n_members | abs_weight |
|----------------------------------|-----------|------------|
| Inflation-Protected Fixed Income | 10        | 1          |

**Replication basket** – VTIP+SCHP+STIP (cor=0.99) basket lead at lag 1: -0.013

| ticker | weight_ols | name                                     | category                 | etf_signal |
|--------|------------|------------------------------------------|--------------------------|------------|
| VTIP   | 0.127      | Vanguard Short-Term Inflation-Protect... | Inflation-Protected F... | FLAT       |
| SCHP   | 0.748      | Schwab U.S. TIPS ETF                     | Inflation-Protected F... | LONG       |
| STIP   | 0.168      | iShares 0-5 Year TIPS Bond ETF           | Inflation-Protected F... | FLAT       |

### Members

|   | ticker | weight | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 4 | TDTF   | 0.1033 | FlexShares iBoxx 5 Year Target Duration... | Inflation-Protected Fixe... | LONG       |
| 6 | TIPX   | 0.1032 | State Street SPDR Bloomberg 1a??10 Year... | Inflation-Protected Fixe... | LONG       |
| 3 | TIP    | 0.1028 | iShares TIPS Bond ETF                      | Inflation-Protected Fixe... | LONG       |
| 8 | SCHP   | 0.1023 | Schwab U.S. TIPS ETF                       | Inflation-Protected Fixe... | LONG       |
| 5 | TDDT   | 0.1007 | FlexShares iBoxx 3 Year Target Duration... | Inflation-Protected Fixe... | LONG       |
| 9 | STIP   | 0.0999 | iShares 0-5 Year TIPS Bond ETF             | Inflation-Protected Fixe... | FLAT       |

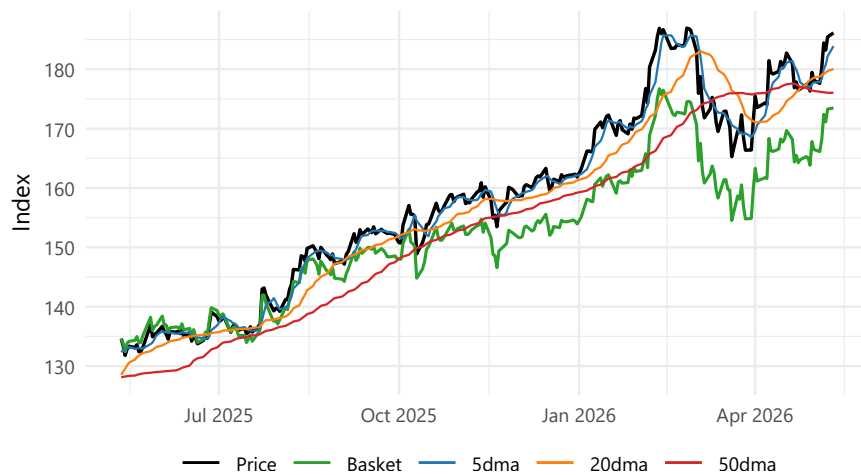
|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 10 | VTIP | 0.0998 | Vanguard Short-Term Inflation-Protected... | Inflation-Protected Fixe... | FLAT |
| 7  | SPIP | 0.0997 | State Street SPDR Portfolio TIPS ETF       | Inflation-Protected Fixe... | LONG |
| 2  | STPZ | 0.0995 | PIMCO 1-5 Year U.S. TIPS Index Exchange... | Inflation-Protected Fixe... | LONG |
| 1  | LTPZ | 0.0886 | Pimco 15 Year U.S. TIPS Index Exchange...  | Inflation-Protected Fixe... | LONG |

## Cluster 20 – Japan Equity (7/9)

**Members:** 9   **Var explained:** 0.888   **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-04-09 – filter **PASS** (126d at date = +57.2%)
- Latest sell\_cross: 2026-04-29
- Current 126d return: +41.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-04-29 more recent than last effective buy 2026-04-09)



### Category breakdown

| category     | n_members | abs_weight |
|--------------|-----------|------------|
| Japan Equity | 7         | 0.7796     |

|                     |   |        |
|---------------------|---|--------|
| Asia-Pacific Equity | 2 | 0.2204 |
|---------------------|---|--------|

**Replication basket** – EWJ+BBJP+VPL (cor=0.97) basket lead at lag 1: -0.039

| ticker | weight_ols | name                            | category            | etf_signal |
|--------|------------|---------------------------------|---------------------|------------|
| EWJ    | 0.481      | iShares MSCI Japan Index Fund   | Japan Equity        | LONG       |
| BBJP   | 0.365      | JPMorgan BetaBuilders Japan ETF | Japan Equity        | LONG       |
| VPL    | 0.072      | Vanguard FTSE Pacific ETF       | Asia-Pacific Equity | LONG       |

## Members

|   | ticker | weight | name                                   | category            | etf_signal |
|---|--------|--------|----------------------------------------|---------------------|------------|
| 1 | EWJ    | 0.1141 | iShares MSCI Japan Index Fund          | Japan Equity        | LONG       |
| 8 | BBJP   | 0.1141 | JPMorgan BetaBuilders Japan ETF        | Japan Equity        | LONG       |
| 9 | HEWJ   | 0.1115 | iShares Currency Hedged MSCI Japan ETF | Japan Equity        | LONG       |
| 6 | DBJP   | 0.1114 | Xtrackers MSCI Japan Hedged Equity ETF | Japan Equity        | LONG       |
| 3 | IPAC   | 0.1113 | iShares Core MSCI Pacific ETF          | Asia-Pacific Equity | FLAT       |
| 5 | FLJH   | 0.1106 | Franklin FTSE Japan Hedged ETF         | Japan Equity        | FLAT       |
| 2 | DXJ    | 0.1097 | WisdomTree Japan Hedged Equity Fund    | Japan Equity        | LONG       |
| 4 | VPL    | 0.1091 | Vanguard FTSE Pacific ETF              | Asia-Pacific Equity | LONG       |
| 7 | EWJV   | 0.1082 | iShares MSCI Japan Value ETF           | Japan Equity        | LONG       |

## Cluster 21 – Broad Commodities (6/9)

**Members:** 9 **Var explained:** 0.913 **Signal:** LONG since 2026-01-08 (84 days, 47.8%)

### Signal logic:

- Latest buy\_cross: 2026-01-08 – filter **PASS** (126d at date = +2.7%)
- Latest sell\_cross: 2025-12-15
- Current 126d return: +61.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-01-08 > last sell at 2025-12-15)





### Category breakdown

| category           | n_members | abs_weight |
|--------------------|-----------|------------|
| Broad Commodities  | 6         | 0.6698     |
| Oil                | 2         | 0.2183     |
| Energy Commodities | 1         | 0.1119     |

**Replication basket** – PDBC+FTGC+DBC (cor=0.98) basket lead at lag 1: 0.072

| ticker | weight_ols | name                                     | category          | etf_signal |
|--------|------------|------------------------------------------|-------------------|------------|
| PDBC   | 0.535      | Invesco Optimum Yield Diversified Com... | Broad Commodities | LONG       |
| FTGC   | -0.029     | First Trust Global Tactical Commodity... | Broad Commodities | FLAT       |
| DBC    | 0.662      | Invesco DB Commodity Index Tracking F... | Broad Commodities | LONG       |

### Members

|   | ticker | weight | name                                       | category          | etf_signal |
|---|--------|--------|--------------------------------------------|-------------------|------------|
| 5 | GSG    | 0.1151 | iShares GSCI Commodity-Indexed Trust Fu... | Broad Commodities | FLAT       |
| 2 | DBC    | 0.1149 | Invesco DB Commodity Index Tracking Fund   | Broad Commodities | LONG       |
| 7 | PDBC   | 0.1149 | Invesco Optimum Yield Diversified Commo... | Broad Commodities | LONG       |
| 8 | COMT   | 0.1148 | iShares GSCI Commodity Dynamic Roll Str... | Broad Commodities | FLAT       |

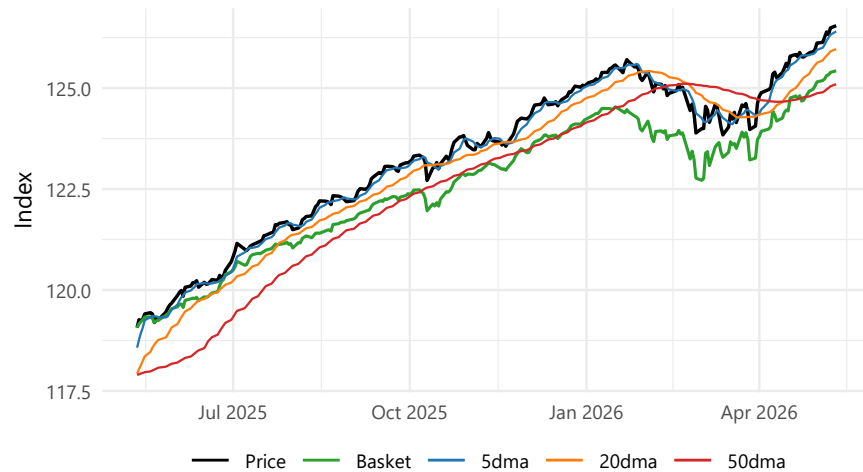
|   |      |        |                                            |                    |      |
|---|------|--------|--------------------------------------------|--------------------|------|
| 9 | DBE  | 0.1119 | Invesco DB Energy Fund                     | Energy Commodities | FLAT |
| 1 | USO  | 0.1098 | United States Oil Fund                     | Oil                | LONG |
| 4 | DBO  | 0.1085 | Invesco DB Oil Fund                        | Oil                | FLAT |
| 6 | DJP  | 0.1051 | iPath Bloomberg Commodity Index Total R... | Broad Commodities  | FLAT |
| 3 | FTGC | 0.1050 | First Trust Global Tactical Commodity S... | Broad Commodities  | FLAT |

### Cluster 22 – Mixed: SRLN, BKLN, FTSL

**Members:** 9   **Var explained:** 0.560   **Signal:** LONG since 2026-04-07 (24 days, 1.2%)

#### Signal logic:

- Latest buy\_cross: 2026-04-07 – filter **PASS** (126d at date = +6.5%)
- Latest sell\_cross: 2026-02-03
- Current 126d return: +6.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-07 > last sell at 2026-02-03)



#### Category breakdown

| category                  | n_members | abs_weight |
|---------------------------|-----------|------------|
| Bank Loan / Floating Rate | 3         | 0.3632     |

|                                   |   |        |
|-----------------------------------|---|--------|
| Interest Rate Hedged Fixed Income | 3 | 0.3245 |
| Floating Rate Fixed Income        | 3 | 0.3123 |

**Replication basket** – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.048

| ticker | weight_ols | name                                     | category                 | etf_signal |
|--------|------------|------------------------------------------|--------------------------|------------|
| FLOT   | 0.292      | iShares Floating Rate Bond ETF           | Floating Rate Fixed I... | FLAT       |
| BKLN   | 0.209      | Invesco Senior Loan ETF                  | Bank Loan / Floating ... | LONG       |
| SRLN   | 0.311      | State Street Blackstone Senior Loan E... | Bank Loan / Floating ... | LONG       |
| FLRN   | 0.127      | State Street SPDR Bloomberg Investmen... | Floating Rate Fixed I... | FLAT       |
| FLTR   | 0.034      | VanEck IG Floating Rate ETF              | Floating Rate Fixed I... | FLAT       |

## Members

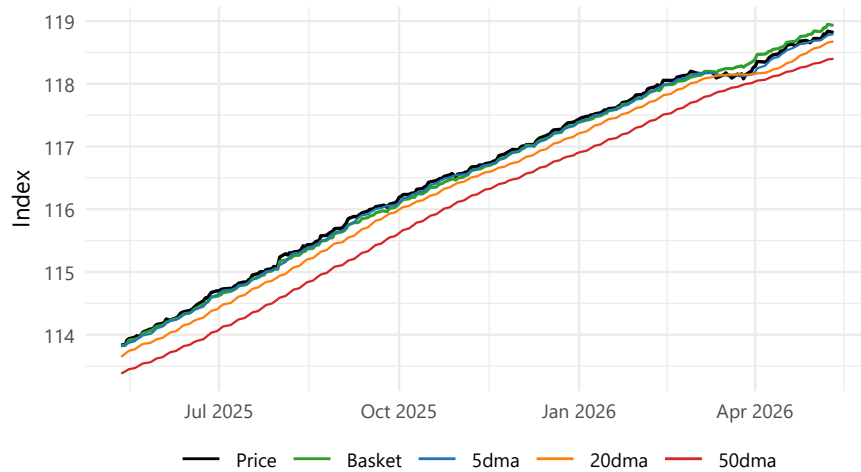
|   | ticker | weight | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 2 | SRLN   | 0.1249 | State Street Blackstone Senior Loan ETF    | Bank Loan / Floating Rate   | LONG       |
| 3 | BKLN   | 0.1213 | Invesco Senior Loan ETF                    | Bank Loan / Floating Rate   | LONG       |
| 1 | FTSL   | 0.1170 | First Trust Senior Loan Fund               | Bank Loan / Floating Rate   | LONG       |
| 5 | LQDH   | 0.1145 | iShares Interest Rate Hedged Corporate ... | Interest Rate Hedged Fix... | LONG       |
| 7 | FLOT   | 0.1081 | iShares Floating Rate Bond ETF             | Floating Rate Fixed Inco... | FLAT       |
| 6 | IGBH   | 0.1057 | iShares Interest Rate Hedged Long-Term ... | Interest Rate Hedged Fix... | LONG       |
| 4 | IGHG   | 0.1043 | ProShares Investment Grade-Interest Rat... | Interest Rate Hedged Fix... | LONG       |
| 8 | FLRN   | 0.1035 | State Street SPDR Bloomberg Investment ... | Floating Rate Fixed Inco... | FLAT       |
| 9 | FLTR   | 0.1008 | VanEck IG Floating Rate ETF                | Floating Rate Fixed Inco... | FLAT       |

## Cluster 23 – Cash / Ultra-Short Fixed Income (9/9)

**Members:** 9 **Var explained:** 0.457 **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2022-10-31 – filter **BLOCKED** (126d at date = -0.5%; needs > 0)
- Latest effective buy: 2022-11-29 – delayed re-arm from blocked cross (126d at fire = +0.0%)
- Latest sell\_cross: 2026-03-24
- Current 126d return: +4.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-03-24 more recent than last effective buy 2022-11-29)



### Category breakdown

| category                        | n_members | abs_weight |
|---------------------------------|-----------|------------|
| Cash / Ultra-Short Fixed Income | 9         | 1          |

**Replication basket** – BIL+SHV+MINT+GBIL+ICSH (cor=0.84) basket lead at lag 1: 0.106

| ticker | weight_ols | name                                     | category                 | etf_signal |
|--------|------------|------------------------------------------|--------------------------|------------|
| BIL    | -0.040     | State Street SPDR Bloomberg 1-3 Month... | Cash / Ultra-Short Fi... | LONG       |
| SHV    | 0.377      | iShares 0-1 Year Treasury Bond ETF       | Cash / Ultra-Short Fi... | FLAT       |
| MINT   | 0.176      | PIMCO Enhanced Short Maturity Active ... | Cash / Ultra-Short Fi... | LONG       |
| GBIL   | 0.118      | Goldman Sachs Access Treasury 0-1 Yea... | Cash / Ultra-Short Fi... | LONG       |
| ICSH   | 0.418      | iShares Ultra Short Duration Bond Act... | Cash / Ultra-Short Fi... | FLAT       |

### Members

|   | ticker | weight | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 5 | GSY    | 0.1359 | Invesco Ultra Short Duration ETF           | Cash / Ultra-Short Fixed... | LONG       |
| 3 | FTSM   | 0.1286 | First Trust Enhanced Short Maturity ETF    | Cash / Ultra-Short Fixed... | FLAT       |
| 7 | SHV    | 0.1261 | iShares 0-1 Year Treasury Bond ETF         | Cash / Ultra-Short Fixed... | FLAT       |
| 6 | ICSH   | 0.1244 | iShares Ultra Short Duration Bond Activ... | Cash / Ultra-Short Fixed... | FLAT       |

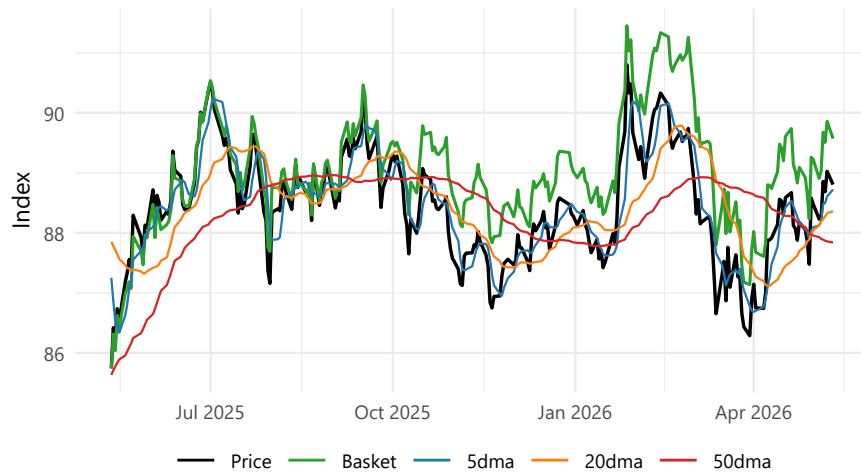
|   |      |        |                                            |                             |      |
|---|------|--------|--------------------------------------------|-----------------------------|------|
| 4 | NEAR | 0.1090 | iShares Short Duration Bond Active ETF     | Cash / Ultra-Short Fixed... | FLAT |
| 2 | MINT | 0.1062 | PIMCO Enhanced Short Maturity Active Ex... | Cash / Ultra-Short Fixed... | LONG |
| 1 | VNLA | 0.1032 | Janus Henderson Short Duration Income E... | Cash / Ultra-Short Fixed... | LONG |
| 8 | BIL  | 0.0879 | State Street SPDR Bloomberg 1-3 Month T... | Cash / Ultra-Short Fixed... | LONG |
| 9 | GBIL | 0.0787 | Goldman Sachs Access Treasury 0-1 Year ... | Cash / Ultra-Short Fixed... | LONG |

### Cluster 24 – Currency (4/6)

**Members:** 6   **Var explained:** 0.742   **Signal:** LONG since 2026-04-30 (7 days, 0.3%)

#### Signal logic:

- Latest buy\_cross: 2026-04-30 – filter **PASS** (126d at date = +0.3%)
- Latest sell\_cross: 2026-03-03
- Current 126d return: +2.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-30 > last sell at 2026-03-03)



#### Category breakdown

| category | n_members | abs_weight |
|----------|-----------|------------|
| Currency | 4         | 0.641      |

|                            |   |       |
|----------------------------|---|-------|
| International Fixed Income | 2 | 0.359 |
|----------------------------|---|-------|

**Replication basket** – BWX+IGOV+FXF (cor=0.98) basket lead at lag 1: 0.031

| ticker | weight_ols | name                                     | category                 | etf_signal |
|--------|------------|------------------------------------------|--------------------------|------------|
| BWX    | 0.269      | State Street SPDR Bloomberg Internati... | International Fixed I... | LONG       |
| IGOV   | 0.315      | iShares International Treasury Bond E... | International Fixed I... | LONG       |
| FXF    | 0.306      | Invesco CurrencyShares Swiss Franc Tr... | Currency                 | LONG       |

### Members

|   | ticker | weight | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 2 | IGOV   | 0.1801 | iShares International Treasury Bond ETF    | International Fixed Inco... | LONG       |
| 1 | BWX    | 0.1789 | State Street SPDR Bloomberg Internation... | International Fixed Inco... | LONG       |
| 4 | FXE    | 0.1701 | Invesco CurrencyShares Euro Currency Tr... | Currency                    | FLAT       |
| 5 | FXF    | 0.1647 | Invesco CurrencyShares Swiss Franc Trust   | Currency                    | LONG       |
| 3 | FXB    | 0.1609 | Invesco CurrencyShares British Pound St... | Currency                    | LONG       |
| 6 | FXV    | 0.1453 | Invesco CurrencyShares Japanese Yen Tru... | Currency                    | FLAT       |

### Cluster 25 – India Equity (3/4)

**Members:** 4 **Var explained:** 0.964 **Signal:** FLAT

#### Signal logic:

- Latest buy\_cross: 2026-05-07 – filter **BLOCKED** (126d at date = -9.2%; needs > 0)
- Latest effective buy: 2026-02-06 – delayed re-arm from blocked cross (126d at fire = +2.9%)
- Latest sell\_cross: 2026-05-04
- Current 126d return: -7.6% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-05-04 more recent than last effective buy 2026-02-06)



### Category breakdown

| category     | n_members | abs_weight |
|--------------|-----------|------------|
| India Equity | 3         | 0.7508     |
| Unknown      | 1         | 0.2492     |

**Replication basket** – IND+INDY+EPI+PIN (cor=1.00) basket lead at lag 1: -0.071

| ticker | weight_ols | name                           | category     | etf_signal |
|--------|------------|--------------------------------|--------------|------------|
| INDA   | 0.250      | iShares MSCI India ETF         | India Equity | FLAT       |
| INDY   | 0.246      | iShares India 50 ETF           | India Equity | FLAT       |
| EPI    | 0.256      | WisdomTree India Earnings Fund | India Equity | LONG       |
| PIN    | 0.249      |                                |              | FLAT       |

### Members

|   | ticker | weight | name                           | category     | etf_signal |
|---|--------|--------|--------------------------------|--------------|------------|
| 1 | INDA   | 0.2527 | iShares MSCI India ETF         | India Equity | FLAT       |
| 3 | EPI    | 0.2493 | WisdomTree India Earnings Fund | India Equity | LONG       |
| 4 | PIN    | 0.2492 |                                |              | FLAT       |

2 INDY 0.2489 iShares India 50 ETF India Equity FLAT

## Cluster 26 – Currency (1/2)

Members: 2 Var explained: 0.624 Signal: FLAT

### Signal logic:

- Latest buy\_cross: 2026-02-05 – filter **BLOCKED** (126d at date = -32.0%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (126d at fire = +0.4%)
- Latest sell\_cross: 2026-04-08
- Current 126d return: -45.4% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-04-08 more recent than last effective buy 2025-04-08)



### Category breakdown

| category            | n_members | abs_weight |
|---------------------|-----------|------------|
| Currency            | 1         | 0.5        |
| Leveraged / Inverse | 1         | 0.5        |



**Replication basket** – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.017

| ticker | weight_ols | name                                  | category            | etf_signal |
|--------|------------|---------------------------------------|---------------------|------------|
| SQQQ   | 0.507      | ProShares UltraPro Short QQQ          | Leveraged / Inverse | FLAT       |
| UUP    | 0.496      | Invesco DB USD Index Bullish Fund ETF | Currency            | LONG       |

## Members

| ticker | weight | name                                  | category            | etf_signal |
|--------|--------|---------------------------------------|---------------------|------------|
| SQQQ   | 0.5    | ProShares UltraPro Short QQQ          | Leveraged / Inverse | FLAT       |
| UUP    | 0.5    | Invesco DB USD Index Bullish Fund ETF | Currency            | LONG       |

## Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

| ticker | name                             | category             | cluster    | cluster_name             | cluster_signal | cluster_nss | entry_date | days | cum_return | prob_sell |
|--------|----------------------------------|----------------------|------------|--------------------------|----------------|-------------|------------|------|------------|-----------|
| MINT   | PIMCO Enhanced Short Maturity... | Cash / Ultra-Shor... | cluster_23 | Cash / Ultra-Short Fi... | FLAT           | 1.000       | 2023-01-30 | 822  | 18.7%      | 0.00      |
| GBIL   | Goldman Sachs Access Treasury... | Cash / Ultra-Shor... | cluster_23 | Cash / Ultra-Short Fi... | FLAT           | 1.000       | 2022-08-18 | 934  | 17.5%      | 0.00      |
| BIL    | State Street SPDR Bloomberg 1... | Cash / Ultra-Shor... | cluster_23 | Cash / Ultra-Short Fi... | FLAT           | 1.000       | 2022-05-19 | 996  | 18.0%      | 0.00      |
| GSY    | Invesco Ultra Short Duration ... | Cash / Ultra-Shor... | cluster_23 | Cash / Ultra-Short Fi... | FLAT           | 1.000       | 2022-12-29 | 842  | 19.5%      | 0.00      |
| VNLA   | Janus Henderson Short Duratio... | Cash / Ultra-Shor... | cluster_23 | Cash / Ultra-Short Fi... | FLAT           | 1.000       | 2026-04-02 | 26   | 0.5%       | 0.03      |
| SRLN   | State Street Blackstone Senio... | Bank Loan / Float... | cluster_22 | Mixed: SRLN, BKLN, FT... | LONG           | 1.000       | 2026-04-07 | 24   | 1.9%       | 0.00      |
| FTSL   | First Trust Senior Loan Fund     | Bank Loan / Float... | cluster_22 | Mixed: SRLN, BKLN, FT... | LONG           | 1.000       | 2026-04-06 | 25   | 1.4%       | 0.00      |
| BKLN   | Invesco Senior Loan ETF          | Bank Loan / Float... | cluster_22 | Mixed: SRLN, BKLN, FT... | LONG           | 1.000       | 2026-04-06 | 25   | 1.3%       | 0.00      |
| LQDH   | iShares Interest Rate Hedged ... | Interest Rate Hed... | cluster_22 | Mixed: SRLN, BKLN, FT... | LONG           | 1.000       | 2026-04-06 | 25   | 1.3%       | 0.01      |
| IGBH   | iShares Interest Rate Hedged ... | Interest Rate Hed... | cluster_22 | Mixed: SRLN, BKLN, FT... | LONG           | 1.000       | 2026-04-08 | 23   | 1.7%       | 0.01      |
| IGHG   | ProShares Investment Grade-In... | Interest Rate Hed... | cluster_22 | Mixed: SRLN, BKLN, FT... | LONG           | 1.000       | 2026-04-07 | 24   | 1.1%       | 0.11      |
| NXTG   | First Trust Indxx NextG ETF      | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-08 | 23   | 21.7%      | 0.00      |
| QTEC   | First Trust NASDAQ-100-Techno... | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-10 | 21   | 31.6%      | 0.00      |
| XSD    | State Street SPDR S&P Semicon... | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-09 | 22   | 58.0%      | 0.00      |
| PTNQ   | Pacer Trendpilot 100 ETF         | Trend-Following E... | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-17 | 16   | 9.9%       | 0.00      |
| FTXL   | First Trust Nasdaq Semiconduc... | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-07 | 24   | 59.1%      | 0.00      |
| PSI    | Invesco Semiconductors ETF       | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-07 | 24   | 56.9%      | 0.00      |
| SOXX   | iShares PHLX SOX Semiconducto... | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-08 | 23   | 43.8%      | 0.00      |
| SMH    | VanEck Semiconductor ETF         | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-08 | 23   | 36.3%      | 0.00      |
| AIQ    | Global X Artificial Intellige... | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-13 | 20   | 24.3%      | 0.00      |
| IYW    | iShares U.S. Technology ETF      | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-10 | 21   | 23.6%      | 0.00      |
| XNTK   | State Street SPDR NYSE Techno... | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-10 | 21   | 24.0%      | 0.00      |
| XLK    | State Street Technology Selec... | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-10 | 21   | 24.7%      | 0.00      |
| IXN    | iShares Global Tech ETF          | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-10 | 21   | 24.5%      | 0.00      |
| PSCT   | Invesco S&P SmallCap Informat... | U.S. Small Cap Te... | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-07 | 24   | 30.0%      | 0.00      |
| QQQ    | Invesco QQQ Trust, Series 1      | U.S. Large Cap Gr... | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-10 | 21   | 16.7%      | 0.00      |
| QTUM   | Defiance Quantum ETF             | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-10 | 21   | 26.7%      | 0.00      |
| PTLC   | Pacer Trendpilot US Large Cap... | U.S. Trend-Follow... | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-20 | 15   | 4.3%       | 0.00      |
| FTEC   | Fidelity MSCI Information Tec... | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-10 | 21   | 22.5%      | 0.00      |
| VGT    | Vanguard Information Tech ETF    | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-10 | 21   | 22.4%      | 0.00      |
| CIBR   | First Trust NASDAQ Cybersecur... | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-17 | 16   | 13.5%      | 0.00      |
| QYLD   | Global X NASDAQ 100 Covered C... | Options Income       | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-08 | 23   | 4.3%       | 0.00      |
| IGM    | iShares Expanded Tech Sector ... | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-09 | 22   | 22.2%      | 0.00      |
| PTF    | Invesco Dorsey Wright Technol... | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-07 | 24   | 35.1%      | 0.00      |
| BOTZ   | Global X Robotics & Artificia... | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-16 | 17   | 12.5%      | 0.00      |
| SKYY   | First Trust Cloud Computing E... | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-17 | 16   | 9.4%       | 0.00      |
| FXL    | First Trust Technology AlphaD... | Technology           | cluster_3  | Technology (34/63)       | LONG           | 1.000       | 2026-04-09 | 22   | 18.0%      | 0.00      |

|      |                                  |                      |           |                          |      |       |            |    |       |      |
|------|----------------------------------|----------------------|-----------|--------------------------|------|-------|------------|----|-------|------|
| XT   | iShares Future Exponential Te... | Technology           | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-13 | 20 | 10.8% | 0.00 |
| TDIV | First Trust NASDAQ Technology... | Technology           | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-13 | 20 | 14.3% | 0.00 |
| ROBO | ROBO Global Robotics and Auto... | Technology           | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-14 | 19 | 14.2% | 0.00 |
| ONEQ | Fidelity Nasdaq Composite Ind... | U.S. Equity          | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-13 | 20 | 13.4% | 0.00 |
| HACK | Amplify Cybersecurity ETF        | Technology           | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-17 | 16 | 8.6%  | 0.00 |
| ICVT | iShares Convertible Bond ETF     | Convertible Bond     | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-08 | 23 | 10.4% | 0.00 |
| QQEW | First Trust Nasdaq-100 Select... | U.S. Equal Weight... | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-16 | 17 | 6.9%  | 0.00 |
| PWB  | Invesco Large Cap Growth ETF     | Growth Equity        | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-10 | 21 | 13.7% | 0.00 |
| SPHB | Invesco S&P 500 High Beta ETF    | U.S. Equity          | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-10 | 21 | 16.4% | 0.00 |
| CWB  | State Street SPDR Bloomberg C... | Convertible Bond     | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-08 | 23 | 10.1% | 0.00 |
| NULG | Nuveen ESG Large-Cap Growth E... | U.S. Large Cap Gr... | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-10 | 21 | 13.7% | 0.00 |
| IWM  | iShares S&P 500 Growth ETF       | U.S. Large Cap Gr... | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-13 | 20 | 11.5% | 0.00 |
| SPYG | State Street SPDR Portfolio S... | U.S. Large Cap Gr... | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-13 | 20 | 11.5% | 0.00 |
| VOOG | Vanguard S&P 500 Growth ETF      | U.S. Large Cap Gr... | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-13 | 20 | 11.5% | 0.00 |
| IUSG | iShares Core S&P U.S. Growth ... | U.S. Large Cap Gr... | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-13 | 20 | 11.2% | 0.00 |
| MGK  | Vanguard Mega Cap Growth ETF     | U.S. Mega Cap Gro... | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-13 | 20 | 11.9% | 0.00 |
| ILCG | iShares Morningstar Large-Cap... | U.S. Large Cap Gr... | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-13 | 20 | 11.5% | 0.00 |
| VUG  | Vanguard Growth ETF              | U.S. Large Cap Gr... | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-13 | 20 | 11.4% | 0.00 |
| BLCN | Siren NexGen Economy ETF         | Technology           | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-15 | 18 | 6.0%  | 0.00 |
| IGV  | iShares Expanded Tech-Softwar... | Technology           | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-21 | 14 | 4.6%  | 0.00 |
| BLOK | Amplify Blockchain Technology... | Technology           | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-13 | 20 | 17.0% | 0.00 |
| SCHG | Schwab U.S. Large-Cap Growth ... | U.S. Large Cap Gr... | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-13 | 20 | 10.3% | 0.00 |
| ARKW | ARK Next Generation Internet ... | Technology           | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-14 | 19 | 13.7% | 0.01 |
| PRNT | 3D Printing (The) ETF            | Technology           | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-16 | 17 | 6.6%  | 0.01 |
| IWY  | iShares Russell Top 200 Growt... | Growth Equity        | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-14 | 19 | 7.7%  | 0.01 |
| VONG | Vanguard Russell 1000 Growth ... | U.S. Large Cap Gr... | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-14 | 19 | 7.3%  | 0.01 |
| IWF  | iShares Russell 1000 Growth F... | U.S. Large Cap Gr... | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-14 | 19 | 7.3%  | 0.02 |
| ARKQ | ARK Autonomous Technology & R... | Technology           | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-14 | 19 | 11.1% | 0.06 |
| IPO  | Renaissance IPO ETF              | IPO Equity           | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-13 | 20 | 7.6%  | 0.09 |
| FDN  | First Trust DJ Internet Index... | Technology           | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-10 | 21 | 8.6%  | 0.18 |
| ARKK | ARK Innovation ETF               | Technology           | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-14 | 19 | 6.9%  | 0.23 |
| FINX | Global X FinTech ETF             | Financials           | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-20 | 15 | -3.4% | 0.44 |
| PNQI | Invesco Nasdaq Internet ETF      | Technology           | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-15 | 18 | -1.6% | 0.55 |
| PBS  |                                  |                      | cluster_3 | Technology (34/63)       | LONG | 1.000 | 2026-04-14 | 19 |       |      |
| EWT  | iShares MSCI Taiwan ETF          | Taiwan Equity        | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-08 | 23 | 27.2% | 0.00 |
| EWY  | iShares MSCI South Korea ETF     | South Korea Equity   | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-09 | 22 | 38.5% | 0.00 |
| ATA  | iShares Asia 50 ETF              | Asia Equity          | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-10 | 21 | 21.0% | 0.00 |
| AAXJ | iShares MSCI All Country Asia... | Asia ex-Japan Equ... | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-13 | 20 | 13.7% | 0.00 |
| EEMA | iShares MSCI Emerging Markets... | Emerging Markets ... | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-13 | 20 | 13.4% | 0.00 |
| EMGF | iShares Emerging Markets Equi... | Emerging Markets ... | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-10 | 21 | 13.3% | 0.00 |
| XSOE | WisdomTree Emerging Markets E... | Emerging Markets ... | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-10 | 21 | 12.5% | 0.00 |
| EEM  | iShares MSCI Emerging Index F... | Emerging Markets ... | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-10 | 21 | 12.1% | 0.00 |
| IEMG | iShares Core MSCI Emerging Ma... | Emerging Markets ... | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-10 | 21 | 11.7% | 0.00 |
| ESGE | iShares ESG Aware MSCI EM ETF    | Emerging Markets ... | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-13 | 20 | 11.2% | 0.00 |
| GEM  | Goldman Sachs ActiveBeta Emer... | Emerging Markets ... | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-10 | 21 | 11.3% | 0.00 |

|      |                                  |                       |           |                          |      |       |            |    |       |      |
|------|----------------------------------|-----------------------|-----------|--------------------------|------|-------|------------|----|-------|------|
| GMF  | State Street SPDR S&P Emergin... | Asia Equity           | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-14 | 19 | 6.1%  | 0.00 |
| DEM  | WisdomTree Emerging Markets H... | Emerging Markets ...  | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-09 | 22 | 7.0%  | 0.00 |
| EEMV | iShares MSCI Emerging Markets... | Emerging Markets ...  | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-10 | 21 | 8.5%  | 0.00 |
| PXH  | Invesco RAFI Emerging Markets... | Emerging Markets ...  | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-10 | 21 | 5.7%  | 0.01 |
| SPEM | State Street SPDR Portfolio E... | Emerging Markets ...  | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-13 | 20 | 5.7%  | 0.02 |
| VWO  | Vanguard FTSE Emerging Market... | Emerging Markets ...  | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-13 | 20 | 5.6%  | 0.02 |
| SCHE | Schwab Emerging Markets Equit... | Emerging Markets ...  | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-13 | 20 | 5.2%  | 0.03 |
| EWX  | State Street SPDR S&P Emergin... | Emerging Markets ...  | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-10 | 21 | 8.5%  | 0.03 |
| FNDE | Schwab Fundamental Emerging M... | Emerging Markets ...  | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-10 | 21 | 4.9%  | 0.03 |
| FEM  | First Trust Emerging Markets ... | Emerging Markets ...  | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-07 | 24 | 11.5% | 0.09 |
| DGS  | WisdomTree Emerging Market Sm... | Emerging Markets ...  | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-10 | 21 | 4.0%  | 0.15 |
| SDIV | Global X SuperDividend ETF       | Global Dividend E...  | cluster_8 | Emerging Markets Equi... | LONG | 1.000 | 2026-04-09 | 22 | 1.2%  | 0.46 |
| EIS  | iShares MSCI Israel ETF          | Israel Equity         | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-10 | 21 | 8.4%  | 0.00 |
| DSI  | iShares ESG MSCI KLD 400 ETF     | Large Cap Blend E...  | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-14 | 19 | 8.6%  | 0.00 |
| IOO  | iShares Global 100 ETF           | Global Equity         | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 9.1%  | 0.00 |
| QAI  | NYLI Hedge Multi-Strategy Tra... | Alternatives          | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-08 | 23 | 4.0%  | 0.00 |
| OEF  | iShares S&P 100 Fund             | U.S. Mega Cap Equ...  | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 9.5%  | 0.00 |
| SUSA | iShares ESG Optimized MSCI US... | U.S. ESG Equity       | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 8.0%  | 0.00 |
| OMFL | Invesco Russell 1000 Dynamic ... | U.S. Multifactor ...  | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-10 | 21 | 7.2%  | 0.00 |
| ESGU | iShares ESG Aware MSCI USA ETF   | U.S. ESG Equity       | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 8.1%  | 0.00 |
| MGC  | Vanguard Mega Cap ETF            | U.S. Mega Cap Equ...  | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 8.7%  | 0.00 |
| IVV  | iShares Core S&P 500 ETF         | U.S. Large Cap Equ... | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.7%  | 0.00 |
| SPY  | State Street SPDR S&P 500 ETF... | U.S. Large Cap Equ... | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.8%  | 0.00 |
| VOO  | Vanguard S&P 500 ETF             | U.S. Large Cap Equ... | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.7%  | 0.00 |
| XLG  | Invesco S&P 500 Top 50 ETF       | U.S. Mega Cap Equ...  | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 9.4%  | 0.00 |
| VV   | Vanguard Large-Cap ETF           | U.S. Large Cap Equ... | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.7%  | 0.00 |
| IWL  | iShares Russell Top 200 ETF      | U.S. Equity           | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 8.2%  | 0.00 |
| ILCB | iShares Morningstar Large-Cap... | U.S. Large Cap Equ... | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.7%  | 0.00 |
| SPTM | State Street SPDR Portfolio S... | U.S. Total Market...  | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.4%  | 0.00 |
| IYY  | iShares Dow Jones U.S. ETF       | U.S. Total Market...  | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.5%  | 0.00 |
| EPS  | WisdomTree U.S. LargeCap Fund    | Large Cap Value E...  | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-10 | 21 | 8.1%  | 0.00 |
| SCHB | Schwab U.S. Broad Market ETF     | U.S. Total Market...  | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.4%  | 0.00 |
| ITOT | iShares Core S&P Total U.S. S... | U.S. Total Market...  | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.4%  | 0.00 |
| SCHX | Schwab U.S. Large-Cap ETF        | U.S. Large Cap Equ... | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.5%  | 0.00 |
| VTI  | Vanguard Total Stock Market E... | U.S. Total Market...  | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.4%  | 0.00 |
| SPYX | State Street SPDR S&P 500 Fos... | U.S. Equity           | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.8%  | 0.00 |
| IWB  | iShares Russell 1000 ETF         | U.S. Large Cap Equ... | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.2%  | 0.00 |
| SPHQ | Invesco S&P 500 Quality ETF      | U.S. Quality / Fa...  | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-10 | 21 | 5.4%  | 0.00 |
| IWV  | iShares Russell 3000 Fund        | U.S. Total Market...  | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.2%  | 0.00 |
| VTHR | Vanguard Russell 3000 ETF        | U.S. Total Market...  | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.3%  | 0.00 |
| VONE | Vanguard Russell 1000 ETF        | U.S. Large Cap Equ... | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.2%  | 0.00 |
| CATH | Global X S&P 500 Catholic Val... | Global Equity         | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 7.3%  | 0.00 |
| TILT | FlexShares Mornigstar US Mark... | U.S. Multifactor ...  | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 6.0%  | 0.00 |
| ACWI | iShares MSCI ACWI ETF            | Global Equity         | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 6.2%  | 0.01 |
| JHML | John Hancock Multifactor Larg... | U.S. Large Cap Equ... | cluster_4 | Mixed: IWB, ITOT, SCHB   | LONG | 0.987 | 2026-04-13 | 20 | 5.6%  | 0.01 |

|      |                                  |                      |            |                        |      |       |            |    |       |      |
|------|----------------------------------|----------------------|------------|------------------------|------|-------|------------|----|-------|------|
| SPGM | State Street SPDR Portfolio M... | Global Equity        | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-10 | 21 | 7.0%  | 0.01 |
| VT   | Vanguard Total World Stock In... | Global Equity        | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-13 | 20 | 6.1%  | 0.01 |
| LRGF | iShares U.S. Equity Factor ETF   | U.S. Multifactor ... | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-13 | 20 | 6.4%  | 0.01 |
| QDF  | FlexShares Quality Dividend I... | Dividend Equity      | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-13 | 20 | 5.3%  | 0.01 |
| CRBN | iShares Low Carbon Optimized ... | Global Equity        | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-13 | 20 | 6.2%  | 0.01 |
| AOA  | iShares Core 80/20 Aggressive... | Asset Allocation     | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-13 | 20 | 4.8%  | 0.01 |
| GSLC | Goldman Sachs ActiveBeta U.S.... | Commodity            | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-13 | 20 | 6.2%  | 0.01 |
| URTH | iShares MSCI World ETF           | Global Equity        | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-13 | 20 | 5.7%  | 0.01 |
| AOR  | iShares Core 60/40 Balanced A... | Asset Allocation     | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-13 | 20 | 3.6%  | 0.02 |
| AOM  | iShares Core 40/60 Moderate A... | Asset Allocation     | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-13 | 20 | 2.4%  | 0.05 |
| QUS  | State Street SPDR MSCI USA St... | U.S. Quality / Fa... | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-14 | 19 | 2.5%  | 0.06 |
| FPXI | First Trust International Equ... | International IPO... | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-10 | 21 | 7.3%  | 0.07 |
| AOK  | iShares Core 30/70 Conservati... | Asset Allocation     | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-13 | 20 | 1.8%  | 0.07 |
| QUAL | iShares MSCI USA Quality Fact... | U.S. Quality / Fa... | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-14 | 19 | 3.6%  | 0.08 |
| DGRW | WisdomTree U.S. Quality Divid... | U.S. Dividend Equ... | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-14 | 19 | 3.2%  | 0.11 |
| XLY  | State Street Consumer Discret... | Consumer Discreti... | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-15 | 18 | 1.0%  | 0.32 |
| FDIS | Fidelity MSCI Consumer Discre... | Consumer Discreti... | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-14 | 19 | 2.4%  | 0.34 |
| VCR  | Vanguard Consumer Discretion ... | Consumer Discreti... | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-14 | 19 | 2.4%  | 0.35 |
| VOX  | Vanguard Communication Servic... | Communication Ser... | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-13 | 20 | 2.3%  | 0.60 |
| FCOM | Fidelity MSCI Communication S... | Communication Ser... | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-13 | 20 | 2.1%  | 0.62 |
| XLC  | State Street Communication Se... | Communication Ser... | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-14 | 19 | -0.8% | 0.74 |
| IYC  | iShares U.S. Consumer Discret... | Consumer Discreti... | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-14 | 19 | -1.0% | 0.82 |
| EWS  | iShares MSCI Singapore ETF       | Singapore Equity     | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-07 | 24 | 1.9%  | 0.83 |
| RTH  | VanEck Retail ETF                | Consumer Discreti... | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-10 | 21 | -0.0% | 0.93 |
| SPLG |                                  |                      | cluster_4  | Mixed: IWB, ITOT, SCHB | LONG | 0.987 | 2026-04-13 | 20 |       |      |
| VPL  | Vanguard FTSE Pacific ETF        | Asia-Pacific Equi... | cluster_20 | Japan Equity (7/9)     | FLAT | 0.947 | 2026-04-10 | 21 | 10.1% | 0.00 |
| EWJ  | iShares MSCI Japan Index Fund    | Japan Equity         | cluster_20 | Japan Equity (7/9)     | FLAT | 0.947 | 2026-04-13 | 20 | 4.4%  | 0.03 |
| BBJP | JPMorgan BetaBuilders Japan E... | Japan Equity         | cluster_20 | Japan Equity (7/9)     | FLAT | 0.947 | 2026-04-30 | 7  | 3.4%  | 0.03 |
| HEWJ | iShares Currency Hedged MSCI ... | Japan Equity         | cluster_20 | Japan Equity (7/9)     | FLAT | 0.947 | 2026-04-08 | 23 | 2.7%  | 0.09 |
| DBJP | Xtrackers MSCI Japan Hedged E... | Japan Equity         | cluster_20 | Japan Equity (7/9)     | FLAT | 0.947 | 2026-04-08 | 23 | 2.5%  | 0.13 |
| EWJV | iShares MSCI Japan Value ETF     | Japan Equity         | cluster_20 | Japan Equity (7/9)     | FLAT | 0.947 | 2026-05-06 | 3  | -0.1% | 0.17 |
| DXJ  | WisdomTree Japan Hedged Equit... | Japan Equity         | cluster_20 | Japan Equity (7/9)     | FLAT | 0.947 | 2026-05-01 | 6  | 4.2%  | 0.23 |
| VLUE | iShares MSCI USA Value Factor... | U.S. Value Factor... | cluster_6  | Mixed: IWD, SCHV, VONV | LONG | 0.919 | 2026-04-09 | 22 | 21.2% | 0.00 |
| FNDX | Schwab Fundamental U.S. Large... | U.S. Fundamental ... | cluster_6  | Mixed: IWD, SCHV, VONV | LONG | 0.919 | 2026-04-09 | 22 | 5.6%  | 0.00 |
| PRF  | Invesco RAFI US 1000 ETF         | U.S. Fundamental ... | cluster_6  | Mixed: IWD, SCHV, VONV | LONG | 0.919 | 2026-04-09 | 22 | 6.4%  | 0.00 |
| IWX  | iShares Russell Top 200 Value... | Value Equity         | cluster_6  | Mixed: IWD, SCHV, VONV | LONG | 0.919 | 2026-04-10 | 21 | 6.3%  | 0.00 |
| NULV | Nuveen ESG Large-Cap Value ETF   | U.S. Large Cap Va... | cluster_6  | Mixed: IWD, SCHV, VONV | LONG | 0.919 | 2026-04-10 | 21 | 6.7%  | 0.00 |
| JVAL | JPMorgan U.S. Value Factor ETF   | U.S. Value Factor... | cluster_6  | Mixed: IWD, SCHV, VONV | LONG | 0.919 | 2026-04-10 | 21 | 9.1%  | 0.00 |
| IWD  | iShares Russell 1000 Value ETF   | U.S. Large Cap Va... | cluster_6  | Mixed: IWD, SCHV, VONV | LONG | 0.919 | 2026-04-10 | 21 | 5.8%  | 0.00 |
| VONV | Vanguard Russell 1000 Value E... | U.S. Large Cap Va... | cluster_6  | Mixed: IWD, SCHV, VONV | LONG | 0.919 | 2026-04-10 | 21 | 5.8%  | 0.00 |
| MGV  | Vanguard Mega Cap Value ETF      | U.S. Mega Cap Val... | cluster_6  | Mixed: IWD, SCHV, VONV | LONG | 0.919 | 2026-04-13 | 20 | 3.5%  | 0.00 |
| ILCV | iShares Morningstar Large-Cap... | U.S. Large Cap Va... | cluster_6  | Mixed: IWD, SCHV, VONV | LONG | 0.919 | 2026-04-13 | 20 | 4.0%  | 0.00 |
| VOOV | Vanguard S&P 500 Value ETF       | U.S. Large Cap Va... | cluster_6  | Mixed: IWD, SCHV, VONV | LONG | 0.919 | 2026-04-14 | 19 | 3.3%  | 0.00 |
| SPYV | State Street SPDR Portfolio S... | U.S. Large Cap Va... | cluster_6  | Mixed: IWD, SCHV, VONV | LONG | 0.919 | 2026-04-14 | 19 | 3.2%  | 0.00 |
| IVE  | iShares S&P 500 Value ETF        | U.S. Large Cap Va... | cluster_6  | Mixed: IWD, SCHV, VONV | LONG | 0.919 | 2026-04-14 | 19 | 3.2%  | 0.00 |

|      |                                  |                      |            |                          |      |       |            |    |       |      |
|------|----------------------------------|----------------------|------------|--------------------------|------|-------|------------|----|-------|------|
| SCHV | Schwab U.S. Large-Cap Value E... | U.S. Large Cap Va... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-10 | 21 | 4.4%  | 0.00 |
| IUSV | iShares Core S&P U.S. Value E... | U.S. Large Cap Va... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-14 | 19 | 3.1%  | 0.00 |
| RWL  | Invesco S&P 500 Revenue ETF      | U.S. Revenue Weig... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-10 | 21 | 4.8%  | 0.01 |
| VTV  | Vanguard Value ETF               | U.S. Large Cap Va... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-13 | 20 | 2.8%  | 0.01 |
| IMCB | iShares Morningstar Mid-Cap E... | U.S. Mid Cap Equi... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-10 | 21 | 5.5%  | 0.04 |
| FEX  | First Trust Large Cap Core Al... | Large Cap Blend E... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-09 | 22 | 4.2%  | 0.06 |
| IWS  | iShares Russell Mid-Cap Value... | U.S. Mid Cap Value   | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-09 | 22 | 4.2%  | 0.06 |
| IWR  | iShares Russell Mid-Cap ETF      | U.S. Mid Cap Equi... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-10 | 21 | 4.9%  | 0.08 |
| ETHO | Amplify Etho Climate Leadersh... | U.S. ESG Equity      | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-09 | 22 | 5.6%  | 0.11 |
| VO   | Vanguard Mid-Cap ETF             | U.S. Mid Cap Equi... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-13 | 20 | 3.3%  | 0.11 |
| SIZE | iShares MSCI USA Size Factor ... | U.S. Size Factor ... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-13 | 20 | 3.4%  | 0.12 |
| DTD  | WisdomTree U.S. Total Dividen... | U.S. Dividend Equ... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-13 | 20 | 2.5%  | 0.12 |
| AIVL | WisdomTree U.S. AI Enhanced V... | U.S. Value Factor... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-10 | 21 | 3.2%  | 0.12 |
| DLN  | WisdomTree U.S. LargeCap Divi... | U.S. Dividend Equ... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-13 | 20 | 2.6%  | 0.12 |
| RSP  | Invesco S&P 500 Equal Weight ... | U.S. Equal Weight... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-14 | 19 | 2.2%  | 0.20 |
| VIG  | Vanguard Div Appreciation ETF    | U.S. Dividend Equ... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-14 | 19 | 2.0%  | 0.25 |
| PFM  | Invesco Dividend Achievers ETF   | U.S. Dividend Equ... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-14 | 19 | 1.9%  | 0.26 |
| FDVV | Fidelity High Dividend ETF       | U.S. Dividend Equ... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-14 | 19 | 2.4%  | 0.27 |
| DIA  | State Street SPDR Dow Jones I... | U.S. Large Cap Eq... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-14 | 19 | 2.4%  | 0.29 |
| JPME | JPMorgan Diversified Return U... | U.S. Equity          | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-07 | 24 | 3.9%  | 0.34 |
| VYM  | Vanguard High Dividend Yield ... | U.S. Dividend Equ... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-10 | 21 | 2.5%  | 0.38 |
| OUSA | ALPS O'Shares U.S. Quality Di... | Dividend Equity      | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-20 | 15 | -0.9% | 0.47 |
| DGRO | iShares Core Dividend Growth ... | U.S. Dividend Equ... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-13 | 20 | 1.2%  | 0.54 |
| MOAT | VanEck Morningstar Wide Moat ... | U.S. Moat Equity     | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-05-06 | 3  | -0.9% | 0.61 |
| PWV  | Invesco Large Cap Value ETF      | Value Equity         | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-02 | 26 | 3.6%  | 0.62 |
| RDVY | First Trust Rising Dividend A... | U.S. Dividend Equ... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-10 | 21 | 3.0%  | 0.70 |
| CFO  | VictoryShares US 500 Enhanced... | U.S. Multifactor ... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-13 | 20 | 0.7%  | 0.72 |
| SDY  | State Street SPDR S&P Dividen... | U.S. Dividend Equ... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-05-01 | 6  | -0.9% | 0.71 |
| CFA  | VictoryShares US 500 Volatili... | U.S. Multifactor ... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-13 | 20 | 0.7%  | 0.73 |
| USMV | iShares MSCI USA Min Vol Fact... | U.S. Low Volatili... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-05-05 | 4  | -0.4% | 0.64 |
| EQAL | Invesco Russell 1000 Equal We... | U.S. Equal Weight... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-08 | 23 | 1.5%  | 0.76 |
| ACWV | iShares MSCI Global Min Vol F... | Global Equity        | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-05-06 | 3  | -0.4% | 0.66 |
| ONEV | State Street SPDR Russell 100... | U.S. Equity          | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-14 | 19 | 0.3%  | 0.85 |
| JPUS | JPMorgan Diversified Return U... | U.S. Multifactor ... | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-08 | 23 | 0.9%  | 0.85 |
| VOE  | Vanguard Mid-Cap Value ETF       | U.S. Mid Cap Value   | cluster_6  | Mixed: IWD, SCHV, VONV   | LONG | 0.919 | 2026-04-09 | 22 | 1.1%  | 0.85 |
| PDBC | Invesco Optimum Yield Diversi... | Broad Commodities    | cluster_21 | Broad Commodities (6/... | LONG | 0.918 | 2025-12-22 | 95 | 39.8% | 0.01 |
| DBC  | Invesco DB Commodity Index Tr... | Broad Commodities    | cluster_21 | Broad Commodities (6/... | LONG | 0.918 | 2025-12-22 | 95 | 38.7% | 0.02 |
| USO  | United States Oil Fund           | Oil                  | cluster_21 | Broad Commodities (6/... | LONG | 0.918 | 2026-01-29 | 70 | 75.2% | 0.30 |
| FXF  | Invesco CurrencyShares Swiss ... | Currency             | cluster_24 | Currency (4/6)           | LONG | 0.873 | 2026-05-05 | 4  | 0.6%  | 0.12 |
| FXB  | Invesco CurrencyShares Britis... | Currency             | cluster_24 | Currency (4/6)           | LONG | 0.873 | 2026-04-13 | 20 | 1.0%  | 0.13 |
| IGOV | iShares International Treasur... | International Fix... | cluster_24 | Currency (4/6)           | LONG | 0.873 | 2026-05-04 | 5  | 1.1%  | 0.19 |
| BWX  | State Street SPDR Bloomberg I... | International Fix... | cluster_24 | Currency (4/6)           | LONG | 0.873 | 2026-05-11 | 0  | 0.0%  | 0.35 |
| CQQQ | Invesco China Technology ETF     | China Equity         | cluster_13 | China Equity (13/15)     | LONG | 0.858 | 2026-04-20 | 15 | 8.0%  | 0.00 |
| CNYA | iShares MSCI China A ETF         | China Equity         | cluster_13 | China Equity (13/15)     | LONG | 0.858 | 2026-04-14 | 19 | 5.8%  | 0.00 |
| ASHR | Xtrackers Harvest CSI 300 Chi... | China Equity         | cluster_13 | China Equity (13/15)     | LONG | 0.858 | 2026-04-13 | 20 | 6.9%  | 0.01 |

|      |                                  |                      |            |                      |      |       |            |    |       |      |
|------|----------------------------------|----------------------|------------|----------------------|------|-------|------------|----|-------|------|
| EWI  | iShares MSCI Hong Kong Index ... | Hong Kong Equity     | cluster_13 | China Equity (13/15) | LONG | 0.858 | 2026-04-08 | 23 | 2.3%  | 0.01 |
| KBA  | KraneShares Bosera MSCI China... | China Equity         | cluster_13 | China Equity (13/15) | LONG | 0.858 | 2026-04-10 | 21 | 9.8%  | 0.01 |
| CNXT | VanEck ChiNext Innovators ETF    | China Equity         | cluster_13 | China Equity (13/15) | LONG | 0.858 | 2026-04-10 | 21 | 17.6% | 0.01 |
| CXSE | WisdomTree China ex-State-Own... | China Equity         | cluster_13 | China Equity (13/15) | LONG | 0.858 | 2026-04-14 | 19 | 3.7%  | 0.17 |
| GXC  | State Street SPDR S&P China E... | China Equity         | cluster_13 | China Equity (13/15) | LONG | 0.858 | 2026-05-04 | 5  | 2.0%  | 0.19 |
| FXI  | iShares China Large-Cap ETF      | China Equity         | cluster_13 | China Equity (13/15) | LONG | 0.858 | 2026-04-30 | 7  | 1.8%  | 0.28 |
| MCHI | iShares MSCI China ETF           | China Equity         | cluster_13 | China Equity (13/15) | LONG | 0.858 | 2026-05-06 | 3  | -0.0% | 0.36 |
| KGRN | KraneShares MSCI China Clean ... | China Equity         | cluster_13 | China Equity (13/15) | LONG | 0.858 | 2026-05-06 | 3  | 1.6%  | 0.62 |
| FV   | First Trust Dorsey Wright Foc... | Momentum Equity      | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-14 | 19 | 8.9%  | 0.00 |
| MTUM | iShares MSCI USA Momentum Fac... | U.S. Momentum Equ... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-08 | 23 | 18.1% | 0.00 |
| PRN  | Invesco Dorsey Wright Industr... | Industrials          | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-08 | 23 | 15.4% | 0.00 |
| FPX  | First Trust US Equity Opportu... | IPO Equity           | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-08 | 23 | 12.4% | 0.00 |
| RPG  | Invesco S&P 500 Pure Growth E... | Growth Equity        | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-09 | 22 | 14.2% | 0.00 |
| PDP  | Invesco Dorsey Wright Momentu... | Momentum Equity      | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-08 | 23 | 9.6%  | 0.00 |
| VFM0 | Vanguard U.S. Momentum Factor... | U.S. Momentum Equ... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-08 | 23 | 10.5% | 0.00 |
| XMMO | Invesco S&P MidCap Momentum E... | U.S. Mid Cap Mome... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-06 | 25 | 13.4% | 0.00 |
| SCHA | Schwab U.S. Small-Cap ETF        | U.S. Small Cap Eq... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-09 | 22 | 8.6%  | 0.00 |
| FTC  | First Trust Large Cap Growth ... | Technology           | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-10 | 21 | 9.2%  | 0.00 |
| IMCG | iShares Morningstar Mid-Cap G... | U.S. Mid Cap Grow... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-09 | 22 | 9.1%  | 0.00 |
| SCHM | Schwab U.S. Mid Cap ETF          | U.S. Mid Cap Equi... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-09 | 22 | 7.2%  | 0.01 |
| PSC  | Principal U.S. Small-Cap ETF     | U.S. Small Cap Eq... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-10 | 21 | 8.0%  | 0.01 |
| IWM  | iShares Russell 2000 Index Fu... | U.S. Small Cap Eq... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-10 | 21 | 9.2%  | 0.01 |
| VTWO | Vanguard Russell 2000 ETF        | U.S. Small Cap Eq... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-10 | 21 | 9.2%  | 0.01 |
| IWO  | iShares Russell 2000 Growth F... | U.S. Small Cap Gr... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-13 | 20 | 10.0% | 0.01 |
| VTWG | Vanguard Russell 2000 Growth ... | U.S. Small Cap Gr... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-13 | 20 | 10.0% | 0.01 |
| IWC  | iShares Microcap ETF             | U.S. Micro Cap Eq... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-09 | 22 | 10.0% | 0.01 |
| IJK  | iShares S&P Mid-Cap 400 Growt... | U.S. Mid Cap Grow... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-09 | 22 | 6.9%  | 0.01 |
| MDYG | State Street SPDR S&P 400 Mid... | U.S. Mid Cap Grow... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-09 | 22 | 6.9%  | 0.01 |
| IVOG | Vanguard S&P Mid-Cap 400 Grow... | U.S. Mid Cap Grow... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-09 | 22 | 6.8%  | 0.02 |
| VOT  | Vanguard Mid-Cap Growth ETF      | U.S. Mid Cap Grow... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-14 | 19 | 4.9%  | 0.02 |
| VIOG | Vanguard S&P Small-Cap 600 Gr... | U.S. Small Cap Gr... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-08 | 23 | 6.6%  | 0.02 |
| QMOM | Alpha Architect U.S. Quantita... | U.S. Momentum Equ... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-08 | 23 | 9.1%  | 0.02 |
| SLYG | State Street SPDR S&P 600 Sma... | U.S. Small Cap Gr... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-08 | 23 | 6.6%  | 0.02 |
| IJT  | iShares S&P SmallCap 600 Grow... | U.S. Small Cap Gr... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-08 | 23 | 6.6%  | 0.02 |
| PAVE | Global X U.S. Infrastructure ... | Industrials          | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-10 | 21 | 4.4%  | 0.02 |
| XSMO | Invesco S&P SmallCap Momentum... | U.S. Small Cap Mo... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-08 | 23 | 7.6%  | 0.02 |
| VBK  | Vanguard Small-Cap Growth ETF    | U.S. Small Cap Gr... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-10 | 21 | 9.9%  | 0.03 |
| XMHQ | Invesco S&P MidCap Quality ETF   | U.S. Mid Cap Qual... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-10 | 21 | 3.6%  | 0.03 |
| ISCG | iShares Morningstar Small-Cap... | Growth Equity        | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-10 | 21 | 7.4%  | 0.05 |
| DWAS | Invesco Dorsey Wright SmallCa... | U.S. Small Cap Mo... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-07 | 24 | 13.7% | 0.06 |
| SMLF | iShares U.S. Small-Cap Equity... | U.S. Small Cap Mu... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-09 | 22 | 6.4%  | 0.06 |
| VXF  | Vanguard Extended Market ETF     | U.S. Extended Mar... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-10 | 21 | 7.2%  | 0.07 |
| IVOO | Vanguard S&P Mid-Cap 400 ETF     |                      | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-10 | 21 | 4.8%  | 0.07 |
| SPMD | State Street SPDR Portfolio S... | U.S. Mid Cap Equi... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-10 | 21 | 4.8%  | 0.07 |
| MDY  | State Street SPDR S&P MIDCAP ... | U.S. Mid Cap Equi... | cluster_2  | Mixed: VB, SCHM, IJH | LONG | 0.849 | 2026-04-10 | 21 | 4.7%  | 0.07 |

|      |                                  |                      |            |                          |      |       |            |    |       |      |
|------|----------------------------------|----------------------|------------|--------------------------|------|-------|------------|----|-------|------|
| IJH  | iShares Core S&P Mid-Cap ETF     | U.S. Mid Cap Equi... | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-04-10 | 21 | 4.7%  | 0.08 |
| VB   | Vanguard Small-Cap ETF           | U.S. Small Cap Eq... | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-04-10 | 21 | 6.1%  | 0.09 |
| PSCI | Invesco S&P SmallCap Industri... | U.S. Small Cap In... | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-04-13 | 20 | 2.7%  | 0.10 |
| IWP  | iShares Russell Midcap Growth... | U.S. Mid Cap Grow... | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-04-15 | 18 | 1.8%  | 0.15 |
| JETS | U.S. Global Jets ETF             | Industrials          | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-05-06 | 3  | -2.3% | 0.17 |
| VIS  | Vanguard Industrials ETF         | Industrials          | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-04-13 | 20 | 2.1%  | 0.24 |
| JHMM | John Hancock Multifactor Mid ... | U.S. Mid Cap Equi... | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-04-10 | 21 | 3.9%  | 0.24 |
| FIDU | Fidelity MSCI Industrials Ind... | Industrials          | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-04-13 | 20 | 2.1%  | 0.26 |
| NUSC | Nuveen ESG Small-Cap ETF         | U.S. Small Cap ES... | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-04-13 | 20 | 3.0%  | 0.28 |
| XLI  | State Street Industrial Selec... | Industrials          | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-04-13 | 20 | 1.3%  | 0.34 |
| FTXR | First Trust Nasdaq Transporta... | Industrials          | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-04-17 | 16 | -0.5% | 0.41 |
| PKB  | Invesco Building & Constructi... | Industrials          | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-04-13 | 20 | -0.2% | 0.63 |
| IYJ  | iShares U.S. Industrials ETF     | Industrials          | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-04-13 | 20 | 0.7%  | 0.65 |
| IYT  | iShares U.S. Transportation E... | Industrials          | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-04-14 | 19 | 1.4%  | 0.82 |
| FXR  | First Trust Industrials Alpha... | Industrials          | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-04-14 | 19 | -0.4% | 0.85 |
| FIW  | First Trust Water ETF            | Utilities            | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-05-08 | 1  | -1.0% | 0.99 |
| XHB  | State Street SPDR S&P Homebui... | Consumer Discreti... | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-05-06 | 3  | -3.9% | 0.98 |
| FXD  | First Trust Cons. Discret. Al... | Consumer Discreti... | cluster_2  | Mixed: VB, SCHM, IJH     | LONG | 0.849 | 2026-05-06 | 3  | -3.2% | 0.99 |
| NURE | Nuveen Short-Term REIT ETF       | Real Estate          | cluster_14 | Real Estate (14/14)      | LONG | 0.816 | 2026-04-13 | 20 | 5.7%  | 0.00 |
| REZ  | iShares Residential and Multi... | Real Estate          | cluster_14 | Real Estate (14/14)      | LONG | 0.816 | 2026-04-14 | 19 | 3.8%  | 0.01 |
| USRT | iShares Core U.S. REIT ETF       | Real Estate          | cluster_14 | Real Estate (14/14)      | LONG | 0.816 | 2026-04-09 | 22 | 5.2%  | 0.11 |
| PSR  | Invesco Active U.S. Real Esta... | Real Estate          | cluster_14 | Real Estate (14/14)      | LONG | 0.816 | 2026-04-09 | 22 | 4.9%  | 0.12 |
| RWR  | State Street SPDR Dow Jones R... | Real Estate          | cluster_14 | Real Estate (14/14)      | LONG | 0.816 | 2026-04-10 | 21 | 4.4%  | 0.16 |
| SCHH | Schwab U.S. REIT ETF             | Real Estate          | cluster_14 | Real Estate (14/14)      | LONG | 0.816 | 2026-04-09 | 22 | 4.4%  | 0.19 |
| ICF  | iShares Select U.S. REIT ETF     | Real Estate          | cluster_14 | Real Estate (14/14)      | LONG | 0.816 | 2026-04-08 | 23 | 5.6%  | 0.19 |
| FREL | Fidelity MSCI Real Estate Ind... | Real Estate          | cluster_14 | Real Estate (14/14)      | LONG | 0.816 | 2026-04-10 | 21 | 4.0%  | 0.26 |
| XLRE | State Street Real Estate Sele... | Real Estate          | cluster_14 | Real Estate (14/14)      | LONG | 0.816 | 2026-04-09 | 22 | 4.3%  | 0.27 |
| VNQ  | Vanguard Real Estate ETF         | Real Estate          | cluster_14 | Real Estate (14/14)      | LONG | 0.816 | 2026-04-10 | 21 | 4.0%  | 0.29 |
| REET | iShares Global REIT ETF          | Real Estate          | cluster_14 | Real Estate (14/14)      | LONG | 0.816 | 2026-04-13 | 20 | 3.4%  | 0.32 |
| IYR  | iShares U.S. Real Estate ETF     | Real Estate          | cluster_14 | Real Estate (14/14)      | LONG | 0.816 | 2026-04-10 | 21 | 3.9%  | 0.32 |
| RWO  | State Street SPDR Dow Jones G... | Real Estate          | cluster_14 | Real Estate (14/14)      | LONG | 0.816 | 2026-04-13 | 20 | 3.0%  | 0.34 |
| INDS | Pacer Industrial Real Estate ... | Real Estate          | cluster_14 | Real Estate (14/14)      | LONG | 0.816 | 2026-04-14 | 19 | 2.0%  | 0.51 |
| SCHD | Schwab US Dividend Equity ETF    | U.S. Dividend Equ... | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-04-21 | 14 | 2.5%  | 0.00 |
| RDIV | Invesco S&P Ultra Dividend Re... | U.S. Dividend Equ... | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-04-21 | 14 | -0.2% | 0.01 |
| SPYD | State Street SPDR Portfolio S... | U.S. Dividend Equ... | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-04-21 | 14 | 1.0%  | 0.01 |
| AML  | Alerian MLP ETF                  | MLP / Energy Infr... | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-04-29 | 8  | 0.6%  | 0.01 |
| RPV  | Invesco S&P 500 Pure Value ETF   | Value Equity         | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-04-20 | 15 | 0.3%  | 0.08 |
| DIV  | Global X Super Dividend ETF      | U.S. Dividend Equ... | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-04-07 | 24 | 1.3%  | 0.12 |
| MLPA | Global X MLP ETF                 | MLP / Energy Infr... | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-04-29 | 8  | -1.5% | 0.03 |
| HDV  | iShares Core High Dividend ETF   | U.S. Dividend Equ... | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-05-01 | 6  | -1.0% | 0.14 |
| SDOG | ALPS Sector Dividend Dogs ETF    | U.S. Dividend Equ... | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-04-07 | 24 | 1.7%  | 0.28 |
| DHS  | WisdomTree U.S. High Dividend... | U.S. Dividend Equ... | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-04-20 | 15 | -0.1% | 0.30 |
| PEY  | Invesco High Yield Equity Div... | U.S. Dividend Equ... | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-04-13 | 20 | 2.3%  | 0.33 |
| SPHD | Invesco S&P 500 High Dividend... | Dividend Equity      | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-05-05 | 4  | -0.5% | 0.30 |
| MLPX | Global X MLP & Energy Infrast... | MLP / Energy Infr... | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-04-30 | 7  | -3.5% | 0.07 |



|      |                                    |                      |            |                          |      |       |            |    |       |      |
|------|------------------------------------|----------------------|------------|--------------------------|------|-------|------------|----|-------|------|
| ONEY | State Street SPDR Russell 100...   | U.S. Dividend Equ... | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-04-10 | 21 | 2.1%  | 0.52 |
| DVY  | iShares Select Dividend ETF        | U.S. Dividend Equ... | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-04-08 | 23 | -0.6% | 0.82 |
| CDC  | VictoryShares US EQ Income En...   | U.S. Defensive Eq... | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-04-09 | 22 | -0.6% | 0.87 |
| EMLP | First Trust North American En...   | MLP / Energy Infr... | cluster_10 | U.S. Dividend Equity ... | LONG | 0.787 | 2026-04-28 | 9  | -0.6% | 0.56 |
| TDDT | FlexShares iBoxx 3 Year Targe...   | Inflation-Protect... | cluster_19 | Inflation-Protected F... | LONG | 0.787 | 2026-04-01 | 27 | 1.2%  | 0.12 |
| STPZ | PIMCO 1-5 Year U.S. TIPS Inde...   | Inflation-Protect... | cluster_19 | Inflation-Protected F... | LONG | 0.787 | 2026-04-02 | 26 | 0.9%  | 0.14 |
| TDTF | FlexShares iBoxx 5 Year Targe...   | Inflation-Protect... | cluster_19 | Inflation-Protected F... | LONG | 0.787 | 2026-04-09 | 22 | 1.0%  | 0.19 |
| TIPX | State Street SPDR Bloomberg 1...   | Inflation-Protect... | cluster_19 | Inflation-Protected F... | LONG | 0.787 | 2026-04-07 | 24 | 0.8%  | 0.21 |
| TIP  | iShares TIPS Bond ETF              | Inflation-Protect... | cluster_19 | Inflation-Protected F... | LONG | 0.787 | 2026-04-10 | 21 | 0.8%  | 0.22 |
| SCHP | Schwab U.S. TIPS ETF               | Inflation-Protect... | cluster_19 | Inflation-Protected F... | LONG | 0.787 | 2026-04-13 | 20 | 0.6%  | 0.23 |
| SPIP | State Street SPDR Portfolio T...   | Inflation-Protect... | cluster_19 | Inflation-Protected F... | LONG | 0.787 | 2026-04-10 | 21 | 0.7%  | 0.27 |
| LTPZ | Pimco 15 Year U.S. TIPS Index...   | Inflation-Protect... | cluster_19 | Inflation-Protected F... | LONG | 0.787 | 2026-05-11 | 0  | 0.0%  | 0.58 |
| ENZL | iShares MSCI New Zealand ETF       | New Zealand Equity   | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-05-06 | 3  | 0.4%  | 0.01 |
| VIDI | Vident International Equity S...   | International Equ... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-09 | 22 | 6.1%  | 0.02 |
| FNDI | Schwab Fundamental Internatio...   | International Equ... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-08 | 23 | 4.2%  | 0.02 |
| IQDF | FlexShares International Qual...   | International Equ... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-10 | 21 | 5.0%  | 0.03 |
| PXF  | Invesco RAFI Developed Market...   | International Equ... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-08 | 23 | 4.2%  | 0.03 |
| GWX  | State Street SPDR S&P Interna...   | International Equ... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-13 | 20 | 5.1%  | 0.03 |
| VEU  | Vanguard FTSE All World Ex US...   | Global ex-US Equi... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-10 | 21 | 5.0%  | 0.04 |
| HFXI | NYLI FTSE International Equit...   | International Equ... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-09 | 22 | 4.7%  | 0.05 |
| VXUS | Vanguard Total International ...   | Global ex-US Equi... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-10 | 21 | 5.1%  | 0.05 |
| IXUS | iShares Core MSCI Total Inter...   | Global ex-US Equi... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-10 | 21 | 5.1%  | 0.06 |
| CWI  | State Street SPDR MSCI ACWI e...   | Global ex-US Equi... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-10 | 21 | 4.6%  | 0.06 |
| SPDW | State Street SPDR Portfolio D...   | Developed ex-US E... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-10 | 21 | 4.6%  | 0.07 |
| SCHF | Schwab International Equity E...   | Developed ex-US E... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-10 | 21 | 4.6%  | 0.07 |
| ACWX | iShares MSCI ACWI ex U.S. ETF      | Global ex-US Equi... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-10 | 21 | 4.7%  | 0.07 |
| VEA  | Vanguard FTSE Developed Marke...   | Developed ex-US E... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-10 | 21 | 4.5%  | 0.08 |
| VSS  | Vanguard FTSE All-World ex-US S... | Global ex-US Smal... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-13 | 20 | 4.8%  | 0.08 |
| FXA  | Invesco CurrencyShares Austra...   | Currency             | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-13 | 20 | 2.1%  | 0.10 |
| EWN  | iShares MSCI Netherlands Inde...   | Netherlands Equity   | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-10 | 21 | 4.9%  | 0.11 |
| IPKW | Invesco International BuyBack...   | International Equ... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-09 | 22 | 2.5%  | 0.13 |
| FNDI | Schwab Fundamental Internatio...   | International Equ... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-10 | 21 | 3.7%  | 0.13 |
| SCZ  | iShares MSCI EAFE Small-Cap E...   | Developed ex-US E... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-10 | 21 | 3.7%  | 0.16 |
| SCHC | Schwab International Small-Ca...   | Developed ex-US E... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-13 | 20 | 3.4%  | 0.20 |
| DLS  | WisdomTree International Smal...   | International Equ... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-13 | 20 | 2.2%  | 0.20 |
| IDV  | iShares International Select ...   | International Div... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-06 | 25 | 4.4%  | 0.26 |
| FXC  | Invesco CurrencyShares Canadi...   | Currency             | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-20 | 15 | -0.2% | 0.27 |
| FEP  | First Trust Europe AlphaDEX F...   | Europe Equity        | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-09 | 22 | 2.7%  | 0.28 |
| EWI  | iShares MSCI Italy ETF             | Italy Equity         | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-08 | 23 | 4.7%  | 0.28 |
| EWX  | iShares MSCI Mexico ETF            | Mexico Equity        | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-05-05 | 4  | 3.4%  | 0.33 |
| VNQI | Vanguard Global ex-U.S. Real ...   | Real Estate          | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-05-04 | 5  | 2.6%  | 0.41 |
| IMTM | iShares MSCI Intl Momentum Fa...   | International Equ... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-10 | 21 | 2.7%  | 0.41 |
| EWG  | iShares MSCI Germany Index Fu...   | Germany Equity       | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-14 | 19 | 0.8%  | 0.41 |
| EWO  | iShares MSCI Austria ETF           | Austria Equity       | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-10 | 21 | 4.2%  | 0.43 |
| VIGI | Vanguard International Divide...   | International Div... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-14 | 19 | 0.5%  | 0.43 |

|      |                                   |                      |            |                          |      |       |            |     |       |      |
|------|-----------------------------------|----------------------|------------|--------------------------|------|-------|------------|-----|-------|------|
| NFRA | FlexShares STOXX Global Broad...  | Global Equity        | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-09 | 22  | -0.0% | 0.46 |
| VYMI | Vanguard International High D...  | International Div... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-09 | 22  | 1.7%  | 0.46 |
| BBAX | JPMorgan BetaBuilders Develop...  | Asia-Pacific Equi... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-30 | 7   | 0.9%  | 0.51 |
| DWX  | State Street SPDR S&P Interna...  | International Div... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-05-04 | 5   | 1.0%  | 0.54 |
| EWL  | iShares MSCI Switzerland ETF      | Switzerland Equity   | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-15 | 18  | -0.2% | 0.57 |
| PID  | Invesco International Dividen...  | International Div... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-30 | 7   | -0.0% | 0.58 |
| IHDG | WisdomTree International Hedg...  | Alternatives         | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-05-06 | 3   | -1.5% | 0.69 |
| RWX  | State Street SPDR Dow Jones I...  | Real Estate          | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-05-06 | 3   | -1.4% | 0.77 |
| IDLV | Invesco S&P International Dev...  | International Equ... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-30 | 7   | -0.6% | 0.78 |
| EWC  | iShares MSCI Canada Index Fund    | Canada Equity        | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-09 | 22  | 3.0%  | 0.81 |
| EZA  | iShares MSCI South Africa Ind...  | South Africa Equi... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-05-08 | 1   | 0.1%  | 0.67 |
| EBND | State Street SPDR Bloomberg E...  | Emerging Markets ... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-05-06 | 3   | 0.0%  | 0.81 |
| EMLC | VanEck J. P. Morgan EM Local ...  | Emerging Markets ... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-05-06 | 3   | -0.0% | 0.84 |
| HDEF | Xtrackers MSCI EAFE High Divi...  | International Equ... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-07 | 24  | 0.5%  | 0.93 |
| EWD  | iShares MSCI Sweden ETF           | Sweden Equity        | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-05-06 | 3   | -2.6% | 0.96 |
| EWU  | iShares MSCI United Kingdom E...  | United Kingdom Eq... | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-30 | 7   | -1.1% | 0.98 |
| IGF  | iShares Global Infrastructure...  | Global Equity        | cluster_1  | Mixed: VEA, SPDW, SCHF   | FLAT | 0.622 | 2026-04-30 | 7   | -1.6% | 0.76 |
| SLX  | VanEck Steel ETF                  | Materials            | cluster_12 | Materials (13/17)        | FLAT | 0.550 | 2026-04-09 | 22  | 10.8% | 0.00 |
| XME  | State Street SPDR S&P Metals ...  | Materials            | cluster_12 | Materials (13/17)        | FLAT | 0.550 | 2026-04-15 | 18  | 7.7%  | 0.04 |
| DBB  | Invesco DB Base Metals Fund       | Industrial Metals    | cluster_12 | Materials (13/17)        | FLAT | 0.550 | 2026-04-10 | 21  | 6.8%  | 0.25 |
| URA  | Global X Uranium ETF              | Materials            | cluster_12 | Materials (13/17)        | FLAT | 0.550 | 2026-04-14 | 19  | 8.3%  | 0.35 |
| COPX | Global X Copper Miners ETF        | Materials            | cluster_12 | Materials (13/17)        | FLAT | 0.550 | 2026-05-08 | 1   | 1.9%  | 0.55 |
| NORW | Global X MSCI Norway ETF          | Norway Equity        | cluster_12 | Materials (13/17)        | FLAT | 0.550 | 2025-12-09 | 104 | 31.8% | 0.63 |
| NANR | State Street SPDR S&P North A...  | Materials            | cluster_12 | Materials (13/17)        | FLAT | 0.550 | 2026-05-11 | 0   | 0.0%  | 0.70 |
| IGE  | iShares North American Natura...  | Energy               | cluster_12 | Materials (13/17)        | FLAT | 0.550 | 2026-05-01 | 6   | 0.1%  | 0.64 |
| MOO  | VanEck Agribusiness ETF           | Consumer Staples     | cluster_12 | Materials (13/17)        | FLAT | 0.550 | 2026-05-06 | 3   | -2.8% | 0.86 |
| PFF  | VanEck Preferred Securities e...  | Preferred Stock      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-14 | 19  | 4.1%  | 0.00 |
| PFF  | iShares Preferred and Income ...  | Preferred Stock      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-15 | 18  | 1.5%  | 0.05 |
| HYEM | VanEck Emerging Markets High ...  | Emerging Markets ... | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-13 | 20  | 1.3%  | 0.12 |
| PSK  | State Street SPDR ICE Preferr...  | Preferred Stock      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-17 | 16  | 0.2%  | 0.41 |
| VRP  | Invesco Variable Rate Preferr...  | Preferred Stock      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-10 | 21  | 0.5%  | 0.45 |
| FPE  | First Trust Preferred Securit...  | Preferred Stock      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-14 | 19  | 0.6%  | 0.46 |
| PGX  | Invesco Preferred ETF             | Preferred Stock      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-17 | 16  | -0.4% | 0.58 |
| HYLS | First Trust Tactical High Yie...  | High Yield Bond      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-09 | 22  | 0.5%  | 0.64 |
| SHYG | iShares 0-5 Year High Yield C...  | High Yield Bond      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-07 | 24  | 0.9%  | 0.76 |
| FALN | iShares Fallen Angels USD Bon...  | High Yield Bond      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-10 | 21  | 0.6%  | 0.84 |
| JNK  | State Street SPDR Bloomberg H...  | High Yield Bond      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-08 | 23  | 0.4%  | 0.87 |
| SJNK | State Street SPDR Bloomberg S...  | High Yield Bond      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-07 | 24  | 0.9%  | 0.89 |
| HYS  | PIMCO 0-5 Year High Yield Cor...  | High Yield Bond      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-08 | 23  | 0.1%  | 0.89 |
| BBHY | JPMorgan BetaBuilders USD Hig...  | High Yield Bond      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-08 | 23  | 0.3%  | 0.90 |
| HYG  | iShares iBoxx \$ High Yield Co... | High Yield Bond      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-08 | 23  | 0.3%  | 0.91 |
| HYLB | Xtrackers USD High Yield Corp...  | High Yield Bond      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-08 | 23  | 0.2%  | 0.92 |
| ANGL | VanEck Fallen Angel High Yiel...  | High Yield Bond      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-09 | 22  | 0.2%  | 0.95 |
| PHB  |                                   |                      | cluster_9  | High Yield Bond (10) ... | LONG | 0.489 | 2026-04-10 | 21  |       |      |
| IWN  | iShares Russell 2000 Value ETF    | U.S. Small Cap Va... | cluster_7  | Mixed: VBR, MDYV, IVOV   | LONG | 0.380 | 2026-04-09 | 22  | 5.8%  | 0.01 |

|      |                                  |                      |            |                        |      |       |            |    |       |      |
|------|----------------------------------|----------------------|------------|------------------------|------|-------|------------|----|-------|------|
| VTWV | Vanguard Russell 2000 Value E... | U.S. Small Cap Va... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-09 | 22 | 5.7%  | 0.01 |
| OMFS | Invesco Russell 2000 Dynamic ... | U.S. Equity          | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-09 | 22 | 6.3%  | 0.02 |
| ISMD | Inspire Small/Mid Cap ETF        | U.S. Small/Mid Ca... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-09 | 22 | 7.5%  | 0.02 |
| EES  | WisdomTree U.S. SmallCap Fund    | Small Cap Value E... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-08 | 23 | 5.1%  | 0.02 |
| VIOO | Vanguard S&P Small-Cap 600 ETF   | U.S. Small Cap Eq... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-09 | 22 | 5.0%  | 0.04 |
| IJR  | iShares Core S&P Small-Cap ETF   | U.S. Small Cap Eq... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-09 | 22 | 5.0%  | 0.05 |
| SPSM | State Street SPDR Portfolio S... | U.S. Small Cap Eq... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-09 | 22 | 5.0%  | 0.05 |
| PRFZ | Invesco RAFI US 1500 Small-Mi... | U.S. Small/Mid Ca... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | 6.8%  | 0.05 |
| FNDA | Schwab Fundamental U.S. Small... | U.S. Small Cap Fu... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | 5.4%  | 0.09 |
| FYX  | First Trust Small Cap Core Al... | U.S. Small Cap Eq... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-08 | 23 | 6.5%  | 0.11 |
| FNX  | First Trust Mid Cap Core Alph... | U.S. Mid/Small Ca... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | 3.9%  | 0.12 |
| RWJ  | Invesco S&P SmallCap 600 Reve... | U.S. Small Cap Eq... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-09 | 22 | 4.6%  | 0.15 |
| SLYV | State Street SPDR S&P 600 Sma... | U.S. Small Cap Va... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | 4.5%  | 0.16 |
| IJS  | iShares S&P SmallCap 600 Valu... | U.S. Small Cap Va... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | 4.5%  | 0.17 |
| RFV  | Invesco S&P MidCap 400 Pure V... | Leveraged / Inver... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | 2.9%  | 0.17 |
| VIOV | Vanguard S&P Small-Cap 600 Va... | U.S. Small Cap Va... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | 4.5%  | 0.18 |
| EZM  | WisdomTree U.S. MidCap Fund      | Mid Cap Value Equ... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | 3.7%  | 0.23 |
| VBR  | Vanguard Small-Cap Value ETF     | U.S. Small Cap Va... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | 3.3%  | 0.32 |
| DES  | WisdomTree U.S. SmallCap Divi... | U.S. Small Cap Di... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-08 | 23 | 3.1%  | 0.39 |
| PSP  | Invesco Global Listed Private... | Global Equity        | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-15 | 18 | -1.1% | 0.42 |
| KCE  | State Street SPDR S&P Capital... | Financials           | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-13 | 20 | 4.7%  | 0.49 |
| RWK  | Invesco S&P MidCap 400 Revenu... | U.S. Mid Cap Equi... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-09 | 22 | 2.7%  | 0.51 |
| RZV  | Invesco S&P SmallCap 600 Pure... | Leveraged / Inver... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-13 | 20 | 3.5%  | 0.58 |
| IJJ  | iShares S&P Mid-Cap 400 Value... | U.S. Mid Cap Value   | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | 2.2%  | 0.65 |
| IVOV | Vanguard S&P Mid-Cap 400 Valu... | U.S. Mid Cap Value   | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | 2.1%  | 0.65 |
| MDYV | State Street SPDR S&P 400 Mid... | U.S. Mid Cap Value   | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | 2.2%  | 0.65 |
| DON  | WisdomTree U.S. MidCap Divide... | U.S. Mid Cap Divi... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-13 | 20 | 0.9%  | 0.72 |
| IAI  | iShares U.S. Broker-Dealers &... | Financials           | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | 4.6%  | 0.76 |
| FYT  | First Trust Small Cap Value A... | Value Equity         | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-07 | 24 | 4.3%  | 0.82 |
| XSLV | Invesco S&P SmallCap Low Vola... | U.S. Small Cap Lo... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-09 | 22 | -0.6% | 0.93 |
| SDVY | First Trust SMID Cap Rising D... | Dividend Equity      | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | 1.0%  | 0.93 |
| KRE  | State Street SPDR S&P Regiona... | Financials           | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-09 | 22 | -1.9% | 0.93 |
| SMDV | ProShares Russell 2000 Divide... | U.S. Dividend Equ... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-09 | 22 | -1.0% | 0.94 |
| REM  | iShares Mortgage Real Estate ... | Real Estate          | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-13 | 20 | 0.1%  | 0.95 |
| KBE  | State Street SPDR S&P Bank ETF   | Financials           | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-09 | 22 | -1.1% | 0.97 |
| FNCL | Fidelity MSCI Financials Inde... | Financials           | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-13 | 20 | -0.8% | 0.99 |
| XLF  | State Street Financial Select... | Financials           | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-13 | 20 | -0.9% | 1.00 |
| REGL | ProShares S&P MidCap 400 Divi... | Dividend Equity      | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | -0.9% | 1.00 |
| IAT  | iShares U.S. Regional Banks E... | Financials           | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-10 | 21 | -1.4% | 1.00 |
| SMMV | iShares MSCI USA Small-Cap Mi... | U.S. Small Cap Lo... | cluster_7  | Mixed: VBR, MDYV, IVOV | LONG | 0.380 | 2026-04-09 | 22 | -1.2% | 1.00 |
| HYD  | VanEck High Yield Muni ETF       | Municipal Bond       | cluster_16 | Municipal Bond (13/13) | FLAT | 0.244 | 2026-04-08 | 23 | 0.9%  | 0.50 |
| HYMB | State Street SPDR Nuveen ICE ... | Municipal Bond       | cluster_16 | Municipal Bond (13/13) | FLAT | 0.244 | 2026-04-09 | 22 | 0.5%  | 0.53 |
| FMB  | First Trust Managed Municipal... | Municipal Bond       | cluster_16 | Municipal Bond (13/13) | FLAT | 0.244 | 2026-04-17 | 16 | 0.0%  | 0.67 |
| NYF  | iShares New York Muni Bond ETF   | Municipal Bond       | cluster_16 | Municipal Bond (13/13) | FLAT | 0.244 | 2026-04-15 | 18 | 0.4%  | 0.69 |
| PZA  | Invesco National AMT-Free Mun... | Municipal Bond       | cluster_16 | Municipal Bond (13/13) | FLAT | 0.244 | 2026-04-10 | 21 | 0.6%  | 0.73 |

|      |                                   |                      |            |                        |      |        |            |     |       |      |
|------|-----------------------------------|----------------------|------------|------------------------|------|--------|------------|-----|-------|------|
| ITM  | VanEck Intermediate Muni ETF      | Municipal Bond       | cluster_16 | Municipal Bond (13/13) | FLAT | 0.244  | 2026-05-08 | 1   | 0.0%  | 0.75 |
| MUB  | iShares National Muni Bond ETF    | Municipal Bond       | cluster_16 | Municipal Bond (13/13) | FLAT | 0.244  | 2026-04-17 | 16  | -0.1% | 0.77 |
| CMF  | iShares California Muni Bond ...  | Municipal Bond       | cluster_16 | Municipal Bond (13/13) | FLAT | 0.244  | 2026-05-06 | 3   | 0.0%  | 0.86 |
| TFI  | State Street SPDR Nuveen ICE ...  | Municipal Bond       | cluster_16 | Municipal Bond (13/13) | FLAT | 0.244  | 2026-05-06 | 3   | -0.0% | 0.89 |
| SUB  | iShares Short-Term National M...  | Municipal Bond       | cluster_16 | Municipal Bond (13/13) | FLAT | 0.244  | 2026-05-11 | 0   | 0.0%  | 0.90 |
| EPI  | WisdomTree India Earnings Fund    | India Equity         | cluster_25 | India Equity (3/4)     | FLAT | 0.084  | 2026-04-17 | 16  | -5.1% | 0.83 |
| EWM  | iShares MSCI Malaysia Index F...  | Malaysia Equity      | cluster_18 | Mixed: ILF, EWZ, ECH   | FLAT | 0.072  | 2026-04-22 | 13  | 3.3%  | 0.00 |
| VNM  | VanEck Vietnam ETF                | Vietnam Equity       | cluster_18 | Mixed: ILF, EWZ, ECH   | FLAT | 0.072  | 2026-04-13 | 20  | 6.5%  | 0.07 |
| TUR  | iShares MSCI Turkey ETF           | Turkey Equity        | cluster_18 | Mixed: ILF, EWZ, ECH   | FLAT | 0.072  | 2026-04-10 | 21  | 3.9%  | 0.31 |
| ECH  | iShares MSCI Chile ETF            | Chile Equity         | cluster_18 | Mixed: ILF, EWZ, ECH   | FLAT | 0.072  | 2026-05-06 | 3   | -2.0% | 0.98 |
| RSX  |                                   |                      | cluster_18 | Mixed: ILF, EWZ, ECH   | FLAT | 0.072  | 2023-07-27 | 699 |       |      |
| IHF  | iShares U.S. Health Care Prov...  | Healthcare           | cluster_11 | Healthcare (16/17)     | LONG | 0.021  | 2026-05-08 | 1   | 0.2%  | 0.00 |
| PSCH | Invesco S&P SmallCap Health C...  | U.S. Small Cap He... | cluster_11 | Healthcare (16/17)     | LONG | 0.021  | 2026-04-13 | 20  | 4.8%  | 0.06 |
| ARKG | ARK Genomic Revolution ETF        | Healthcare           | cluster_11 | Healthcare (16/17)     | LONG | 0.021  | 2026-04-15 | 18  | 0.0%  | 0.45 |
| FXH  | First Trust Health Care Alpha...  | Healthcare           | cluster_11 | Healthcare (16/17)     | LONG | 0.021  | 2026-04-10 | 21  | 0.5%  | 0.78 |
| IBB  | iShares Biotechnology ETF         | Healthcare           | cluster_11 | Healthcare (16/17)     | LONG | 0.021  | 2026-05-06 | 3   | -2.2% | 0.81 |
| LDUR | PIMCO Enhanced Low Duration A...  | Fixed Income         | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-04-07 | 24  | 0.4%  | 0.29 |
| CMBS | iShares CMBS Bond ETF             | Mortgage-Backed F... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-08 | 1   | -0.1% | 0.41 |
| IAGG | iShares International Aggrega...  | International Fix... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-08 | 1   | -0.3% | 0.42 |
| BNDX | Vanguard Total International ...  | International Fix... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-08 | 1   | -0.3% | 0.41 |
| SPSB | State Street SPDR Portfolio S...  | Fixed Income         | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-04-09 | 22  | 0.1%  | 0.71 |
| IGLB | iShares 10 Year Investment Gr...  | Investment Grade ... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-08 | 1   | -0.4% | 0.76 |
| USIG | iShares Broad USD Investment ...  | Investment Grade ... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-06 | 3   | -0.2% | 0.81 |
| VCLT | Vanguard Long-Term Corporate ...  | Investment Grade ... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-08 | 1   | -0.4% | 0.78 |
| GTO  | Invesco Total Return Bond ETF     | Active Fixed Inco... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-07 | 2   | 0.0%  | 0.80 |
| FLCO | Franklin Investment Grade Cor...  | Active Fixed Inco... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-07 | 2   | 0.3%  | 0.81 |
| SPLB | State Street SPDR Portfolio L...  | Investment Grade ... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-08 | 1   | -0.3% | 0.78 |
| CORP | PIMCO Investment Grade Corpor...  | Investment Grade ... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-06 | 3   | -0.2% | 0.83 |
| GNMA | iShares GNMA Bond ETF             | Mortgage-Backed F... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-07 | 2   | -0.1% | 0.79 |
| BAB  | Invesco Taxable Municipal Bon...  | Taxable Municipal... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-11 | 0   | 0.0%  | 0.65 |
| IGIB | iShares 5-10 Year Investment ...  | Investment Grade ... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-07 | 2   | 0.1%  | 0.87 |
| SPIB | State Street SPDR Portfolio I...  | Investment Grade ... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-06 | 3   | -0.1% | 0.90 |
| LQD  | iShares iBoxx \$ Investment Gr... | Investment Grade ... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-08 | 1   | -0.2% | 0.85 |
| AGGY | WisdomTree Yield Enhanced U.S...  | Aggregate Fixed I... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-08 | 1   | -0.3% | 0.84 |
| VCIT | Vanguard Intermediate-Term Co...  | Investment Grade ... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-07 | 2   | 0.1%  | 0.90 |
| IUSB | iShares Core Universal USD Bo...  | Aggregate Fixed I... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-08 | 1   | -0.2% | 0.87 |
| BSV  | Vanguard Short-Term Bond ETF      | Fixed Income         | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-06 | 3   | -0.1% | 0.91 |
| QLTA | iShares Aaa A Rated Corporate...  | Investment Grade ... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-08 | 1   | -0.2% | 0.90 |
| FBND | Fidelity Total Bond ETF           | Active Fixed Inco... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-08 | 1   | -0.3% | 0.87 |
| VMBS | Vanguard Mortgage-Backed Secu...  | Mortgage-Backed F... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-11 | 0   | 0.0%  | 0.86 |
| BOND | PIMCO Active Bond Exchange-Tr...  | Active Fixed Inco... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-11 | 0   | 0.0%  | 0.91 |
| MBB  | iShares MBS ETF                   | Mortgage-Backed F... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-11 | 0   | 0.0%  | 0.87 |
| GVI  | iShares Intermediate Governme...  | Fixed Income         | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-11 | 0   | 0.0%  | 0.93 |
| AGG  | iShares Core U.S. Aggregate B...  | Aggregate Fixed I... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-11 | 0   | 0.0%  | 0.89 |
| BND  | Vanguard Total Bond Market ETF    | Aggregate Fixed I... | cluster_5  | Mixed: BND, AGG, SCHZ  | LONG | -0.177 | 2026-05-11 | 0   | 0.0%  | 0.90 |

Sid Mathur's Thematic ETF Screener – Daily Report

|      |                                  |                      |            |                          |      |        |            |    |       |      |
|------|----------------------------------|----------------------|------------|--------------------------|------|--------|------------|----|-------|------|
| SPAB | State Street SPDR Portfolio A... | Aggregate Fixed I... | cluster_5  | Mixed: BND, AGG, SCHZ    | LONG | -0.177 | 2026-05-11 | 0  | 0.0%  | 0.89 |
| SCHZ | Schwab US Aggregate Bond ETF     | Aggregate Fixed I... | cluster_5  | Mixed: BND, AGG, SCHZ    | LONG | -0.177 | 2026-05-11 | 0  | 0.0%  | 0.93 |
| SPMB | State Street SPDR Portfolio M... | Mortgage-Backed F... | cluster_5  | Mixed: BND, AGG, SCHZ    | LONG | -0.177 | 2026-05-11 | 0  | 0.0%  | 0.89 |
| AGZ  | iShares Agency Bond ETF          | Agency Fixed Inco... | cluster_5  | Mixed: BND, AGG, SCHZ    | LONG | -0.177 | 2026-05-11 | 0  | 0.0%  | 0.98 |
| SIVR | abrdn Physical Silver Shares ... | Silver               | cluster_15 | Mixed: DBP, GLTR, GDXJ   | FLAT | -0.193 | 2026-05-11 | 0  | 0.0%  | 0.15 |
| SLV  | iShares Silver Trust             | Silver               | cluster_15 | Mixed: DBP, GLTR, GDXJ   | FLAT | -0.193 | 2026-05-11 | 0  | 0.0%  | 0.15 |
| PPLT | abrdn Physical Platinum Share... | Platinum             | cluster_15 | Mixed: DBP, GLTR, GDXJ   | FLAT | -0.193 | 2026-05-11 | 0  | 0.0%  | 0.50 |
| SILJ | Amplify Junior Silver Miners ... | Materials            | cluster_15 | Mixed: DBP, GLTR, GDXJ   | FLAT | -0.193 | 2026-05-11 | 0  | 0.0%  | 0.52 |
| SIL  | Global X Silver Miners ETF       | Materials            | cluster_15 | Mixed: DBP, GLTR, GDXJ   | FLAT | -0.193 | 2026-05-11 | 0  | 0.0%  | 0.59 |
| XLP  | State Street Consumer Staples... | Consumer Staples     | cluster_17 | Consumer Staples (6/1... | LONG | -0.232 | 2026-05-05 | 4  | -0.8% | 0.00 |
| KXI  | iShares Global Consumer Stapl... | Consumer Staples     | cluster_17 | Consumer Staples (6/1... | LONG | -0.232 | 2026-05-06 | 3  | -1.6% | 0.02 |
| IYK  | iShares U.S. Consumer Staples... | Consumer Staples     | cluster_17 | Consumer Staples (6/1... | LONG | -0.232 | 2026-05-06 | 3  | -0.2% | 0.01 |
| VDC  | Vanguard Consumer Staples ETF    | Consumer Staples     | cluster_17 | Consumer Staples (6/1... | LONG | -0.232 | 2026-05-01 | 6  | -1.7% | 0.04 |
| FSTA | Fidelity MSCI Consumer Staple... | Consumer Staples     | cluster_17 | Consumer Staples (6/1... | LONG | -0.232 | 2026-05-01 | 6  | -1.8% | 0.04 |
| UUP  | Invesco DB USD Index Bullish ... | Currency             | cluster_26 | Currency (1/2)           | FLAT | -1.000 | 2026-04-27 | 10 | -0.4% | 0.47 |

## Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

| ticker | name                               | category                 | cluster_id | cluster_size | etf_signal | max_corr_to_others | median_corr_to_others | abs_pc1_loading | compo |
|--------|------------------------------------|--------------------------|------------|--------------|------------|--------------------|-----------------------|-----------------|-------|
| RSX    |                                    |                          | cluster_18 | 11           | LONG       | 0.371              | 0.052                 | 0.00021         |       |
| UNG    | United States Natural Gas Fund ... | Natural Gas              | cluster_18 | 11           | FLAT       | 0.492              | 0.081                 | 0.00017         |       |
| GBIL   | Goldman Sachs Access Treasury 0... | Cash / Ultra-Short Fi... | cluster_23 | 9            | LONG       | 0.692              | 0.040                 | 0.00007         |       |
| VNLA   | Janus Henderson Short Duration ... | Cash / Ultra-Short Fi... | cluster_23 | 9            | LONG       | 0.635              | 0.221                 | 0.00034         |       |
| FXV    | Invesco CurrencyShares Japanese... | Currency                 | cluster_24 | 6            | FLAT       | 0.732              | 0.094                 | 0.00022         |       |
| DBA    | Invesco DB Agriculture Fund        | Agricultural Commodit... | cluster_18 | 11           | FLAT       | 0.591              | 0.174                 | 0.00038         |       |
| FTSM   | First Trust Enhanced Short Matu... | Cash / Ultra-Short Fi... | cluster_23 | 9            | FLAT       | 0.787              | 0.130                 | 0.00030         |       |
| BIL    | State Street SPDR Bloomberg 1-3... | Cash / Ultra-Short Fi... | cluster_23 | 9            | LONG       | 0.825              | 0.021                 | 0.00000         |       |
| SHV    | iShares 0-1 Year Treasury Bond ... | Cash / Ultra-Short Fi... | cluster_23 | 9            | FLAT       | 0.825              | 0.089                 | 0.00022         |       |
| NEAR   | iShares Short Duration Bond Act... | Cash / Ultra-Short Fi... | cluster_23 | 9            | FLAT       | 0.720              | 0.182                 | 0.00041         |       |
| MINT   | PIMCO Enhanced Short Maturity A... | Cash / Ultra-Short Fi... | cluster_23 | 9            | LONG       | 0.656              | 0.188                 | 0.00047         |       |
| ICSH   | iShares Ultra Short Duration Bo... | Cash / Ultra-Short Fi... | cluster_23 | 9            | FLAT       | 0.793              | 0.162                 | 0.00038         |       |
| TUR    | iShares MSCI Turkey ETF            | Turkey Equity            | cluster_18 | 11           | LONG       | 0.487              | 0.219                 | 0.00056         |       |
| GSY    | Invesco Ultra Short Duration ETF   | Cash / Ultra-Short Fi... | cluster_23 | 9            | LONG       | 0.793              | 0.179                 | 0.00043         |       |
| LDUR   | PIMCO Enhanced Low Duration Act... | Fixed Income             | cluster_5  | 58           | LONG       | 0.844              | 0.124                 | 0.00033         |       |
| FXF    | Invesco CurrencyShares Swiss Fr... | Currency                 | cluster_24 | 6            | LONG       | 0.818              | 0.203                 | 0.00047         |       |
| CMBS   | iShares CMBS Bond ETF              | Mortgage-Backed Fixed... | cluster_5  | 58           | LONG       | 0.876              | 0.160                 | 0.00030         |       |
| LMBS   | First Trust Low Duration Opport... | Mortgage-Backed Fixed... | cluster_5  | 58           | FLAT       | 0.836              | 0.231                 | 0.00044         |       |
| SHM    | State Street SPDR Nuveen ICE Sh... | Municipal Bond           | cluster_16 | 13           | FLAT       | 0.877              | 0.250                 | 0.00044         |       |
| VNM    | VanEck Vietnam ETF                 | Vietnam Equity           | cluster_18 | 11           | LONG       | 0.452              | 0.334                 | 0.00082         |       |
| PPLT   | abrdn Physical Platinum Shares ... | Platinum                 | cluster_15 | 13           | LONG       | 0.750              | 0.269                 | 0.00072         |       |
| SUB    | iShares Short-Term National Mun... | Municipal Bond           | cluster_16 | 13           | LONG       | 0.877              | 0.263                 | 0.00051         |       |
| KURE   | KraneShares MSCI All China Heal... | China Equity             | cluster_13 | 15           | FLAT       | 0.804              | 0.271                 | 0.00070         |       |
| DBB    | Invesco DB Base Metals Fund        | Industrial Metals        | cluster_12 | 17           | LONG       | 0.751              | 0.357                 | 0.00082         |       |
| EIDO   | iShares MSCI Indonesia ETF         | Indonesia Equity         | cluster_18 | 11           | FLAT       | 0.514              | 0.374                 | 0.00105         |       |
| EPHE   | iShares MSCI Philippines ETF       | Philippines Equity       | cluster_18 | 11           | FLAT       | 0.578              | 0.409                 | 0.00105         |       |
| KBA    | KraneShares Bosera MSCI China A... | China Equity             | cluster_13 | 15           | LONG       | 0.855              | 0.254                 | 0.00070         |       |
| FLTR   | VanEck IG Floating Rate ETF        | Floating Rate Fixed I... | cluster_22 | 9            | FLAT       | 0.864              | 0.332                 | 0.00070         |       |
| ECH    | iShares MSCI Chile ETF             | Chile Equity             | cluster_18 | 11           | LONG       | 0.637              | 0.429                 | 0.00116         |       |
| CNXT   | VanEck ChiNext Innovators ETF      | China Equity             | cluster_13 | 15           | LONG       | 0.905              | 0.242                 | 0.00062         |       |

## Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

|                          | scope    | parameter                | value   |
|--------------------------|----------|--------------------------|---------|
| short_ma                 | screen   | short_ma                 | 5       |
| buy_ma                   | screen   | buy_ma                   | 50      |
| sell_ma                  | screen   | sell_ma                  | 20      |
| vol_window               | screen   | vol_window               | 20      |
| horizon                  | screen   | horizon                  | 5       |
| cluster_smoothing_window | screen   | cluster_smoothing_window | 3       |
| sell_cushion_k           | screen   | sell_cushion_k           | 0.5     |
| sell_cushion_sd_window   | screen   | sell_cushion_sd_window   | 20      |
| min_hold_days            | backtest | min_hold_days            | 30      |
| vol_window1              | backtest | vol_window               | 60      |
| cov_window               | backtest | cov_window               | 90      |
| cap                      | backtest | cap                      | 0.2     |
| target_vol               | backtest | target_vol               | 0.1     |
| rebal_freq               | backtest | rebal_freq               | monthly |
| cost_bps                 | backtest | cost_bps                 | 10      |
| periods_per_year         | backtest | periods_per_year         | 252     |
| abs_mom_window           | backtest | abs_mom_window           | 252     |
| abs_mom_threshold        | backtest | abs_mom_threshold        | 0       |
| exit_on_mom_loss         | backtest | exit_on_mom_loss         | FALSE   |
| rearm_blocked_buys       | backtest | rearm_blocked_buys       | TRUE    |